NEW EDITION

विश्वविद्यालय एवं स्वशासी महाविद्यालयों के विद्यार्थियों हेतु

VVP विश्वविद्यालय प्रकाशन (रजि.)

ग्वालियर द्वारा प्रकाशित

20 प्रश्न®

(परीक्षोपयोगी महत्वपूर्ण मॉडल प्रश्नोत्तर)

A reliable book based on the prescribed unified syllabus as per the new National Education Policy.

असली पुस्तक की पहचान

Business Mathematics

[Minor-1]

B.Com. First Year

प्रकाशक

VVP विश्वविद्यालय प्रकाशन (रजिस्टर्ड)

महादजी पार्क, हरिदर्शन विद्यालय मार्ग, ग्वालियर 474 009 (मध्यप्रदेश)

According to New Education Policy

VVP (Regd.)

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BUSINESS MATHEMATICS

VVP (Regd.)

MINOR - 1

B. COM. FIRST YEAR

New Syllabus

Business Mathematics

[Maximum Marks : 30+70]

Unit–I. Ancient Vedic and other calculaton methods of India, practical use of Vedic mathematics in the current context.

Methods and practice of quick calculation of addition, multiplication, division, square and square root of numbers through Vedic mathematics, method of quick verification of answers from Digit Sum.

Unit–II. Rules for signs in Algebra and pratice, Rules for calculation (BODMAS) and practice, Simultaneous Equation- Meaning, Characteristic, types, calculations (with word problems).

Unit–III. Theory of indices (preliminary knowledge only formulae), Logarithms and Antilogarithms- principles and calculations, Percentage.

Unit–IV. Ratio, Proportion, Discount, Brokerage.

Unit–V. Commission, Average, Profit and Loss.

Unit–VI. Simple interest, Compound interest.

Assessment and Evaluation

Suggested Continuous Evaluation Methods		
Maximum Marks : 100 Continuous Comprehensive Evaluation (CCE): 30 Marks University Exam (UE): 70 Marks		
Internal Assessment: Continuous Comprehensive Evaluation (CCE): 30	**Class Test Assignment/Presentation/Quiz/ Peer Teaching**	**30** Marks
External Assessment: University Exam: 70	Section (A) : Multiple Choice Questions Section (B) : Short Questions Section (C) : Long Questions	**70** Marks
		Total Marks 100

Note : In the Examination, some short answer questions will also be asked. So the students are advised to go through all the long answer questions thoroughly as given in this book and write down the answer as much as needed/asked in the question paper.

BUSINESS MATHEMATICS

UNIT-I

Long and Short Answer type Questions :

ANCIENT VEDIC MATHEMATICS

Q.1. What are the various Ancient Vedic calculation methods of India?

Or

Throw some light on Ancient Vedic calculation methods of India.

Ans. Ancient India made significant contributions to mathematics, particularly through the development of the decimal number system and the concept of zero as a placeholder. This allowed for efficient representation and manipulation of numbers of varying magnitudes. Beyond these fundamental concepts, ancient Indians also developed sophisticated methods for performing various mathematical operations, especially in relation to arithmetic, algebra, and geometry. These methods are broadly associated with what is now known as Vedic Mathematics.

Some of key aspects of ancient Indian calculation methods of Vedic Mathematics:

(i) Vedic Mathematics is an ancient Indian system using special "sutras" or formulas for performing mathematical operations quickly and efficiently.

(ii) The system was popularized by Swami Bharati Krishna Tirtha in the mid-20th century, who published a book titled "Vedic Mathematics", based on his study of ancient Indian scriptures.

(iii) Vedic Maths comprises 16 main sutras and 13 sub-sutras that provide shortcut techniques for mental calculations and problem-solving across various areas of mathematics, including arithmetic, algebra, and geometry.

Some of the notable sutras and their applications include:

1. **Ekadhikena Purvena (By One More than the Previous One):** Used for multiplying numbers whose last digits sum to a power of 10, and also for squaring numbers ending in 5. For example, to multiply 63×67, since $3 + 7 = 10$, you can multiply $6 \times (6 + 1)$ to get 42 and then multiply 3×7 to get 21, resulting in 4221.

2. **Nikhilam Navatashcaramam Dashatah (All From Nine and the Last From Ten):** This sutra is used for calculating complements, performing subtraction, multiplication by numbers close to a base, and division by numbers less than a base.
3. **Urdhva-Tiryagbyham (Vertically and Crosswise):** This is a general multiplication technique that involves multiplying numbers by breaking them into components and performing calculations vertically and diagonally.
4. **Paravartya Yojayet (Transpose and Apply):** This sutra is particularly useful for division, especially when the divisor is close to a base number.
5. **Shunyam Saamyasamuccaye (When the Sum is the Same, that Sum is Zero):** This principle finds applications in solving linear equations.

Benefits of Vedic mathematics :

(i) Speed and Efficiency : Vedic Maths offers significantly faster calculation methods compared to traditional mathematics, which is beneficial in competitive exams and daily life.

(ii) Reduced Errors : The simplified nature of the methods minimizes the chances of committing errors during calculations.

(iii) Improved Mental Agility : Regular practice with Vedic Maths enhances cognitive skills like memory, concentration, and logical thinking.

(iv) Versatility : These techniques can be applied to a wide range of mathematical problems, from basic arithmetic to complex algebraic equations.

(v) Boosts Confidence : The ability to perform rapid mental calculations instills confidence and reduces math anxiety.

Q.2. What are the practical applications of Vedic mathematics in the current context?

Ans. Vedic mathematics, an ancient Indian system, emphasizes mental calculations and offers unique techniques to solve mathematical problems quickly and efficiently. These techniques, based on 16 main sutras and 13 sub-sutras, can be applied across various branches of mathematics, including arithmetic, algebra, geometry, and calculus.

This is how Vedic mathematics finds practical application in today's world:

Competitive examinations:

(i) Vedic Maths helps students solve arithmetic, algebra, and geometry problems with greater ease and confidence, boosting their performance in competitive exams like IIT-JEE, NEET, SAT, and GMAT, CAT, UPSC, Banking exams, and Olympiads.

(ii) It significantly reduces the time required for calculations, which is a crucial advantage in timed tests.

Mental math and everyday life:

(i) Vedic Maths promotes mental arithmetic, allowing individuals to perform calculations quickly and accurately without relying heavily on calculators or spreadsheets.

(ii) It can be applied in real-life scenarios like budgeting, splitting bills, calculating discounts, estimating distances and costs, and managing daily finances.

(iii) This ability to calculate mentally enhances cognitive functions like memory, concentration, and problem-solving skills.

Education:

(i) Vedic Maths offers a different and engaging approach to learning mathematics, which can be particularly beneficial for students who struggle with traditional methods.

(ii) It simplifies complex mathematical operations, making them easier to grasp and apply.

(iii) Vedic Maths can be incorporated into school curricula as a supplementary tool to enhance students' understanding and speed in solving mathematical problems.

(iv) It fosters creativity and promotes a deeper appreciation for the logic and patterns within mathematics.

Professions:

(i) In fast-paced work environments, especially in fields like finance, engineering, and data analysis, the ability to perform quick mental calculations is a significant advantage.

(ii) Vedic Maths techniques enhance efficiency during routine tasks, decision-making, and financial calculations.

(iii) It can streamline tasks like budgeting, forecasting, preparing financial reports, and analyzing data.

Computer science and algorithm development:

(i) While Vedic mathematics was developed long before the advent of computer science, some principles can inspire the development of more efficient algorithms for tasks like multiplication, sorting, and searching.

(ii) The emphasis on pattern recognition and elegant solutions can lead to innovative approaches in algorithm design.

(iii) Researchers are exploring the potential of integrating Vedic Maths with machine learning and AI algorithms for faster processing times and enhanced computational efficiency.

Vedic mathematics offers a valuable set of tools that can enhance mental agility, improve problem-solving skills, and boost confidence in mathematics across various domains, including education, competitive exams, professional roles, and even the evolving field of computer science. It provides an alternative and often more intuitive approach to traditional methods, enabling individuals to engage with mathematics in a more enjoyable and efficient way.

METHODS OF QUICK CALCULATIONS

Q.3. What are the methods of quick calculations for :

(a) Addition

(b) Substraction

(c) Multiplication

(d) Division

Ans. Methods of quick calculations : Imagine how mathematics would be easy and interesting when we have the ability to calculate the problems in a matter of seconds using some Maths tricks. There are different kinds of arithmetic operation like addition, subtraction, division, multiplication, squaring, roots, powers, logarithms, divisions, etc. Here are some of the best tricks, which will help students to perform arithmetic calculations easily.

1. **Method for Addition :** With the help of basic principles of tens and unit places, the addition of two-digit numbers is performed by: +30 +4

- Take 43 + 34
- Split the second number into tens and unit places. 34 = 30 + 4
- Finish the ten's addition. 43 + 30 = 73
- Finally, add the remaining unit place digit. 73 + 4 = 77.

2. **Method for Subtraction :** Here is an example that requires a lot of borrowing :

- Consider two numbers say 1000 and 676
- Subtract 1 from both the numbers; we get 999 and 675
- Then subtract 675 from 999, we get 324
- So, 1000 – 676 = 324.

3. **Quick Multiplication method by Breaking Down Numbers:**

- Let's try the numbers 24 and 16
- First split the number 24, which gives 4 × 6
- Then multiply 6 with 16, we get 96
- Finally multiply the number, 96 × 4 = 384
- So, the multiplication of two numbers 24 × 16 gives the solution 384.

4. **Multiplied By 15 :**

- Consider the multiplication of two numbers say 56 and 15
- Now add zero at the end of the first number, it becomes 560
- Divide that number by 2; we get 560/2 = 280
- Add the resultant number with 560, so 560 + 280 = 840.
- So the answer for 56 and 15 is 840.

5. **Multiplication of Two-Digit Numbers :** If anyone of the given numbers is an even number, then follow the steps to solve

- Consider an example, 18 × 37
- Here 18 is an even number, then divide the first number in half, so that 18/2 = 9
- Then double the second number. 37 × 2 = 74
- Finally, multiply the resultant numbers. It becomes 74 × 9 = 666

6. **Maths Division Method :** The numbers that can be evenly divided by certain numbers are:

- If a number is an even number and ends in 0, 2, 4, 6 or 8, it is divided by 2.
- A number is divisible by 3 if the sum of the digits is divisible by 3. Consider the number 12 = 1 + 3 and 3 is divisible by 3.
- A number is divisible by 4 if the last two digits are divisible by 4. Example: 9312. Here the last two digits are 12, and 12 is divisible by 4.
- If the last digit is 0 or 5, it is divisible by 5
- If a number is divisible by 2 and 3, then it is divisible by 6, since 6 is the product of 2 and 3.
- If the number is divisible by 8, the last three digits of the numbers are divisible by 8.
- If a number is divisible by 9, the sum of the digits is divided by 9. Let us consider the example, 4518 = 4 + 5 + 1 + 8 = 18, which is divisible by 9.
- If the final digit of the number is 0, it is divisible by 10.

7. **Method to Calculate Squares ending with digit 5 :**

- Let's consider the number 75 to find its square.
- Start writing the answer of last two digits number that is 25 because any number that ends with 5 is 25
- Take the first digit of the number 75. That is 7 and take the number that follows 7 is 8.
- Now, multiply 7 and 8, we get the number 56.

Square root of 144:

- Pair the digits from the right-hand side: **1 44**
- The unit place of 144 is 4. So, either the square root will have 2 or 8 at the unit place
- Now, considering the first digit, 1, it is the square of 1. Thus, the first digit of the square root of 144 is 1.
- Since 1 is the smallest of the square. Hence, the square root of 144 will take a smaller number between 2 and 8. Thus, it will be 2.
- Hence, the required square root is 12.

More:

- Square root of 169 = 13
- Square root of 225 = 15
- Square root of 324 = 18

Square Root Tricks of 4-Digit Numbers : To find the square root of small numbers like 4, 9, 16, 25, etc. is an easy task. Because we already know from the multiplication table of 1 to 10, the number when multiplied by itself gives the squares, in a two-digit form. But if the number is three-digit or four-digit, then it is difficult to find the root of these numbers, because we cannot remember the table for higher numbers. Let us find out the trick to determine the root of large numbers.

Example 1: Suppose we need to find the square root of large numbers such as 4489.

- **Step 1:** The unit digit in this number is 9, which can be a unit digit of its square root number such as 3 or 7. Because 3^2 is 9 and 7^2 is 49.
- **Step 2:** Now let us consider the first two digits that is 44 which comes between the squares of 6 and 7 because $6^2 < 44 < 7^2$
- **Step 3:** We can assume that the ten's digit of the square root of 4489 is the lowest among the two numbers i.e. 6 and we need to find the unit digit of the square root of the number 4489.
- **Step 4:** Now, we need to find between 63 or 67 which is the square root of 4489.
- **Step 5:** Since the ten's digit is 6 and the next number is 7, we need to multiply both the numbers like 6 × 7 = 42 and since 42 is less than 44.

- **Step 6:** Square root of 4489 will be the bigger number between 63 and 67 i.e. 67.

Therefore, $\sqrt{4489} = 67$

Example 2: Let us have a look at another example, the square root of 7056.

Here is the step by step method :

- Now, consider the unit digit that is 6. Which all numbers have the unit digit 6 on their square roots. That are 4 and 6 because 4^2 is 16 and 6^2 is 36
- Now let's consider the first two digits that is 70 which comes between the squares of 8 and 9 because of $8^2 < 70 < 9^2$
- We can assume that the ten's digit of the square root of the 7056 is the lowest among the two numbers that is 8
- Now, we need to find the unit digit of the square root of the number 7056. For that, we need between 84 and 86 which is the square root of 7056
- Since the ten's digit is 8 and the next number is 9, we need to multiply both the numbers like 8 × 9 = 72 and since 72 is bigger than 70
- Square root of 7056 will be the lesser number between 84 and 86 that is 84

Therefore, $\sqrt{7056} = 84$

Square Root of 5-Digit Numbers

To find the square root of a 5-digit number, follow the below steps with the help of an example.

Example: Find the square root of 40401.

- **Step 1:** Pair the digits from right to left, 404 01.
- **Step 2:** Check the unit digit of the number and compare it with the above table. Here, the unit digit is 1, thus possible unit digits for the square root of 40401 is 1 and 9. But the square root of 1 is 1, thus unit digit will have 1.
- **Step 3:** Now, the first three digits are 404. It will lie between 20^2 and 21^2. Thus, 400 < 404 < 441. Hence, the square root of first three digit will be near to 20.
- **Step 4:** Hence, the required square root is 201

Square Root Table From 1 to 50

We can also memorise the square root table from numbers 1 to 50, to solve problems based on them.

- Finally, write the number 56 in the prefix and combine with 25 what we already wrote.
- So, the answer is 5625.
- Squares Ending in 5: $n5 = n(n + 1)5^2 = n(n + 1)25$, where n is the first digit.
- Example: Let's consider the number 75 to find its square. Here n = 7,

  So, $75 = 7(7 + 1)25 = (7 \times 8)\ 25 = 5625$.

8. Method to Multiply by 2 and 4 : When a number is multiplied by 2 or 4, then the last digit of the resulting value will be an even number always.

Examples:

- $19 \times 2 = 38$
- $19 \times 4 = 76$

9. Multiplication by 5 : When a number is multiplied by 5, then the resulting value will either end with 0 or 5.

Examples:

- $11 \times 5 = 55$
- $8 \times 5 = 40$
- $121 \times 5 = 605$

10. Multiplication by 10 : When a number is multiplied by 10, then the resulting value ends with 0 always.

Examples:

- $5 \times 10 = 50$
- $10 \times 10 = 100$
- $11 \times 10 = 110$
- $17 \times 10 = 170$
- $211 \times 10 = 2110$

SQUARE AND SQUARE ROOTS

Q.4. What is the method of quick calculation of Square and Square Roots?

Ans. Square root tricks are those tricks that are helpful in solving square root related questions. Knowing square root tricks to find the square root of numbers proves to be very helpful when you are solving complex equations which will not take much time for getting solved. Tips and tricks help us to solve mathematical problems easily and quickly.

Hence, we have some useful tips to find the square root of a given number even without using a calculator.

What is Square Root?:

The square root of a number is the value which when multiplied to itself gives the original number. Suppose, 5 when multiplied by 5 results in 25. So we can say, 5 is the square root value of 25. Similarly, 4 is the root value of 16, 6 is the root value 36, 7 is the root value of 49, etc. Since square represents the area of a square which is equal to 'side x side', therefore, square root represents the length of the side of the square. The symbol of the square root is denoted by '√'. Hence, square root numbers are represented as √4, √5, √8, √9, etc.

Tricks to Find Square Root :

The tricks to find the square root of perfect squares is very easy to remember. We need to keep in mind the unit place digit of squares of numbers from 1 to 10. Find the unit digits we get after squaring the numbers from 1 to 10 in the below table.

Number	Unit digit of Number^2
1	1
2	4
3	9
4	16
5	25
6	36
7	49
8	64
9	81
10	100

Hence, from the above table we can figure out if any perfect square ends with the above digits at the unit place, then its square root will have the same respective number at the unit place.

For example, the square root of 81 is 9.

Square root Tricks of 3-digit Numbers

The square root of a three-digit number is always a two-digit number. Let us learn to find the square root with an example.

Here is the list available:

Number	Square Root Value (√)
1	1
2	1.414
3	1.732
4	2
5	2.236
6	2.449
7	2.646
8	2.828
9	3
10	3.162
11	3.317
12	3.464
13	3.606
14	3.742
15	3.873
16	4
17	4.123
18	4.243
19	4.359
20	4.472
21	4.583
22	4.69
23	4.796
24	4.899
25	5
26	5.099
27	5.196
28	5.292
29	5.385
30	5.477
31	5.568
32	5.657
33	5.745

34	5.831
35	5.916
36	6
37	6.083
38	6.164
39	6.245
40	6.325
41	6.403
42	6.481
43	6.557
44	6.633
45	6.708
46	6.782
47	6.856
48	6.928
49	7
50	7.071

DIGIT SUM

Q.5. What is the method of verification of answers from Digit Sum?

Ans. This method is not used for quick calculations in vedic maths but it helps in verifying the answers. This technique has a wonderful application for the students giving competitive exams as they are already provided four alternatives to every answer.

The digit-sum method can be used to check answers involving multiplication, division, addition, substraction, squares, square roots, cube roots etc. The procedure is very simple. We have to convert any given number into a single digit by repetitive adding of all the digits of that number.

Example : Find the digit sum of 2468539

Ans. The number is 2468539

We add all the digits of that number 2+4+6+8+5+3+9 = 37

Add 3+7=10 and 1+0=1

Example : Find the digit sum of 67769539

Ans. 6+7+7+6+9+5+3+9=51

5+1=6

Now we shall solve a variety of examples involving different arithmetical operations.

Example : Verify 467532 multiplied by 107777 equals 50389196364

Ans. First calculate the digit sum of multiplicand and then calculate the digit-sum of multiplier.

The digit sum of 467532 is 9

The digit sum of 107777 is 2

When we multiply 9 x 2 we get answer 18

Again the digit sum of 18 is 9

Now we will get the digit sum of the product 50389196364, it comes out 9.

Hence we assume that the answer is correct.

Example : Verify whether 2308682040 divided by 36524 equals 63210.

Ans. We will use the same formula that we learnt in school. Dividend=Divisor x Quotient + Remainder

2308682040 = 36524 × 63210 + remainder (0) we have written the reminder 0 because we assume it as 0.

But, here we will use the digit-sum 6=2 × 3 +0

6=6 We assume the answer is correct.

ONE MORE SHORT CUT WHILE CALCULATING DIGIT SUM OF A NUMBER, WE CAN ELIMINATE ALL THE NINES AND THE NUMBERS THAT ADD-UP TO NINE. SUPPOSE YOU HAVE TWO 9'S IN A NUMBER AND 3 AND 6 THEN 2 AND 7 WE CAN ELIMINATE ALL THESE WHILE TOTALLING.

This elimination will have no effect on the final result.

Example : Find the digit-sum of 6372819923.

6+3+7+2+8+1+9+9+2+3 = 50 and 5+0=5

Short-cut: Eliminate 9's and numbers that add up to nine. Eliminate 6,3 7,2 8,1 9

What is left 2+3=5

NOTE: IF THE DIGIT SUM OF A NUMBER IS 9, ELIMINATE 9 AND THE DIGIT SUM BECOMES "0"

Example : MULTIPLICATION :

Q.6. Verify whether 999816 multiplied by 727235 is 727101188750

Ans. The digit sum of 999816 can be instantly found by eliminating 9's and 8 and 1.The remaining is 6

The digit sum of 727235 is 8

The digit sum of 727101188750 is 2

The digit sum doesn't match, hence the answer is wrong.

UNIT-2

Long and Short Answer type Questions :

RULES FOR SIGNS IN ALGEBRA

Q.1. What are the sign rules in Algebra?

Or

What are the laws for the signs in Algebra?

Ans. The law or sign rule indicates the sign that prevails when operations of two equal or different signs are performed, and is applied differently for various mathematical operations:

Sign rule for addition and subtraction :

- When two positive numbers are added, the result will have a positive sign.

  3 + 5 = 8

- When two numbers are added or subtracted, one with a positive sign and the other with a negative sign, the result will have the sign of the largest number.

  5 – 6 = –1

- When two numbers are added or subtracted, one with a negative sign and the other with a positive sign, the result will have the sign of the largest number.

  – 7 + 4 = –3

- When two negative numbers are subtracted, the result will have a negative sign.

  – 5 – 4 = –9

Simplifying the above:

positive y positive = positive

positive $\pm$ negative = largest number

negative $\pm$ positive = largest number

negative y negative = negative

Signs rule for multiplication and division

- When two positive numbers are multiplied or divided, the result will have a positive sign.

$$(+) \times (+) = +(+) \div (+) = +$$

- When two numbers are multiplied or divided, one with a positive sign and the other with a negative sign, the result will have a negative sign.

$$(+) \times (-) = -(+) \div (-) = -$$

- When two numbers are multiplied or divided, one with a negative sign and the other with a positive sign, the result will have a negative sign.

$$(-) \times (+) = -(-) \div (+) = -$$

- When two negative numbers are multiplied or divided, the result will have a positive sign.

$$(-) \times (-) = +(-) \div (-) = +$$

BODMAS

Q.2. What are the rules of BODMAS?

Or

How calculations are performed based on BODMAS?

Ans. The BODMAS rule is used to evaluate mathematical expressions and to deal with complex calculations in a much easier and standard way.

BODMAS Definition :

According to the BODMAS rule, to solve any arithmetic expression, we first solve the terms written in brackets, and then we simplify exponential terms and move ahead to division and multiplication operations, and then, in the end, work on the addition and subtraction. Following the order of operations in the BODMAS rule, always results in the correct answer. Simplification of terms inside the brackets can be done directly. This means we can perform the operations inside the bracket in the order of division, multiplication, addition, and subtraction. If there are multiple brackets in an expression, all the same types of brackets can be solved simultaneously.

For example, $(14 + 19) \div (13 - 2) = 33 \div 11 = 3$.

Observe the table given below to understand the terms and operations denoted by the BODMAS acronym in the proper order.

B	[{( )}]	Brackets
O	x^2	Order of Powers or Roots
D	÷	Division
M	×	Multiplication
A	+	Addition
S	−	Subtraction

When to Use BODMAS?

BODMAS is used when there is more than one operation in a mathematical expression. There is a sequence of certain rules that need to be followed when using the BODMAS method. This gives a proper structure to produce a unique answer for every mathematical expression.

Conditions to follow:

- If there is any bracket, open the bracket, then add or subtract the terms. $a + (b + c) = a + b + c$, $a + (b - c) = a + b - c$
- If there is a negative sign just open the bracket, multiply the negative sign with each term inside the bracket. $a - (b + c) = a - b - c$
- If there is any term just outside the bracket, multiply that outside term with each term inside the bracket. $a(b + c) = ab + ac$

Easy Ways to Remember the BODMAS Rule :

The simple rules to remember the BODMAS rule are given below:

- Simplify the brackets first.
- Solve all exponential terms.
- Perform division or multiplication (go from left to right)
- Perform addition or subtraction (go from left to right)

Example 1: Simplify the expression by using the BODMAS rule: [18 –2(5 + 1)] ÷ 3 + 7

Solution: The given expression is $[18 - 2(5 + 1)] \div 3 + 7$

We begin with solving the innermost bracket first. Starting with $5 + 1 = 6$. Thus, $[18 - 2(6)] \div 3 + 7$

Next, we work with the order. thereby multiplying 2 (6) or $2 \times 6 = 12$. Thus, $[18 - 12] \div 3 + 7$

There is one bracket left, $[18 - 12] = 6$. So, $6 \div 3 + 7$

After B and O comes D, hence, $6 \div 3 = 2$. So, $2 + 7$

And finally, addition, $2 + 7 = 9$

$\therefore$ The expression is simplified and the answer is 9.

Example 2: Evaluate using the order of operations using the BODMAS rule: $(1 + 20 - 16 \div 4^2) \div ((5 - 3)^2 + 12 \div 2)$

Solution:

Step 1: First, we need to simplify the innermost bracket, $(1 + 20 - 16 \div 4^2) \div (2^2 + 12 \div 2)$

Step 2: Now we have to evaluate exponents, $(1 + 20 - 16 \div 16) \div (4 + 12 \div 2)$

Step 3: Now, we need to divide 16 by 16 and 12 by 2 inside the brackets, and we get, $(1 + 20 - 1) \div (4 + 6)$

Step 4: Add 1 to 20 and 4 to 6, $(21 - 1) \div 10$

Step 5: Subtract 1 from 21 to solve the bracket, we get, $20 \div 10$

Step 6: Divide 20 by 10 to get the final answer, we get, 2. $\therefore (1 + 20 - 16 \div 4^2) \div ((5 - 3)^2 + 12 \div 2) = 2$

Example 3: Simplify the expression by using the BODMAS rule: $(9 \times 3 \div 9 + 1) \times 3$

Solution:

Step 1: Using Bodmas Rule (left to right whichever operations come first we will follow that). Here, first we need to multiply 9 by 3 in the given expression, $(9 \times 3 \div 9 + 1) \times 3$, and we get, $(27 \div 9 + 1) \times 3$

Step 2: Now, we need to divide 27 by 9 inside the bracket, and we get, $(3 + 1) \times 3$

Step 3: Remove the parentheses after adding 3 and 1, we get, 4×3

Step 4: Multiply 4 by 3 to get the final answer, which is 12.

$\therefore (9 \times 3 \div 9 + 1) \times 3 = 12$

Example 4: Solve the given expression applying the BODMAS rule: $[50-\{3\times(9+7)\}]$

Solution: To solve this expression, $[50-\{3\times(9+7)\}]$, we will use the following steps:

Step 1: Solve the innermost bracket by adding 9 to 7, that is, 16. So, the simplified expression is $[50-\{3\times16\}]$

Step 2: Multiply 3 by 16, to get $[50-48]$

Step 3: Subtract 48 from 50 to get the final answer, i.e, 2.

SIMULTANEOUS EQUATIONS

Q.3. Give the meaning of simultaneous equations and also explain the methods of solving simultaneous equations.

Or

Describe Simultaneous equations and discuss the method of solving it.

Or

Give the definition of Simultaneous equation and discuss the various methods of solving these equations.

Or

What do you mean by Simultaneous equations?

Or

Define Simultaneous Equation.

Ans. Meaning of Simultaneous equation : Two (or more) unknowns are called a system of simultaneous equations if they are simultaneously satisfied by the same set of values of the unknowns. If the unknowns occur only in the first powers, the equations are called simple simultaneous equation.

For example the equation :

$$x + 2y = 5$$
$$3x - y = 8$$

From a pair of simple simultaneous equation if the unknowns are x and y and both the equations are simultaneously satisfied only by the set of values $x = 3$, $y = 1$ which are called the solutions or roots of the equations.

To find the solutions of a set of simultaneous equations, the number of given equations must be equal to the number of unknown quantities involved in the question.

Methods of solving Equations : To solve simultaneous equation there are means to know that pair of value of unknown quantities which may satisfy all the given equations.

There are two following methods of solving simultaneous equations :

(1) Graphical Representation Method : In a linear equation involving two variables there are many values of x and y which satisfy the equation. If we plot various ordered pairs of x and y and draw a line passing through those points, it will be graphs of the line.

In order to draw the graph of a line, first of all, we prepare a table, in which the values of x and y satisfy the equation. We plot at least three such pairs of x and y and join them. Two points determine the line, and if, the third point also lies on the line it shows, that the procedure is correct.

The graph of a linear equation is straight line. If two lines are not parallel, they will intersect at a point. The co-ordinates of the point of intersection will satisfy both the equations. The abscissa of the point of intersection will give the value of x, while the ordinate will give the value of y.

(2) Arithmetical Method : Questions on simultaneous equations are generally solved by this method. The values obtained by these methods are mostly correct.

The following are the methods of solving the simultaneous equations, under this method :

(i) Elimination Method : The process by which we get rid of the unknown quantities is called elimination. Under this method, the equations are multiplied by such number so that the multiplies of any one unknown quantity in both the equations become equal. When the co-efficient of one unknown are equal and unlike in sign, they are added and if the co-efficients are equal and like in sign are subtracted, so that the equation may be such in which there is only one unknown quantity. On solving this equation we can obtain the value of one unknown quantity.

In this way the value so obtained of one unknown quantity is substituted in any of the equations to obtain the value of the second unknown quantity.

Example : Solve the following equation by elimination method :

$$3x + 2y = 9$$
$$x + 3y = 10$$

Solution: We have,

$$3x + 2y = 9 \quad \ldots(1)$$
$$x + 3y = 10 \quad \ldots(2)$$

Multiplying (2) by 3

$$3x + 9y = 30$$

From (1) $\quad 3x + 2y = 9$

Subtracting $\quad 7y = 21$

$$\therefore \quad y = \frac{21}{7} = 3$$

Substituting y – 3 in (2) we get,

$$x + 3 \times 3 = 10$$

Or $\quad x + 9 = 10$

$$\therefore \quad x = 1$$

$\therefore$ The required solutions are $x = 1, y = 3$

(ii) Substitution Method : Under this method, the value of the variables is obtained from one of the equations and it is substituted in the other equation. Thus, we get an equation involving one variable, whose value is determined. Substituting this value in any one of the equation the value of the other variable is obtained.

Example : Solve by substitution method :

$$3x + 2y = 9 \quad \ldots(i)$$
$$x + 3y = 10 \quad \ldots(ii)$$

From equation (ii) we get the value of x

$$x = 10 - 3y$$

Now by substituting the value of x in equation (i)

$$3(10 - 3y) + 2y = 9$$
$$30 - 9y + 2y = 9$$
$$30 - 7y = 9$$
$$-7y = 9 - 30$$
$$-7y = -21$$
$$y = \frac{21}{7} = 3$$

By putting the value of y in equation (ii), we get :

$$x + 3y = 10$$
$$x + (3 \times 3) = 10$$
$$x + 9 = 10$$
$$x = 10 - 9 = 1$$
$$x = 1, y = 3$$

Ans.

(iii) Comparative Method : Under this method by putting the value of one unknown variable with the value of another unknown variable and thereafter equating these values a new equation is formed. On solving this, the value so obtained, is the required value of one unknown variable. By substituting this value in any other equation we can know the required value of the other variable.

Example : Solve the following equation by comparative method :

$$3x - y = 23$$
$$4x + 3y = 48$$

Solution: $\quad 3x - y = 23 \Rightarrow 3x = y + 23$

$$\Rightarrow \quad x = \frac{y + 23}{3}$$

$$4x + 3y = 48 \Rightarrow 4x = 48 - 3y$$

$$\Rightarrow \qquad x = \frac{48-3y}{4}$$

Therefore, $$x = \frac{y+23}{3}$$

and $$x = \frac{48-3y}{4}$$

$$\Rightarrow \qquad \frac{y+23}{3} = \frac{48-3y}{4}$$

$$4(y+23) = 3(48-3y)$$

$$\Rightarrow \qquad 4y+92 = 134-9y$$

$$4y+9y = 144-92$$

$$\Rightarrow \qquad 13y = 52$$

$$\Rightarrow \qquad y = \frac{52}{13} = 4$$

By substituting this value of y in equation (i) we get

$$3x-4 = 23$$

$$\therefore \qquad 3x = 27$$

Or $$x = 9$$

Hence, $x = 9, y = 4$ are the values of given equation.

(iv) Method of cross multiplication : The solution of the pair of linear simultaneous equations

$$a_1x + b_1y + c_1 = 0 \qquad \text{...(1)}$$

$$a_2x + b_2y + c_2 = 0 \qquad \text{...(2)}$$

are given by

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1} \qquad \text{...(3)}$$

Where $a_1b_2 - a_2b_1 \neq 0$

The relation (3) is obtained by the method (or rule) of cross multiplication, which is shown below.

$$\begin{array}{ccc} b_1 & \times & c_1 \\ b_2 & & c_2 \end{array} \qquad \begin{array}{ccc} c_1 & \times & a_1 \\ c_2 & & a_2 \end{array} \qquad \begin{array}{ccc} a_1 & \times & b_1 \\ a_2 & & b_2 \end{array}$$

Explanation : Write the co-efficients $(a_1b_1c_1)$ and $(a_2b_2c_2)$ twice as shown above and cancel the first and the last column. For the denominator $b_1c_2 - b_2c_1$ of the first term in (3) write the second and third columns :

$$\begin{array}{ccc} b_1 & \times & c_1 \\ b_2 & & c_2 \end{array}$$

and multiply crosswise as indicated above by the arrows. The descending arrow gives the product b_1c_2 and the ascending arrow the product b_2c_1. Now subtract the second product b_2c_1 from the first b_1c_2 to obtain $b_1c_2 - b_2c_1$.

Similarly, we can find the other denominators $c_1a_1 - c_2a_1$ and $a_1b_2 - a_2b_1$ from the 3rd, 4th and the 4th, 5th columns respectively.

After obtaining the relation (3) as above, the required solutions would be

$$x = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1} \text{ and } y = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \text{ where } a_1b_2 - a_2b_1 \neq 0$$

Q.4. Solve the following equation by Cross Multiplication Method :

$$\mathbf{5x + 2y = 17 \; ; \; 2x - 5y = 1}$$

Solution: The given equation may be written as

$$5x + 2y - 17 = 0$$

and $$2x - 5y - 1 = 0$$

Here $a_1 = 5, \quad b_1 = 2, \quad c_1 = -17$

$a_2 = 2, \quad b_2 = -5, \quad c_2 = -1$

$\therefore$ By cross multiplication, we have

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\Rightarrow \qquad \frac{x}{2(-1)-(-5)(-17)} = \frac{y}{(-17)2-(-1)5} = \frac{1}{5(-5)-2\times 2}$$

$$\Rightarrow \qquad \frac{x}{-2-85} = \frac{y}{-34+5} = \frac{1}{-25-4}$$

$$\Rightarrow \qquad \frac{x}{-87} = \frac{y}{-29} = \frac{1}{-29}$$

$$\Rightarrow \qquad \frac{x}{87} = \frac{y}{29} = \frac{1}{29}$$

$$\Rightarrow \qquad \frac{x}{3} = \frac{y}{1} = \frac{1}{1}$$

$$\therefore \qquad x = 3 \; ; \; y = 1$$

Hence the required solution is $x = 3, y = 1$ **Ans.**

Q.5. Solve : $\frac{x}{4} + \frac{y}{5} + 1 = 23$

$$\frac{x}{5} + \frac{y}{4} = 23$$

Solution: We have

$$\frac{x}{4} + \frac{y}{5} + 1 = 23$$

and $\frac{x}{5} + \frac{y}{4} = 23$

Or $\frac{5x + 4y}{20} = 23 - 1$

and $\frac{4x + 5y}{20} = 23$

Or $5x + 4y = 440$ ...(1)

and $4x + 5y = 460$ ...(2)

Multiplying (1) by 4 and (2) by 5 we get

$$20x + 16y = 1760$$
$$20x + 25y = 2300$$
$$- \quad - \quad\quad -$$

Subtracting, $-9y = -540$

Or $y = 60$

Substituting $y = 60$ in (1), $5x + 240 = 440$

Or $5x = 440 - 240 = 200$ $\therefore x = 40$

Hence, the required solutions are $x = 40$, $y = 60$. **Ans.**

Q.6. The age of a man is three times the sum of the ages of his two sons, and five years hence his age will be double the sum of their ages. Find his present age.

Solution: Let the present age of the man be x years and the sum of the present ages of the two sons be y years.

Then by the given conditions,

$$x = 3y \quad ...(1)$$

and $x + 5 = 2(y + 5 + 5)$ ...(2)

From (2), $x + 5 = 2y + 20$

Putting $x = 3y$, $3y + 5 = 2y + 20$

Or $3y - 2y = 20 - 5$

$$y = 15$$

$\therefore$ From (1) $x = 3y = 3 \times 15 = 45$ $\therefore y = 15$

Hence the required present age of the man = 45 years. **Ans.**

Q.7. Solve the following equations :

$$x + y + z = 7 \quad ...(1)$$
$$5x + 4y - 3z = 1 \quad ...(2)$$
$$6x - 3y + 2z = 8 \quad ...(3)$$

Solution: Here by eliminating z, we get two simultaneous equations in x and y.

By multiplying equation (1) by 3, we get :

$$3x + 3y + 3z = 21 \quad ...(4)$$

By adding equation (2) and (4), we get :

$$8x + 7y = 22 \quad ...(5)$$

Again, multiplying equation (1) by 2 we get :

$$2x + 2y + 2z = 14 \quad ...(6)$$

By subtracting equation (6) from equation (3) we get :

$$4x - 5y = 6 \quad ...(7)$$

As such we get two equations in which there are two unknown quantities. Now we can get the values of x and y by solving equations (5) and (7).

By multiplying equation (7) by 2 we get :

$$8x - 10y = 12 \quad ...(8)$$

Subtracting equation (8) from equation (5), we get :

$$17y = 22 + 12 = 34$$
$$17y = 34$$
$$y = \frac{34}{17} \text{ or } 2$$

By putting the value of y in equation (5), we get :

$$8x + 7 \times 2 = 22$$
$$8x = 22 - 14 = 8$$
$$x = 1$$

By putting the value of x and y in equation (1), we get :

$$1 + 2 + z = 7$$
$$z = 7 - 3 = 4$$

Q.8. Solve the following Equations :

$$3x - 2y = xy \; ; \; 5x + 6y = 4xy$$

Solution: Dividing the given equations by xy, We get : $\frac{3}{y} - \frac{2}{x} = 1$ ...(1)

$$\frac{5}{y} + \frac{6}{x} = 4 \quad ...(2)$$

Multiplying (1) by 3 we get,

$$\frac{9}{y} - \frac{6}{x} = 3 \quad ...(3)$$

By Adding (2) and (3) we get

$$\frac{14}{y} = 7$$

$$\Rightarrow \quad 7y = 14$$

$$\Rightarrow \quad y = \frac{14}{7} = 2$$

Putting the value of y in (1) we get

$$\frac{3}{2} - \frac{2}{x} = 1$$

$$\Rightarrow \quad 3x - 4 = 2x$$

$$\Rightarrow \quad 3x - 4 = 2x$$

$$\Rightarrow \quad x = 4$$

Hence the required solution is $x = 4, y = 2$ **Ans.**

Q.9. Solve the following Equations :

$$\frac{x}{4} - \frac{y}{6} = 0 \quad ...(1)$$

$$6x + 4y = 48 \quad ...(2)$$

Solution: The L.C.M. of the numbers in denominator of equation (1) is 12. Multiply each term of equation (1) by 12, and we get :

$$\frac{3x - 2y}{12} = 0$$

$$3x - 2y = 0$$

$$3x - 2y \text{ Or } x = \frac{2y}{3}$$

By putting the value of x in equation (2), we get :

$$6\left\{\frac{2y}{3}\right\} + 4y = 48$$

$$\frac{12y}{3} + 4y = 48$$

$$4y + 4y = 48$$

$$8y = 48$$

$$y = \frac{48}{8} = 6$$

By putting the value of y in equation (3) we get :

$$3x - 2y = 0$$

$$3x - 2 \times 6 = 0$$

$$3x = 12$$

$$x = \frac{12}{3} \text{ Or } x = 4$$

$$x = 4, \text{ and } y = 6$$ **Ans.**

Q.10. Solve the following Equation :

$$x + y = 5xy$$

$$2x + 3y = 12xy$$

Solution:

$$x + y = 5xy \quad ...(1)$$

$$2x + 3y = 12xy \quad ...(2)$$

By dividing equation (1) and (2) by xy on both sides we get :

$$\frac{x}{xy} + \frac{y}{xy} = \frac{5xy}{xy}$$

$$\frac{1}{y} + \frac{1}{x} = 5$$

$$\frac{2x}{xy} + \frac{3y}{xy} = \frac{12xy}{xy}$$

$$\frac{2}{y} + \frac{3}{x} = 12$$

Suppose : $\frac{1}{x} = a \text{ and } \frac{1}{y} = b$

$$b + a = 5 \quad ...(3)$$

$$2b + 3a = 12 \quad ...(4)$$

By Multiplying equation (3) by 2, we get

$$2b + 2a = 10 \quad ...(5)$$

On subtracting equation (5) from equation (4)

$$2b + 3a = 12$$

$$2b + 2a = 10$$

$$- \quad - \quad -$$

$$a = 2$$

By substituting this value of a in equation (3), we get,

$$b + a = 5$$
$$b + 2 = 5$$
$$b = 3$$

$$\therefore \quad x = \frac{1}{a} = \frac{1}{2} \text{ and } y = \frac{1}{b} = \frac{1}{3}$$

$$x = \frac{1}{2} \text{ and } y = \frac{1}{3}$$ **Ans.**

Q.11. Solve the following equation :

$$\mathbf{x + 0.2y = 0.6} \quad ...(1)$$
$$\mathbf{3.4x - 0.01 = 0.02y} \quad ...(2)$$

Solution: For making the coefficient of x same in both the equations, multiplying equation (1) by 3.4 and equation (2) by 1, we get :

$$3.4x + 0.68y = 2.04 \quad ...(3)$$
$$3.4x - 0.02y = 0.01 \quad ...(4)$$

On subtracting equation (4) from equation (3)

$$3.4x + 0.68y = 2.04$$
$$3.4x - 0.02y = 0.01$$
$$- \quad + \quad -$$

$$0.70y = 2.03$$
$$y = \frac{2.03}{0.70}$$

Or
$$y = 2.90$$

By putting the value of y in equation (1), we get

$$x + 0.2y = 0.6$$
$$\Rightarrow \quad x + 0.2\,(2.90) = 0.6$$
$$\Rightarrow \quad x + 0.58 = 0.6$$
$$\Rightarrow \quad x = 0.6 - 0.58$$
$$\Rightarrow \quad x = 0.02$$
$$x = 0.02 \text{ and } y = 2.90$$

Q.12. Solve the equation :

$$\frac{3}{x} + \frac{2}{y} = 1, \quad \frac{5}{x} + \frac{4}{y} = \frac{13}{8}$$

Solution:

In the above equations substituting $\frac{1}{x}$ by a and $\frac{1}{y}$ by b, we get :

$$3 \times \frac{1}{x} + 2 \times \frac{1}{y} = 1$$

And
$$5 \times \frac{1}{x} + 4 \times \frac{1}{y} = \frac{13}{8}$$

Or
$$3a + 2b = 1 \quad ...(1)$$

and
$$5a + 4b = \frac{13}{8} \quad ...(2)$$

Multiplying equation (1) by 2.

$$6a + 4b = 2 \quad ...(3)$$

Subtracting equation (2) from equation (3) :

$$6a + 4b = 2$$
$$5a + 4b = \frac{13}{8}$$
$$- \quad - \quad -$$

$$a = \frac{3}{8}$$

Putting the value of a in equation (1) :

$$3 \times \left(\frac{3}{8}\right) + 2b = 1$$
$$\Rightarrow \quad \frac{9}{8} + 2b = 1$$
$$\Rightarrow \quad 2b = 1 - \frac{9}{8}$$
$$\Rightarrow \quad 2b = -\frac{1}{8}$$
$$\Rightarrow \quad b = -\frac{1}{16}$$

Now, $x = \frac{1}{a}$ and $y = \frac{1}{b}$

$$\therefore \quad x = \frac{8}{3} \text{ and } y = -16$$ **Ans.**

Q.13. Solve the Equation :

$$\mathbf{3x - 2y = xy; \; 5x + 6y = 4xy}$$

Solution: Equations given :

$$3x - 2y = xy$$
$$5x + 6y = 4xy$$

By dividing the given equations by xy

$$\frac{3}{y} - \frac{2}{x} = 1 \quad ...(1)$$

$$\frac{5}{y} + \frac{6}{x} = 4 \quad ...(2)$$

Multiplying equation (1) by 3, we get

$$\frac{9}{y} - \frac{6}{x} = 3 \quad ...(3)$$

Adding equations (2) and (3)

$$\frac{5}{y} + \frac{6}{x} = 4$$
$$\frac{9}{y} - \frac{6}{x} = 3$$

$$\frac{14}{y} = 7$$

$\Rightarrow$ $7y = 14$

$\Rightarrow$ $y = 2$

Putting the value of y in equation (1) –

$$\frac{3}{2} - \frac{2}{x} = 1$$

$\Rightarrow$ $3x - 4 = 2x$

$\Rightarrow$ $3x - 2x = 4$

$\Rightarrow$ $x = 4$

Thus, $x = 4, y = 2$ **Ans.**

Q.14. The sum of two numbers is 150. The double of the first number and five times of the second number are equal to 450. Find the numbers.

Solution : Suppose the two numbers are x and y. According to the first condition the sum of two numbers :

$$x + y = 150 \quad ...(1)$$

and according to the second condition, the sum of the double of the first number and five times of the second number –

$$2x + 5y = 450 \quad ...(2)$$

Multiplying equation (1) by 2, we get :

$$2x + 2y = 300 \quad ...(3)$$

Subtracting equation (3) from equation (2) :

$$3y = 150$$

$\therefore$ $y = \frac{150}{3} = 50$

Putting this value of y in equation (1), we get,

$$x + 50 = 150$$

$\therefore$ $x = 150 - 50 = 100$

Thus the numbers are 100 and 50. **Ans.**

Q.15. A number of two digits is 7 times the sum of the digits and the sum of the reciprocals of the digits is 9 times the reciprocal of their product. Find the number.

Solution: Suppose the digit of the unit is x and the digit of tens is y.

Or The number $= 10y + x$

The sum of the digits $= x + y$

7 times of the sum of the digits $= 7(x + y)$

$\therefore$ According to the fist condition of the question

$$10y + x = 7(x + y)$$

$\Rightarrow$ $10y + x = 7x + 7y$

$\Rightarrow$ $10y + x - 7x - 7y = 0$

$\Rightarrow$ $3y - 6x = 0$

$\Rightarrow$ $3(y - 2x) = 0$

$\Rightarrow$ $y - 2x = 0 \quad ...(1)$

Now, the sum of the reciprocals of the digits $\frac{1}{x} + \frac{1}{y}$

Product of the reciprocals of the digits $= \frac{1}{xy}$

9 times of the product of reciprocals of the digits $= \frac{9}{xy}$

Therefore, according to the second condition of the question,

$$\frac{1}{x} + \frac{1}{y} = \frac{9}{xy} \Rightarrow \frac{y + x}{xy} = \frac{9}{xy}$$

$\Rightarrow$ $y + x = 9 \quad ...(2)$

(on multiplying by xy on both the sides)

Subtracting equation (1) from equation (2) :

$$y - 2x = 0$$
$$y + x = 9$$
$$- \quad - \quad -$$

$$-3x = -9$$
$$x = 3$$

Or

By substituting the value of x in equation (1) we get :

$$y - 2 \times 3 = 0$$
$$\Rightarrow \quad y - 6 = 0$$
$$\therefore \quad y = 6$$

Hence, the number is $= 10 \times 6 + 3 = 60 + 3 = 63$ **Ans.**

Q.16. Ram says to Shyam 'I am twice as old as you were when I was as old as you are now.' The sum of their present ages is 63 years. Find their present ages.

Solution: Suppose the present age of Ram is x years and the present age of Shyam is y years.

Hence, the sum of their present ages :

$$x + y = 63 \qquad ...(1)$$

Suppose Ram's age equal to the present age of Shyam was a years before.

$$\therefore \quad y + a = x$$
$$\Rightarrow \quad a = x - y$$

Or Ram's age equal to y years was before (x − y) years then the present age.

$\therefore$ Shyam's age before x − y years $= y - (x - y)$

$$= y - x + y = 2y - x$$

According to the problem $x = 2(2y - x)$

$$\Rightarrow \quad x = 4y - 2x$$
$$\Rightarrow \quad x + 2x - 4y = 0$$
$$\Rightarrow \quad 3x - 4y = 0 \qquad ...(2)$$

Multiplying equation (1) by 4, we get

$$4x + 4y = 252 \qquad ...(3)$$

Adding equation (2) and (3)

$$3x - 4y = 0$$
$$4x + 4y = 252$$

$$7x = 252$$

$$\therefore \quad x = \frac{252}{7} = 36$$

Putting the value of x in equation (1) :

$$36 + y = 63$$
$$\therefore \quad y = 63 - 36$$
$$y = 27 \text{ years}$$

Ram's age is 36 years and Shyam's age is 27 years **Ans.**

Q.17. Find the fraction which is equal to $\frac{1}{2}$ when both its numerator and denominator are increased by 1 and which is equal to $\frac{2}{3}$ when both are increased by 4.

Solution: Suppose the fraction is $\frac{x}{y}$

Adding 1 to numerator and denominator

$$\frac{x+1}{y+1} = \frac{1}{2}$$
$$\Rightarrow \quad 2(x + 1) = y + 1$$
$$\Rightarrow \quad 2x + 2 = y + 1$$
$$\Rightarrow \quad 2x - y = 1 - 2$$
$$\Rightarrow \quad 2x - y = -1 \qquad ...(1)$$

Adding 4 numerator and denominators the fraction will be :

$$\frac{x+4}{y+4} = \frac{2}{3}$$
$$\Rightarrow \quad 3(x + 4) = 2(y + 4)$$
$$\Rightarrow \quad 3x + 12 = 2y + 8$$
$$\Rightarrow \quad 3x - 2y = 8 - 12$$
$$\Rightarrow \quad 3x - 2y = -4 \qquad ...(2)$$

To solve equations (1) and (2), multiplying equation (1) by 2 we get,

$$4x - 2y = -2 \qquad ...(3)$$

Subtracting equation, (2) from equation (3) :

$$4x - 2y = -2$$
$$3x - 2y = -4$$
$$- \quad + \quad +$$

$$x = 2$$

Putting the value of x in equation (1) :

$$2 \times 2 - y = 1$$
$$4 - y = -1$$
$$-y = -1 - 4$$
$$-y = -5$$
$$y = 5$$

Or

The fraction will be $\frac{2}{5}$ **Ans.**

Q.18. 16 kg wheat and 6 kg rice cost Rs. 29.24 kg wheat and 15 kg rice cost Rs. 52.50. Find the cost of wheat and rice per kg.

Solution: Let price of 1 Kg wheat = x and that of rice = y

Cost of 16 Kg wheat = 16x and rice = 6y

The cost of wheat and rice = 16x + 6y

Cost of 24 Kg Wheat = 24x, cost of 15 kg rice = 15y

Total cost of wheat and rice = 24x + 15y

According to problem :

$$16x + 6y = 29 \quad \text{...(1)}$$
$$24x + 15y = 52.50 \quad \text{...(2)}$$

Multiplying equation (1) by 3 and equation (2) by 2, we get

$$48x + 18y = 87 \quad \text{...(3)}$$
$$48x + 30y = 105 \quad \text{...(4)}$$

Subtracting equation (3) from equation (4),

$$12y = 18$$

$\Rightarrow$ $$y = \frac{18}{12} = 1.25$$

Putting the value of y in equation (1),

$$16x + 6y = 29$$

$\Rightarrow$ $$16x + 6(1.5) = 29$$

$\Rightarrow$ $$16x + 9 = 29$$

$\Rightarrow$ $$16x = 29 - 9$$

$\Rightarrow$ $$x = \frac{20}{16} = 1.5$$

Hence the cost of Rice per Kg is Rs. 1.50 and that of wheat is Rs. 1.25 per Kg. **Ans.**

Q.19. Two numbers are such that 4 times of the smaller is 5 less than 3 times of the larger but sum of the numbers is greater than 6 times their difference by 6. Find the numbers.

Solution: Let larger number = x and smaller number = y

According to first condition,

3 (larger number) – 4 (smaller number) = 5

$\Rightarrow$ $$3x - 4y = 5 \quad \text{... (i)}$$

and sum of the numbers = 6 (difference of the numbers) + 6

$\Rightarrow$ $$x + y = 6(x - y) + 6$$

$\Rightarrow$ $$x + y = 6x - 6y + 6$$

$\Rightarrow$ $$x + y - 6x + 6y = 6$$

$\Rightarrow$ $$-5x + 7y = 6 \quad \text{... (ii)}$$

Multiply equation (i) by 5 and equation (ii) by 3,

$$15x - 20y = 25$$
$$\underline{-15x + 21y = 18} \quad \text{... (iii)}$$
$$y = 43, \text{ Add} \quad \text{... (iv)}$$

Substitute this value of y in equation (i),

$$3x - 4 \times 43 = 5$$

$\therefore$ $$x = \frac{5 + 172}{3} = \frac{177}{3} = 59$$

Ans. The number are 59 and 43.

Verification : $3 \times 59 - 4 \times 43 = 177 - 172 = 5$

$(59 + 43) - 6(59 - 43) = 102 - 96 = 6$

Q.20. The sum of 2 digits of a number is 15. If 9 is added to the number, the resulting number is equal to the original number with the digits interchanged. Find the number.

Solution: Let the digit at the unit place be x and the digit at the tenth place be y, i.e., the number is 10y + x.

Hence, the sum of the digits,

$$y + x = 15 \quad \text{... (i)}$$

When 9 is added to the number the digits are reversed, i.e. the number becomes 10x + y

Hence, Original number + 9 = New number

$\Rightarrow$ $$10y + x + 9 = 10x + y$$

or $$10y + x - 10x = -9$$

or $$9y - 9x = -9$$

or $$y - x = -1, \text{ dividing by 9, both sides}$$

or $$x - y = 1 \quad \text{... (ii)}$$

Adding equations (i) and (ii),

$$2x = 16$$

$$x = \frac{16}{2} = 8$$

$\therefore$

Substitute this value of x in equation (i),

$$y + 8 = 15$$

$$y = 15 - 8 = 7$$

$\therefore$

Thus the required number is $10 \times 7 + 8 = 70 + 8 = 78$.

Q.21. Let sum of the digits of a number consisting of two digits is 9. If the digits be reversed, the resulting number will be greater than original number by 45. Find the number.

Solution: Let the digit at the unit place be x and the digit at the tenth place be y, i.e., the number is $10y + x$.

Hence, the sum of the digits,

$$y + x = 9 \quad \text{... (i)}$$

New number by reversing the digits $= 10x + y$

Hence New number = Original number + 45

$\Rightarrow$ $(10x + y) = (10y + x) + 45$

or $10x + y - 10y - x = 45$

or $9x - 9y = 45 \quad \text{... (ii)}$

Solving equations (i) and (ii),

$$x = 7, y = 2$$

Hence, required number $= 2 \times 10 + 7 = 20 + 7 = 27$.

Q.22. A number consists of three digits. It remains unchanged if the digits in hundredth and units place are interchanged. The sum of the three digits is 17 and the difference of the digits in tens and units place is 2. Find the number.

Solution: The number remains unchanged if the digit in hundredth and unit place are interchanged, implies that the digit at unit place and hundredth place are same. Let

p = digit at the unit place,

q = digit at the tenth place,

p = digit at the hundredth place

Hence, sum of the digits,

$$p + q + p = 17$$

$$2p + q = 17$$

the difference between the digits at tenth place and unit place,

$$q - p = 2 \quad \text{... (ii)}$$

Subtracting equation (ii) from equation (i),

$$3p = 17 - 2 = 15$$

$\therefore$ $$p = \frac{15}{3} = 5$$

Substitute this value of p in equation (ii),

$$q = 2 + p = 2 + 5 = 7$$

Thus, the required number $= 100 \times 5 + 10 \times 7 + 5$

$$= 500 + 70 + 5 = 575.$$

UNIT-2

Long and Short Answer type Questions :

THEORY OF INDICES

Q.1. What are Indices? Write down the rules of Indices.

Or

What are the Laws of Indices (Index)?

Ans. Indices (index) in Maths is the power or exponent which is raised to a number or a variable. For example, in number 2^4, 4 is the index of 2. The plural form of index is indices. In algebra, we come across constants and variables. The constant is a value which cannot be changed. Whereas a variable quantity can be assigned any number or we can say its value can be changed. In algebra, we deal with indices in terms of numbers. Let us learn the laws/rules of the indices along with formulas with examples.

Definition of Index:

A number or a variable may have an index. Index of a variable (or a constant) is a value that is raised to the power of the variable. The indices are also known as **powers or exponents**. It shows the number of times a given number has to be multiplied. It is represented in the form:

$\mathbf{a^m = a \times a \times a \times ...... \times a}$ **(m times)**

Here, a is the base and m is the index.

The index says that a particular number (or base) is to be multiplied by itself, the number of times equal to the index raised to it. It is a compressed method of writing big numbers and calculations.

Example: $2^3 = 2 \times 2 \times 2 = 8$

In the example, 2 is the base and 3 is the index.

Laws of Indices : There are some fundamental rules or laws of indices which are necessary to understand before dealing with indices. These laws are used while performing algebraic operations on indices and while solving the algebraic expressions, including it.

Rule 1: If a constant or variable has index as '0', then the result will be equal to one, regardless of any base value.

$\mathbf{a^0 = 1}$

Example: $5^0 = 1$, $12^0 = 1$, $y^0 = 1$

Rule 2: If the index is a negative value, then it can be shown as the reciprocal of the positive index raised to the same variable.

$\mathbf{a^{-p} = 1/a^p}$

Example: $5^{-1} = ?$, $8^{-3} = 1/8^3$

Rule 3: To multiply two variables with the same base, we need to add its powers and raise them to that base.

$\mathbf{a^p.a^q = a^{p+q}}$

Example: $5^2.5^3 = 5^{2+3} = 5^5$

Rule 4: To divide two variables with the same base, we need to subtract the power of denominator from the power of numerator and raise it to that base.

$\mathbf{a^p/a^q = a^{p-q}}$

Example: $10^4/10^2 = 10^{4-2} = 10^2$

Rule 5: When a variable with some index is again raised with different index, then both the indices are multiplied together raised to the power of the same base.

$\mathbf{(a^p)^q = a^{pq}}$

Example: $(8^2)^3 = 8^{2.3} = 8^6$

Rule 6: When two variables with different bases, but same indices are multiplied together, we have to multiply its base and raise the same index to multiplied variables.

$\mathbf{a^p.b^p = (ab)^p}$

Example: $3^2.5^2 = (3 \times 5)^2 = 15^2$

Rule 7: When two variables with different bases, but same indices are divided, we are required to divide the bases and raise the same index to it.

$\mathbf{a^p/b^p = (a/b)^p}$

Example: $3^2/5^2 = (?)^2$

Rule 8: An index in the form of a fraction can be represented as the radical form.

$\mathbf{a^{p/q} = {}^q?a^p}$

Example: $6^{1/2} = ?6$

LOGARITHMS & ANTILOGARITHMS

Q.2. What are Logarithms? Explain the method of finding Log.

Or

What do you understand by Logarithms, Antilogarithms, Characteristics and Mantissa.

Or

Explain Logarithms and describe the rules of finding out Characteristics and Mantissa.

Or

Explain Logarithms and describe the rules of finding out values.

Ans. Logarithms are of great significance in mathematical numeration. They simplify calculations and enable us to obtain results of such quantities as are otherwise difficult to solve.

The logarithm of a given number to a certain base is the power to which the base must be raised in order to obtain that number. Thus the logarithm (or simply log) of 4096 to base 8 is 4. It means that if 8 (the base) is raised to the power 4 we shall get the required number $8 \times 8 \times 8 \times 8 = 4096$.

The logarithm of a number consists of two parts – (i) the characteristics, and (ii) the mantissa. The number which refers to the integral power of 10 is called the characteristics of the logarithm. The decimal part is called the Mantissa of the logarithm.

Finding the Characteristics : The following rules are applicable for determining the characteristic of a number :

(i) The characteristic of any number greater than unity is positive and is less by one than the number of digits to the left of the decimal point.

(ii) The characteristic of a number less than unity is negative and is greater by one than the number of zeros which follow the decimal point. The characteristic for some of the numbers is given below :

Number	Characteristic
8	0
82	1
827	2
8275	3
82758	4
.8	1 or $\bar{1}$
.08	–2 or $\bar{2}$
.008	–3 or $\bar{3}$
.0008	–4 or $\bar{4}$

Conventionally the minus sign of the characteristics written at the top of the figure and is read as 1 bar, 2 bar etc.

Finding the Mantissa : The fraction value is known as mantissa of logarithm. The mantissa value is determined by consulting specially prepared log tables. Two things must be remembered about mantissa :

(i) Mantissa is always positive.

(ii) Mantissa is not affected by the position of decimal point. Thus mantissa of 625; 6.25; 625;.0625 will be the same.

In order to find the mantissa of a given number we should reduce it to 4 digits by approximation, if necessary, because beyond this we cannot consult log table. The first two digits are seen in left–hand vertical column and for the third digit the corresponding horizontal column is read. To the figure thus obtained, the quantity appearing under the mean difference column is added to adjust for the fourth digit. Thus if we want to obtain logarithm of 1267.8 we shall approximate it to four digits i.e., 1268. The characteristic of this number is 3. For mantissa consult log table. Now 12 in column 6 is 1004. To this add the 8th mean difference i.e., 27. So the required logarithm is 3.1031.

The following are the logs of a few numbers :

Number	Logarithm
5732	3.7582
573	2.7582
57.3	1.7582
5.73	0.7582
.573	$\bar{1}.7582$
.057	$\bar{2}.7559$
.008	$\bar{3}.9031$

Finding the Antilogarithm

If for a certain logarithm we wish to find out its natural number the table of antilogarithm shall have to be consulted. For finding out antilogarithm we consult the table only for the mantissa part, i.e., the digits after the decimal point. The procedure for consulting antilog tables is the same as for log tables. The place of decimal is determined by the characteristic part of the given number, i.e. the number of digits given before the decimal point. We add one to the characteristic for finding out the natural number.

Some important formulae.

(i) $m \times n = \text{Antilog}(\log_m - \log_n)$

(ii) $\frac{m}{n} = \text{Antilog}(\log_m + \log_n)$

(iii) $m^n = \text{Antilog}(n \log m)$

(iv) $\sqrt{m} = \frac{1}{2}\log m$

Q.3. With the help of Logarithm table find the value of —
7.2 × 8.3 × .94

Solution:

Suppose $y = 7.2 \times 8.3 \times .94$

$\Rightarrow \quad \log y = \log 7.2 + \log 8.3 + \log .94$

$\Rightarrow \quad y = \text{antilog}(\log 7.2 + \log 8.3 + \log .94)$

$= \text{antilog}(1.7764 - 1 + .9731)$

$= \text{antilog}(1.7495) = 56.16$ **Ans.**

Q.4. Find the value of the following using Logarithm table— (.325)⁹.

Solution:

Suppose $y = (.325)^9$

$\log y = 9 \log 325$

$y = \text{Antilog}(9 \log 325)$

$= \text{Antilog}(9 \times 1.5119)$

$= \text{Antilog}[(9(-1 + .119))]$

$= \text{Antilog}(-9 + 4.6071)$

$= \text{Antilog}(-5 + .6071)$

$= .00004047$ **Ans.**

Q.5. Find the values with the help of logarithms :
(i) 39. 8 ÷ 19.9 (ii) 80.17 ÷ 25.46
(iii) 1.005 ÷ .039 (iv) 34.5 ÷ 0.0625

Solution: (i) 39.8 ÷ 19.9

Taking log.

$\log y \Rightarrow \log\left(\frac{39.8}{19.9}\right)$

$\log y \Rightarrow \log 39.9 - \log 19.9$

$\log y \Rightarrow 1.5999 - 1.2989 = 0.301$

Taking Antilog $= y = \text{A.L.}(0.301) = 2.00$ **Ans.**

(ii) 80.17 ÷ 25.46

Taking log, $\log y \Rightarrow \log\left(\frac{80.17}{25.46}\right)$

$\log y \Rightarrow \log 80.17 - \log 25.46$

$\log y \Rightarrow 1.9040 - 1.4058 = 0.4982$

Taking Antilog

$y = \text{A.L.}\ (0.4982) = 3.149$ **Ans.**

(iii) $1.005 \div .039$

Taking log,

$$\log y \Rightarrow \left(\frac{1.005}{.039}\right)$$

$$\log y \Rightarrow \log (1.005) - \log .039$$

$$\log y \Rightarrow 0.0021 - \bar{2}.5911$$

$$\log y \Rightarrow 0.0021 + 2 - .5911 = 1.4110$$

$$y \Rightarrow \text{Antilog } 1.4110 = 25.76$$ **Ans.**

(iv) $34.5 \div 0.0625$

Taking log ;

$$\log y \Rightarrow \left(\frac{34.5}{0.0625}\right) = \log 34.5 - \log 0.0625$$

$$\Rightarrow 1.5378 - \bar{2}.7959 = 1.5378 + -.7959$$

$$\Rightarrow 3.5378 - .7959 = 2.7419$$

$$y \Rightarrow \text{Antilog } 2.7419$$

$$\Rightarrow 552$$ **Ans.**

Q.6. Find the value of the following by logarithms :

(i) $\sqrt{12765.4}$ (ii) $\sqrt[7]{0.00487}$ (iii) $\sqrt[7]{\dfrac{1}{1.235}}$

Solution: (i) Let $y = \sqrt{12765.4}$

$$y = (12765.4)^{1/2}$$

$$\log y = \log (12765.4)^{1/2}$$

$$\log y = \frac{1}{2} \log (12765.4)$$

$$\log y = \frac{1}{2} \log (4.1059)$$

$$y = \text{Antilog } (2.05295)$$

$$y = 112.9$$ **Ans.**

(ii) Let $y = \sqrt[7]{.00487}$

$\Rightarrow$ $y = (.00487)^{1/7}$

$$\Rightarrow \quad \log y = \frac{1}{7} \log .00487$$

$$\Rightarrow \quad y = \text{Antilog}\left(\frac{\log .00487}{7}\right) = \text{Antilog}\left(\frac{\bar{3}.6875}{7}\right)$$

$$\Rightarrow \quad y = \text{Antilog}\left(\frac{-3 + .6875}{7}\right)$$

$$= \text{Antilog}\left(\frac{-7 + 4 + .6875}{7}\right)$$

$$= \text{Antilog}\left(-\frac{7}{7} + \frac{4.6875}{7}\right)$$

$$= \text{Antilog } (-1 + .6696)$$

$$= \text{Antilog } (\bar{1}.6696) = .4673$$ **Ans.**

(iii) Let $y = \sqrt[7]{\dfrac{1}{1.235}} = \left(\dfrac{1}{1.235}\right)^{1/7}$

$$\log y = \frac{1}{7} \log \left(\frac{1}{1.235}\right)$$

$$= \frac{1}{7} (\log 1 - \log 1.235)$$

$$= \frac{1}{7} (0. - 0.0916)$$

$$= -0.0131 = \bar{1}.9869$$

$$y = \text{Antilog } (\bar{1}.9869) = 0.9703$$ **Ans.**

Q.7. Find the value of :

$$\mathbf{\frac{824.523 \times 4.025}{8.2035}}$$

Solution: Suppose $y = \dfrac{824.523 \times 4.025}{8.2035}$

$$\log y = 824.5 + \log 4.025 - \log 8.204$$

$$\log y = 2.9162 + .6048 - .9140$$

$$\log y = 3.5210 - .9140$$

$$y = \text{Antilog } (2.6070)$$

$$y = 404.6$$ **Ans.**

Q.8. Evaluate using log tables :

$$\mathbf{\frac{(0.7634)^{1/3}}{\sqrt{272.3 \times 15.2}}}$$

Solution:

$$\frac{(0.7634)^{1/3}}{\sqrt{272.3 \times 15.2}} = \text{A.L.}\left[\frac{1}{3}\log 0.7634 - \frac{1}{2}(\log 272.3 + \log 15.2)\right]$$

$$= \text{A.L.}\left[\frac{1}{3}(\bar{1}.8827) - \frac{1}{2}(2.4351 + 1.1818)\right]$$

$$= \text{A.L.}\left[\frac{1}{3}(-1 + .8827) - \frac{1}{2}(3.6169)\right]$$

$$= \text{A.L.}\left[\frac{1}{3}(-3 + 2.8827) - 1.80845\right]$$

$$= \text{A.L.}(-1 + .9609 - 1.80845)$$

$$= \text{A.L.}(-2.80845 + .9609)$$

$$= \text{A.L.}(\bar{2}.15245)$$

$$= \text{A.L.}(\bar{2}.1524) = .004204$$ **Ans.**

Q.9. Find the value of – $\sqrt[7]{0.00487}$

Solution: Suppose $y = \sqrt[7]{0.00487}$

$$y = (0.00487)^{1/7}$$

Taking Log $\log y = \frac{1}{7}\log 0.00487$

$$y = \text{Antilog}\left(\frac{3.6884}{7}\right)$$

$$y = \text{Antilog}\left(\frac{-3 + .6884}{7}\right)$$

$$y = \text{Antilog}\left(\frac{-7 + 4 + .6884}{7}\right)$$

$$y = \text{Antilog}\left(\frac{-7}{7} + \frac{4.6884}{7}\right)$$

$$y = \text{Antilog}(-1 + .6698)$$

$$y = 0.4675$$ **Ans.**

Q.10. Find the value, using Logarithm tables :

$$\frac{25.25 \times (0.00345) \times (345)^2}{\sqrt{405.30}}$$

Solution: Suppose $x = \dfrac{25.25 \times (0.00345) \times (345)^2}{\sqrt{405.30}}$

$$\text{Log } x = \log 25.25 + \log 0.00345 + 2\log 345$$

$$-\frac{1}{2}\log 405.30$$

$$= 1.4023 + \bar{3}.5378 + 2(2.5378)$$

$$-\frac{1}{2} \times 2.6078$$

$$= 1.4023 - 3 + 0.5378 + 5.0756 - 1.3039$$

$$= 7.0157 - 4.3039$$

$$= 2.7118$$

$$x = \text{Antilog}(2.7118)$$

$$x = 515$$ **Ans.**

Q.11. Find the value of the following using Logarithm table :

$\sqrt[5]{2.415} \div (0.824)^4$

Solution:

Suppose $y = \sqrt[5]{2.415} \div (0.824)^4$

$$= (2.415)^{1/5} + (0.824)^4$$

$$\Rightarrow \log y = \frac{1}{5}\log 2.415 - 4\log 0.824$$

$$\Rightarrow \log y = \frac{1}{5}(0.3829) - 4(\bar{1}.9159)$$

$$= \frac{1}{5}(0.3829) - (-4 + 3.6636)$$

$$= 0.766 + 4 - 3.6636$$

$$= 0.766 + .3364 = .4130$$

$$y = \text{A.L.}(.4130) = 2.588$$ **Ans.**

Q.12. Prove that :

$x^{\log y - \log z} \cdot y^{\log z - \log x} \cdot z^{\log x - \log y} = 1$

Solution:

Suppose – $A = x^{\log y - \log z} \cdot y^{\log z - \log x} \cdot z^{\log x - \log y}$

$$\log A = (\log y - \log z)\log x + (\log z - \log x)\log y + (\log x - \log y)\log z$$

$$= \log y \log x - \log z \log x + \log z \log y - \log x \log y + \log x \log z - \log y \log z$$

$$= 0$$

$$\Rightarrow A = a^{\circ} = 1 \qquad \{\because \log_a 1 = 0, a \neq 0\}$$

Q.13. Simplify without use of Log tables :

$$\frac{8 \log 2 - 2 \log 4}{\log 2}$$

Solution: $\dfrac{8 \log 2 - 2 \log 4}{\log 2}$

$$= \frac{\log 2^8 \div 4^2}{\log 2} \quad \text{(Rule of Logarithm)}$$

$$= \frac{\log (2^8 \div 4^2)}{\log 2} \quad \text{(By Rule of Logarithm)}$$

$$= \log \left\{\frac{2^8 \div (2^2)^2}{\log 2}\right\}$$

$$= \frac{\log (2^{8-4})}{\log 2} = \frac{\log (2^4)}{\log 2}$$

$$= \frac{4 \log 2}{\log 2} = 4$$ **Ans.**

Q.14. Find the value of the following :

63.42 × 7.639

Ans. $63.42 \times 7.639 = \text{A.L. } (\log 63.42 + \log 7.639)$

$= \text{A.L. } (1.8022 + 0.8830)$

$= \text{A.L. } (2.6852) = 484.4$ **Ans.**

Q.15. Find the value of .0009328 ÷ 2614

Ans. $= \text{A.L. } (\text{Log } .0009328 \div 2614)$

$= \text{A.L } (\overline{4}.9698 - \overline{1}.4173)$

$= \text{A.L. } -4 + .9698 + 1 - .4173)$

$= \text{A.L. } (\overline{3}.5525) = .003569$ **Ans.**

Q.16. Find the value of : $\dfrac{5.856 \times 0.007161}{0.06253}$

Ans. $= \text{Log } 5.856 + \text{Log } 0.007161 - \log 0.06253$

$= 0.7676 + \overline{3}.8550 - \overline{2}.7961$

$= 0.7676 + (-3 + .8550) - (-2 + .7961)$

$= 0.7676 - 3 + .8550 + 2 - .7961$

$= -3 + 1.6226 + 2 - .7961$

$= -1 + 1.6226 - .7961$

$= -1 + .8265$

$= \overline{1}.8265, \text{ A.L. } (\overline{1}.8265)$

$= 0.6707$ **Ans.**

Q.17. If Log4 = 0.6021, find Log 16 + Log 256.

Solution: (i) $\text{Log } 16 = \text{Log } 4^2 = 2 \text{ Log } 4 = 2 \times 0.6021 = 1.2042$

(ii) $\text{Log } 256 = \text{Log} 4^4 = 4 \text{ Log } 4 = 4 \times 0.6021 = 2.4084$ **Ans.**

Q.18. If $a^2 + b^2 = 7ab$, Prove that,

$$\text{Log}\left(\frac{a+b}{3}\right) = \frac{\text{Log } a + \text{Log } b}{2}$$

Solution: $a^2 + b^2 = 7ab$

Adding 2ab on both the side, we get,

$$a^2 + b^2 + 2ab = 7ab + 2ab$$

$$(a+b)^2 = 9ab$$

$$(a+b)^2 = 3^2 ab$$

$$\frac{(a+b)^2}{3^2} = ab, \text{ or } \left(\frac{(a+b)}{3}\right)^2 = ab$$

Taking Logarithrns –

$$\text{Log} \frac{(a+b)^2}{3} = \text{Log } ab$$

$$2\text{Log}\left(\frac{a+b}{3}\right) = \text{Log } a + \text{Log } b$$

$$\text{Log}\left(\frac{a+b}{3}\right) = \frac{\text{Log } a + \text{Log } b}{2}$$ ∵ **Proved**

PERCENTAGE

Q.19. What do you understand by 'Percentage'? Discuss the various rules regarding calculations of percentage.

Ans. Meaning of Percentage : The word 'Percent' is made of two words 'per' and 'cent'. 'Per' means each and 'cent' means hundred. As such 'Percent' means 'on or for each hundred' or 'in each hundred'. It is usually denoted by symbol '%' e.g., $\frac{80}{100}$ is called '80 percent or 80%.' 80 is called the rate percent.

Percentage can be defined as :

(i) A percentage is a fraction of which denominator is 100 and the numerator is a definite quantity which is known as the rate percent.

(ii) A number expressed as a fraction of 100 is called a percentage.

(iii) A percentage is a ratio, the consequent of which is 100 and the antecedent is the rate percent.

Rules regarding calculation of Percentage :

(1) To calculate percentage of a number, the number is multiplied by the rate percent and the result is divided by 100.

$$\text{Percentage of a number} = \frac{\text{Number} \times \text{Rate percent}}{100}$$

(2) To convert percentage into fraction, remove the sign % and divide the rate percent by 100.

$$\text{Fraction} = \frac{\text{Rate Percent}}{100}$$

(3) To change percentage into a decimal fraction, the rate percent is divided by 100 and the sign % is removed.

$$\text{Decimal fraction (Number)} = \frac{\text{Rate Percent}}{100}$$

(4) To change a decimal number into percentage, the decimal number is multiplied by 100.

$$\text{Percentage} = \text{Decimal Number} \times 100$$

(5) To change or convert a fraction into percentage, the fraction is multiplied by 100.

$$\text{Percentage} = \text{Fraction} \times 100$$

(6) If we have to obtain one number as a percentage of other, we divide the first number by the second number and multiply the result by 100.

$$\text{Percentage} = \frac{\text{Number whose percentage is to be obtained}}{\text{other number with which percentage is to be found out}} \times 100$$

Q.20. In a school upto secondary classes, out of the total enrollment 55% are in primary classes and 25% in middle classes. It there are 320 students in secondary classes, find the total enrollment of the school.

Solution: Let the total enrollment of the school be 100.

Number of students in Primary classes = 55

Number of students in Middle classes = 25

Number of students in Secondary classes = 100 – (55 + 25)

$= 100 - 80 = 20$

If number of students in secondary classes is 20, then the total = 100

If number of students in secondary class is 1, then the total $= \frac{100}{20}$

If number of students in secondary classes is 320, then the total

$$= \frac{100}{20} \times 320 = 1600 \qquad \textbf{Ans.}$$

Q.21. The daily wages of a casual labour have increased by 20% this year. If the present daily wages of a casual labour is Rs. 30, find the wage before increase.

Solution: Let the wages of a casual labour be Rs. x

$$\text{Increase in wages} = 20\% \text{ of } x$$

$$= \frac{20}{100} \times x = \text{Rs. } \frac{x}{5}$$

$$\therefore \quad \text{New wages} = x + \frac{x}{5} = \text{Rs. } \frac{6x}{5}$$

$$\text{But New wages} = \text{Rs. } 30$$

$$\therefore \quad \frac{6x}{5} = 30$$

$$\text{Or} \quad x = \frac{30 \times 5}{6} = \text{Rs. } 25 \qquad \textbf{Ans.}$$

Q.22. In an election 4% votes are invalid of the votes polled. One candidate gets 55% of the valid votes and wins the election by a margin of 240 votes. Find the number of votes polled.

Solution: Suppose the total number of votes polled is 1000.

$$\text{Invalid votes} = 1000 \times \frac{4}{100} = 40$$

$$\text{Valid votes} = 1000 - 40 = 960$$

Now, number of votes of the winning candidate

$$= 960 \times \frac{55}{100}$$

$$= 528 \text{ votes}$$

Number of votes of the defeated candidate

$$= \frac{960 \times 45}{100} = 432 \text{ votes}$$

$\therefore$ Difference = 528 – 432 = 96 votes

When difference in votes is 96 then total votes polled = 1000

$\therefore$ When difference in votes is 1 then total votes polled

$$= \frac{1000}{96}$$

$\therefore$ When difference in votes is 240 then total votes polled

$$= \frac{1000 \times 240}{96} = 2500 \text{ votes} \qquad \textbf{Ans.}$$

Q.23. In an examination 70% examinees passed in Commercial Mathematics and 65% in Financial Accounts, 15% failed in both the subjects. If 2700 examinees passed in both the subjects, find the total number of examinees.

Solution: Let the number of examinees be 100

No. of examinees failed in Commercial Maths 100 – 70 = 30

No. of examinees failed in Financial Accounts 100 – 65 = 35

No. of examinees failed in both the subject = 15

Total No. of examinees who failed = 30 + 35 – 15 = 50.

$\therefore$ 50 Examinees passed if total is 100

$\therefore$ 1 Examinees passed if total is $\frac{100}{50}$

$\therefore$ 2700 Examinees passed if total is $\frac{100 \times 2700}{50}$

= 5400 Examinees. **Ans.**

Q.24. The annual turnover of a firm has decreased by 2.5% for consecutive years. If its turnover is Rs. 1,18,638 in the 3rd year, find out his turnover in the first year of these three years.

Solution: Let the turnover for the first year be P_1, in the second year be P_2, and that of third year be P_3. Then,

$$P_2 = P_1\left(1 - \frac{r}{100}\right), P_3 = P_2\left(1 - \frac{r}{100}\right)$$

$$\Rightarrow \quad P_3 = P_1\left(1 - \frac{r}{100}\right)^2$$

Where r = rate percentage of decrease per year

$$\Rightarrow \quad \text{Rs. } 1,18,638 = P_1\left(1 - \frac{2.5}{100}\right)^2 = P_1\left(\frac{97.5}{100}\right)^2$$

$$\Rightarrow \quad P_1 = \frac{118638 \times (100)^2}{(97.5)^2}$$

= Rs. 1,24,800 **Ans.**

Q.25. The Sales of this year is 10% less in comparison to the sales of last year. By how much percent the sales should be increased so that the sales of next year equates with the sales of last year?

Solution: Let the sales of last year be Rs. 100

Hence, Sales of this year = Rs. 100 – 10 = Rs. 90

Let the required increase in the sales of this year be x% then Rs. 90 + x% of 90 = Rs. 100

$$\Rightarrow \quad 90 + \frac{90x}{100} = 100$$

$$\Rightarrow \quad \frac{90x}{100} = 100 - 90 \Rightarrow \frac{90x}{100} = 10$$

$$\therefore \quad x = \frac{100 \times 10}{90} = \frac{1000}{90} = 11\frac{1}{9}$$

Hence, the sales of this year should be increased by $11\frac{1}{9}$%. **Ans.**

Q.26. In an Examination Mohan gets 30% marks of the total marks and fails by 20 marks, whereas Sohan gets 40% marks of the total marks and his marks are 40 more than minimum passing marks. Find- (a) maximum marks of examination, and (b) minimum passing marks.

Solution: Let Maximum Marks be x.

Minimum Passing Marks :

(i) Mohan $x \times \frac{30}{100} + 20$ or $\frac{30x}{100} + 20$

(ii) Sohan $x \times \frac{40}{100} - 40$ or $\frac{40x}{100} - 40$

In both the circumstances the minimum pass marks are equal hence the equation will be :

$$\frac{40x}{100} - \frac{40}{1} = \frac{30x}{100} + \frac{20}{1}$$

$$\frac{40x - 4000 = 30x + 2000}{100}$$

Or $40x - 4000 = 30x + 2000$ Or $40x - 30x = 2000 + 4000$

Or $10x = 6000$ Or $x = \frac{6000}{10} = 600$

(a) Total Marks are 600

(b) Minimum Passing Marks

$= 600 \times \frac{30}{100} + 20 = 200$ Or $600 \times \frac{40}{100} - 40 = 200$

Minimum Passing Marks = 200 **Ans.**

Q.27. Reduction by 20% in the price of tea enables a person to purchase 4 kg. tea more for Rs. 80. What is the reduced price and the price reduction?

Solution : Due to reduction in price of tea by 20% the purchaser saves $80 \times \frac{20}{100}$ = Rs. 16 in exchange of which he gets 4 kg. tea more.

$\therefore$ Reduced Price of tea = $\frac{16}{4}$ = Rs. 4 per kg.

$\therefore$ When reduced Price is Rs. 80, the price before reduction is Rs. 100

$\therefore$ When reduced Price is Rs. 1 the price before reduction

$$= \text{Rs. } \frac{100}{80}$$

$\therefore$ When reduced Price is Rs. 4, the price before reduction

$$= \text{Rs. } \frac{100 \times 4}{80}$$

$$= \text{Rs. 5 per kg.}$$

Ans.

Q.28. The population of a village is 3,500. 5/7 of the population are men and the remaining are women. Among men 40% are married. Find the percentage of married women.

Solution : Total Population is 3500

$$\text{Population of Men} = \frac{5}{7} \text{ of } 3500 = 2500 \text{ Men}$$

$$\text{Population of Women} = 3500 - 2500 = 1000 \text{ Women}$$

$$\text{Married men} = 40\% \text{ of } 2500 = \frac{2500 \times 4}{100} = 1000$$

Married women=(on assumption one man and his one wife)

Hence, percentage of married

$$\text{Women} = \frac{1000 \times 100}{100} = 1{,}000$$

As such 100% are married women **Ans.**

UNIT-4

Long and Short Answer type Questions :

RATIO

Q.1. What is 'Ratio'? Mention its main Characteristics.

Or

What do you understand by 'Ratio'? Give its main features.

Or

What do you mean by 'Ratio'? Explain and also give its essential features. State the points to be remembered in working out problems.

Ans. Ratio is the comparative relationship between two quantities of the same kind expressed in the same unit. The comparison is made to consider what multiple or part or parts, one quantity is of the others.

In other words, the ratio of a number with another number is that third number which multiplied by second number gives first number.

For example.

The ratio of 48 to 6 is $\frac{48}{6}$, i.e. 8

Since, $8 \times 6 = 48$

Here 48 is first number, 6 is second number and 8 is third number.

The ratio of a and b is written as a ; b (read as a is to b). The quantities a and b are called the terms of the ratio a : b of which the first term, a is called the antecedent and the second term, b is called the consequent.

To find what multiple or part a is of b, we divide a by b. Therefore, the ratio a: b is expressed by the fraction $\frac{a}{b}$.

Thus, the ratio of 4 kg. of tea to 7 kg of tea is 4 : 7 or $\frac{4}{7}$. The ratio of 5 cms to 30mm is $\frac{5cm}{3cm}$. i.e., $\frac{5}{3}$ or 5 : 3, since 30mm = 3 cms.

Main Features of a Ratio are :

(i) Ratio is the relation between two quantities of the same kind.

(ii) The terms of a ratio must be in the same unit.

(iii) A ratio expresses the number of times that one quantity contains another. Therefore, every ratio is an abstract quantity, no unit is associated with it.

(iv) Ratio is calculated by dividing one number by another number.

(v) A ratio remains unaltered even if both numerator and denominator is multiplied or divided by any number.

Points to remember in working out problems :

(i) Reduce the two quantities in the same unit.

(ii) When the quantity is increased by a given ratio, multiply the quantity by the greater ratio.

Example : Production of fans which was 500 in the previous year has increased during the current year in the ratio 5 : 7, find the new production.

$$\text{New Production} = 500 \times \frac{7}{5} = 700$$

(iii) When a quantity is decreased in a certain ratio multiply the lesser ratio.

Example : Manufacturing cost of a fan which was Rs. 90 in the previous year, has decreased in the 5 : 4 find the new cost.

$$\text{New Cost} = \text{Rs. } 90 \times \frac{4}{5} = \text{Rs. } 72$$

(iv) When both increase and decrease of quantity are present in a problem, multiply the quantity by the greater ratio in case of increase and then multiply the result by the lesser ratio to obtain the final result.

Example : Production of fans which was 70 previously, has increased in the ratio 7 : 9 at first, but subsequently decreased in the ratio 5 : 4, find the new production.

$$\text{New Production} = 70 \times \frac{9}{7} \times \frac{4}{5} = 72$$

Properties of Ratio : The various properties of ratio are :

(i) Ratio of Equality : If a = b, the ratio a : b is said to be a ratio of equality i.e., 3 : 3

(ii) Ratio of greater inequality : If a > b, the ratio a : b is said to be a ratio of greater inequality i.e., 5 : 3

(iii) Ratio of Lesser inequality : If a < b, the ratio a : b is said to be a ratio of lesser inequality i.e. 3 : 5. The inverse ratio of a to b is b : a. Thus 3 : 7 is the inverse ratio of 7 : 3.

Q.2. The sum of two numbers is 40 and difference is 4. What is the ratio between the numbers?

Solution: Suppose two numbers are x and y.

The sum of numbers $= x + y = 40$ ...(1)

The difference of numbers $= x - y = 4$ ...(2)

By adding equations (1) and (2) we get –

$$2x = 44$$

$$\therefore \quad x = 22$$

By putting the value of x in equation (1)

$$22 + y = 40$$

$$y = 40 - 22 = 18$$

$\therefore$ Ratio between x and y –

$$x : y = 22 : 18$$

or $\quad x : y = 11 : 9$ **Ans.**

Q.3. A, B and C play cricket. The runs made by A and B are in ratio of 3 : 2 and runs made by B and C are also in Ratio of 3 : 2. They make together 342 runs. Find the runs made by each.

Solution:

$A : B = 3 : 2 = 9 : 6$ $\quad$ A's runs $= 342 \times \frac{9}{19} = 162$ runs

$B : C = 3 : 2 = 6 : 4$ $\quad$ B's runs $= 342 \times \frac{6}{19} = 108$ runs

$\therefore A : B : C = 9 : 6 : 4$ $\quad$ C's runs $= 342 \times \frac{4}{19} = 72$ runs **Ans.**

Q.4. Distribute Rs. 500 among A, B, C and D in such a way that A and B together get thrice than C and D get together, B should get 4 times than C and C should get $1\frac{1}{2}$ times than D.

Solution: Let D's share = 1

$$\text{C's share} = 1\frac{1}{2} = \frac{3}{2},$$

and $\quad \text{B's share} = 4 \times \frac{3}{2} = 6$

$$\text{A's share + B's share} = 3\,(\text{C's share + D's share})$$

$\Rightarrow$ A's share = 3 (C's share + D's share) – B's share

$$= 3\left(\frac{3}{2} + 1\right) - 6$$

$$3 \times \frac{5}{2} - 6 = \frac{15}{2} - 6 = \frac{15 - 12}{2} = \frac{3}{2}$$

Hence, the ratio for Division

$$A : B : C : D = \frac{3}{2} : 6 : \frac{3}{2} : 1 = 3 : 12 : 3 : 2$$

$\text{A's share} = 500 \times \frac{3}{20} = \text{Rs. } 75$ **Ans.**

$\text{B's share} = \frac{500 \times 12}{20} = \text{Rs. } 300$ **Ans.**

$\text{C's share} = \frac{500 \times 3}{20} = \text{Rs. } 75$ **Ans.**

$\text{D's share} = \frac{500 \times 2}{20} = \text{Rs. } 50$ **Ans.**

Q.5. (A) In a club, 20 of the members play carom, 16 play table-tennis and 24 play badminton. Find the ratio of the number of members who play

(i) Carom to the number of those who play table-tennis.

(ii) Carom to the number of those who play badminton.

(iii) Table–tennis to the number of those who play badminton.

(B) If the remaining 40 members do not play any game. Find the ratio of the members of players and non–players.

Solution: (A) (i) Ratio between the number of those who play Carom and who play table–tennis = 20 : 16.

H.C.F. of 20 and 16 = 4

$\therefore$ The required ratio $= \frac{20}{4} : \frac{16}{4} = 5 : 4$

(ii) Ratio between the number of those who play carom and who play badminton = 20 : 24

H.C.F. of 20 and 24 = 4

$\therefore$ The required ratio $= \frac{20}{4} : \frac{24}{4} = 5 : 6$

(iii) Ratio between the number of those who play table tennis and who play badminton = 16 : 24

H.C.F. of 16 and 24 = 8

$\therefore$ The required ratio $= \frac{16}{8} : \frac{24}{8} = 2 : 3$

(B) Total players = 20 + 16 + 24 = 60 ; Non players = 40

Ratio between players and non–players = 60 : 40

H.C.F. of 60 and 40 = 20

$\therefore$ The required ratio $= \frac{60}{20} : \frac{40}{20} = 3 : 2$

Q.6. A student bought books, note–books, and pencils from a stationer. If the ratio of the number of books, to the number of note–books is the same as the ratio of the number of note books to the number of pencils, find the number of note–books, if the books and pencils are 20 and 5 respectively.

Solution:

Number of books = 20

Number of Pencils = 5

Let the number of note–books be x, then

$$20 : x = x : 5$$

$\therefore$ $x \times x = 20 \times 5$ or $x^2 = 100$

or $x = \sqrt{100} = 10$

$\therefore$ The number of note books = 10

Q.7. Divide Rs. 52 among A, B and C in ratio of $\frac{1}{2} : \frac{1}{3} : \frac{1}{6}$.

Solution: Ratio of division among A, B and C $= \frac{1}{2} : \frac{1}{3} : \frac{1}{6}$

Ratio with common denominator

$$= \frac{1}{2} \times 6 : \frac{1}{3} \times 6 : \frac{1}{6} \times 6$$

[L.C.M. of 2,3, 6 is 6]

or $3 : 2 : 1$

Sum of the ratios = 6

A's share $= \frac{52 \times 3}{6} =$ Rs. 26 **Ans.**

B's share $= \frac{52 \times 2}{6} =$ Rs. 17.33 **Ans.**

C's share $= \frac{52 \times 1}{6} =$ Rs. 8.67 **Ans.**

Q.8. The total salary of two employees is Rs. 345. The proportion of their salaries is 7 : 8. Find the salary of each. What reduction of equal amount should be made in their salaries, so that this ratio becomes 5 : 6?

Solution:

Sum of the terms in the ratio = 7 + 8 = 15

Salary of first employee = Rs. $345 \times \frac{7}{15}$ = Rs. 161

Salary of second employee = Rs. $345 \times \frac{8}{15}$ = Rs. 184

Let a reduction of Rs. x be made in their salaries,

then :

$$\frac{161 - x}{184 - x} = \frac{5}{6}$$

$\Rightarrow$ $6(161 - x) = 5(184 - x)$

$\Rightarrow$ $966 - 6x = 920 - 5x$

$\Rightarrow$ $-6x + 5x = 920 - 966$

$\Rightarrow$ $x = -46$

$x = \text{Rs. } 46$

or

i.e., a reduction of Rs. 46 should be made in the salary of each employee so that the ratio between their incomes becomes 5 : 6. **Ans.**

Q.9. In a box by collecting one rupee coins, 50 paise coins and 25 paise coins, in the ratio of 4 : 5 : 6, the amount becomes Rs. 64. Find the number of coins of each denominations.

Solution:

Ratio of coins in respect of their values 4 : 2.50 : 1.50

As such the total amount Rs. 4 + 2.50 + 1.50 = Rs. 8

When total amount is Rs. 8 the number of one rupee coins = 4

∴ When total amount is Rs. 64 the number of one rupee coins

$$= \frac{4 \times 64}{8} = 32$$

∴ When total amount is Rs. 8 the number of 50 paise coins = 5

∴ When total amount is Rs. 64 the number of 50 paise coins

$$= \frac{5 \times 64}{8} = 40$$

When total amount is Rs. 8 the number of 25 paise coins = 6

∴ When total amount is Rs. 64 the number of 25 paise coins

$$= \frac{6 \times 64}{8} = 48$$

The number of coins of all kinds are 32, 40, 48 respectively. **Ans.**

Q.10. The difference of bonus amount of two workers is Rs. 36. Their ratio of Salary is 2 : 5. Find their bonus amount. How much of bonus amount should be paid to each so that their ratio may be established as 1 : 2?

Solution:

The difference of their ratio is 5 – 2 = 3

When the difference is 3, the first term is 2

∴ When the difference is 36, the first term is $\frac{2}{3} \times 36 = 24$

When the difference is 3, the second term is 5

∴ When the difference is 36, the second term is $\frac{5}{3} \times 36 = 60$

Suppose each worker is given x salary more so as to make ratio 1 : 2 therefore the equation will be –

$$\frac{24 + x}{60 + x} = \frac{1}{2}$$

By cross multiplication $48 + 2x = 60 + x$

$$2x - x = 60 - 48$$

$$x = 12$$

$$\left.\begin{aligned} 24 + 12 &= 36 \\ 60 + 12 &= 72 \end{aligned}\right\} \text{Therefore Rs. 12 bonus should be given more}$$ **Ans.**

Q.11. What should be deducted from the numerator and denominator of the ratio 5 : 6, so that the resulting ratio is 8 : 11?

Solution: Let the number deducted be x

Then $\frac{5 - x}{6 - x} = \frac{8}{11}$

$\Rightarrow$ $11(5 - x) = 8(6 - x)$

$\Rightarrow$ $55 - 11x = 48 - 8x$

$\Rightarrow$ $-11x + 8x = 48 - 55$

$\Rightarrow$ $-3x = -7$

$\Rightarrow$ $x = \frac{7}{3}$

Therefore, 7 should be duducted from the numerator and 3 from the denomirator **Ans.**

Q.12. Which is the greatest ratio?

$$\frac{7}{3} ; \frac{25}{7} ; \frac{35}{9} ; \frac{19}{2}$$

Solution: To make comparison of two or more ratios, they are converted into the fraction of common denominator. Therefore to make the common denominator–

L.C.M. of 3, 7, 9 and 2 = $3 \times 7 \times 9 \times 2 = 126$

Therefore, the fraction will be–

$$\frac{7 \times 42}{126} ; \frac{25 \times 18}{126} ; \frac{35 \times 14}{126} ; \frac{19 \times 63}{126}$$

Or $$\frac{294}{126} ; \frac{450}{126} ; \frac{490}{126} ; \frac{1197}{126}$$

It is clear that the fraction of the highest numerator is the greatest.

Therefore the ratio 19 : 2 or $\frac{19}{2}$ is the greatest. **Ans.**

Q.13. Two clerks A and B had their incomes as Rs. 75 and Rs. 145 respectively, were given an increment in their income. If

their income has been increased by same proportion then what is the income of B if income of A is Rs. 90?

Solution:

A's old salary = Rs. 75

His new salary = Rs. 90

Increment in salary = Rs. 90 – 75 = Rs. 15

$$\text{Ratio of increment} = \frac{15}{75}$$

Now let the ratio of increment in B's income

$$B = \frac{x}{145}$$

$$\text{According to question,} \quad \frac{15}{75} = \frac{x}{145}$$

$$x = \frac{15 \times 145}{75} = \text{Rs. } 29$$

$\therefore$ B's income = Rs. 145 + Rs. 29 = Rs. 174 **Ans.**

Q.14. An Employer reduces the number of his worker in the ratio of 9 : 8 and increases their salary in the ratio of 14 : 15. Find out what ratio the wages bill increases or decreases, and find the difference of the bill amount if was previously Rs. 1890.

Solution:

Ratio of Employees = 9 : 8

Ratio of the rate of Salary = 14 : 15

Ratio of total Salary

= (14 × 9) : (15 × 8)

Or 126 : 120 or 21 : 20

Therefore, the wages bill decreases in the ratio of 21 : 20

$\therefore$ The amount of the new bil

$$= 1890 \times \frac{20}{21} = \text{Rs.}1800$$

Difference in the amount of the bill = Rs.1890 – 1800

= Rs. 90 **Ans.**

Q.15. The ratio of railway fare between two stations for I, II and III class was 10 : 8 : 3 and during the year, the number of passengers of I, II and III class between these two stations were 3 : 4 : 10. During the year, the sale of tickets to passengers traveling between these two stations was of Rs. 8050. What amount was realised by the sale of second class tickets?

Solution:

Ratio of Fare = 10 : 8 : 3

Ratio of Passengers = 3 : 4 : 10

$\therefore$ Ratio of sale of Tickets = (10 × 3) : (8 × 4) : (3 × 10)

= 30 : 32 : 30

$$\therefore \text{Sale of second class Tickets} = \frac{8050 \times 32}{92} = \text{Rs. } 2800$$ **Ans.**

Q.16. The Annual Income of A and B is in the ratio of 4 : 3 and their expenses are in the ratio of 3 : 2. If at the end of the year each of them saves Rs. 600, find the annual income of each.

Solution: Ratio of Annual Income of A and B = 4 : 3

$\therefore$ Ratio of Annual Expenses of A and B = 3 : 2

Ratio of Annual Saving of A and B = 1 : 1

When A Saves Re. 1, his income is Rs. 4

$\therefore$ When A Saves Rs. 600 his income will be Rs.4×600 = Rs.2400

When B Saves Re. 1, his income is Rs. 3

$\therefore$ When B Saves Rs. 600 his income will be Rs. 3×600= Rs. 1800

Ans.

Q.17. What is meant by Profitability Ratio? Mention the various ratios which are used in its analysis.

Or

Write short notes on: (i) Operating Ratio (ii) Sacrificing Ratio (iii) Gaining Ratio.

Ans. Meaning of Profitability Ratio : The primary object of each enterprise is to earn profit and therefore, every enterprise tries to obtain more profit not in absolute form but also in relative form. That is in comparison to the capital etc., profit should be quite sufficient. The efficiency of obtaining maximum profit by any business organisation is known as 'Profitability'. Thus profitability ratios measure the profit earning capacity of an enterprise. Various ratios are computed in the evaluation of profitability are as under :

(a) Gross Profit Ratio : This ratio measures a relationship between Gross Profit and Net Sales. The purpose of computing this ratio is to indicate :

(a) The extent to which cost of goods sold is covered by the sales and (b) the cushion available to meet the operating expenses (other than the cost of goods sold) and non–operating expenses, to declare dividend and to build up reserves for growth.

This ratio is computed as under :

$$\textit{Gross Profit Ratio} = \frac{\textit{Gross Profit}}{\textit{Net Sales}} \times 100$$

(b) Net Profit Ratio : This ratio measures a relationship between Net Profit and Net Sales. The purpose of commuting this ratio is to indicate (i) the extent to which operating cost is covered by the sales, and (ii) the cushion available to meet the non–operating expenses, to declare dividend and to build up reserves for growth. This ratio is computed as under :

$$\text{Net Profit Ratio} = \frac{\text{Net Profit}}{\text{Net Sales}} \times 100$$

(c) Operating Ratio : An operating ratio relates the operating cost to net sales. The main objective of computing this ratio is to determine the operational efficiency with which production and/or purchase and selling operations are carried on.

$$\text{Operating Ratio} = \frac{\text{Operating Expenses}}{\text{Sales}} \times 100$$

(1) Sacrificing Ratio : The ratio in which the old partners have agreed to sacrifice their shares in profit in favour of the new partner is called the sacrificing ratio and is calculated by taking out the difference between old profit sharing ratio and the new profit sharing ratio. Unless given otherwise, the ratio of sacrifice is the same as old profit sharing ratio.

Sacrificing Ratio = Old Ratio – New Ratio

(2) Gaining Ratio : The ratio in which the continuing partners acquire the outgoing (retired or deceased) partner's share called as gaining ratio. This ratio is calculated by taking out the difference between new profit sharing ratio and old profit sharing ratio. Unless given otherwise, the gaining ratio is the same as old profit sharing ratio.

Gaining Ratio = New Ratio – Old Ratio

Q.18. A and B are partners sharing the profits in the ratio of 5 : 4. They admit C for $\frac{1}{3}$ share. Calculate new profit sharing ratio.

Solution: C's share $= \frac{1}{3}$

Remaining share $= 1 - \frac{1}{3} = \frac{2}{3}$

A's share is $\frac{5}{9}$ of $\frac{2}{3}$ or $\frac{5}{9} \times \frac{2}{3} = \frac{10}{27}$

B's share is $\frac{4}{9}$ of $\frac{2}{3}$ or $\frac{4}{9} \times \frac{2}{3} = \frac{8}{27}$

C's share is $\frac{1}{3}$ of $\frac{9}{9}$ or $\frac{1}{3} \times \frac{9}{9} = \frac{9}{27}$

New Ratio is $\frac{10}{27} : \frac{8}{27} : \frac{9}{27}$ or $10 : 8 : 9$

Sacrificing Ratio = Old Ratio – New Ratio

A's sacrifice $\frac{5}{9} - \frac{10}{27} = \frac{15-10}{27} = \frac{5}{27}$

B's sacrifice $\frac{4}{9} - \frac{8}{27} = \frac{12-8}{27} = \frac{4}{27}$

Hence Sacrificing Ratio is 5 : 4 **Ans.**

Q.19. X and Y are partners sharing profits and losses in the ratio of 7 : 5. They admit Z as a partner who acquires his share as $\frac{1}{12}$ from X and $\frac{1}{8}$ from Y. Calculate the new profit sharing ratio and the sacrificing ratio.

Solution:

	X	Y
(A) Their existing shares	$\frac{7}{12}$	$\frac{5}{12}$
(B) Share acquired by Z	$\frac{1}{12}$	$\frac{1}{8}$
(C) New Share of old partners (A) – (B)	$\frac{6}{12}$	$\frac{7}{24}$

(D) Share of incoming partner (Z) $= \frac{1}{12} + \frac{1}{8} = \frac{5}{24}$

(E) New ratio of X, Y and Z $= \frac{12}{24} : \frac{7}{24} : \frac{5}{24} = 12 : 7 : 5$

(F) Sacrificing ratio = Share acquired from one existing partner : Share acquired from another existing partner

$= \frac{1}{12} : \frac{1}{8} = 2 : 3$ **Ans.**

Q.20. X, Y and Z are partners sharing profits and losses in the ratio of $\frac{1}{2} : \frac{3}{10} : \frac{1}{5}$. Calculate the new profit sharing ratio and gaining ratio (a) If X retires, (b) If Y retires, (c) If Z retires.

Solution: Let us convert the old ratio given in the form of fraction into one consisting of whole numbers

$$\frac{1}{2} : \frac{3}{10} : \frac{1}{5} = \frac{5:3:2}{10} \text{ or } 5:3:2$$

New ratio : In such a case, new ratio will be the same as the old ratio existed between the remaining partners since the remaining partners continue to share the future profits and losses in the old ratio as existed immediately before the retirement of a partner unless agreed otherwise. Thus, the new ratio will be as under :

(a) Between Y & Z = 3 : 2
(b) Between X & Z = 5 : 2
(c) Between X & Y = 5 : 3

Gaining Ratio : In such a case, gaining ratio will be the same as the old ratio existed between the remaining partners since the remaining partners gain in their old ratio unless agreed otherwise. Thus, the gaining ratio will be as under :

(a) Between Y & Z = 3 : 2
(b) Between X & Z = 5 : 2
(c) Between X & Y = 5 : 3

Q.21. X, Y and Z are partners sharing profits and losses in the ratio of $\frac{1}{2} : \frac{1}{8} : \frac{3}{8}$. Z retires and surrenders $\frac{4}{9}$th of his share in favour of X and remaining in favour of Y. Calculate the new profit sharing ratio and gaining ratio.

Solution:

Calculation of New share :

	X	Y
(a) Their existing share	$\frac{1}{2}$	$\frac{1}{8}$
(b) Share surrendered by z	$\frac{4}{9} \times \frac{3}{8} = \frac{12}{72}$	$\frac{5}{9} \times \frac{3}{8} = \frac{15}{72}$
(c) New share of remaining Partners (a + b)	$\frac{48}{72}$	$\frac{24}{72}$

(d) New ratio of X and Y = 48 : 24 or 2 : 1 **Ans.**

Calculation of Gaining Ratio :

Gaining Ratio = Share surrendered in favour of one remaining partner i.e., X : Share surrendered in favour of another remaining partner i.e., Y

$= \frac{12}{72} : \frac{15}{72}$

= 12 : 15 or 4 : 5

PROPORTION

Q.22. What do you mean by a proportion? Explain in brief.

Ans. A statement (or an equation) which represents two ratios that are equal is called proportion.

Let the two ratios $\frac{a}{b}$ and $\frac{c}{d}$ be equal. Then equation $\frac{a}{b} = \frac{c}{d}$ is a proportion.

$\frac{a}{b} = \frac{c}{d}$ means that the ratio from c to d is same as the ratio from a to b.

The equality of two ratios is also written as ': :'. For example,

$$\frac{a}{b} : : \frac{c}{d}$$

or

$$a : b : : c : d$$

Extremes and Means : The four terms a, b, c, d in the proportion a : b : : c : d are called proportional which are also known as first term, second term, third term and fourth term.

The terms a and d are called the extremes and the terms b and c are called the means.

Fourth proportional : In the proportion a : b : : c : d fourth term d is said to be the fourth proportional of the first three terms.

Product : According to the definition

$$\frac{a}{b} = \frac{c}{d}$$

$$\Rightarrow \quad ad = bc$$

i.e., the product of the extreme is equal to the product of the means. Hence, if any three terms in a proportion are known, the fourth may be determined by the relation ad = bc. This is why, this rule is called Rule of three.

Q.23. Differenciate between a Ratio and a Proportion?

Or

Give the differences between a Ratio and a Proportion.

Or

What qualities distinguishes a Proportion from a Ratio?

Ans. Difference between Ratio and Proportion : Ratio and proportion may be distinguished as follows :

(1) There are two terms in a ratio; there are four terms in a proportion.

(2) In ratio the two quantities must be of same kind; in proportion all the four quantities are not of same kind but the first two are of one kind and the last two may be of another kind.

(3) In proportion, the product of extremes is equal to the product of the means; there is no such rule in a ratio.

(4) Ratio is a comparison of two quantities of same kind; proportion is a comparison of two ratios.

Q.24. What do you mean by a Continued Proportion? Give example.

Ans. Continued Proportion : Three quantities a, b, c of same kind are said to be in continued ratio, if the ratio of first and second is same as the ratio of second and third and their proportion is called continued proportion, i.e., if $a : b = b : c$, then a, b, c are in continued ratio and the proportion $a : b :: b : c$ is continued proportion. Here second term is called mean proportional of first and third and third; term is called third proportional of the first two. For example,

Let us consider three numbers 20, 40, 80 where the ratio of 20 and 40 is same as the ratio of 40 and 80. Thus, the continued proportion is

$$\frac{20}{40} = \frac{40}{80} \text{ Or } \frac{1}{2} = \frac{2}{4}$$

or

$$1 : 2 :: 2 : 4$$

According to the rule.

the product of means = the product of extremes

$$2 \times 2 = 1 \times 4$$

i.e., (mean proportional)^2 = (first term × third term)

Q.25. Find mean proportional of 4 and 16.

Solution:

Let the mean proportional of 4 and 16 be x. Then

$$4 : x :: x : 16$$

$$\Rightarrow \quad x^2 = 4 \times 16 = 64$$

$$\therefore \quad x = \sqrt{64} = 8$$

Ans.

Q.26. The salary of A, B and C are in continued proportion. If the salary of B and C are respectively Rs. 250 and Rs. 1,250, find the salary of A.

Solution:

Let the salary of A be Rs. x. Then

$$x : 250 :: 250 : 1,250$$

$$\Rightarrow \quad x \times 1,250 = 250 \times 250$$

$$\therefore \quad x = \frac{250 \times 250}{1250} = 50$$

A's Salary = Rs. 50

Ans.

Q.27. What is the significance of Direct Proportion? Explain.

Ans. Direct Proportion : If in a proportion the two quantities of different kinds are so related that an increase (or decrease) in the value of one is accompanied by an increase (or decrease) in the same ratio, then these two quantities are called in direct ratio or directly proportional to each other and this proportion is known as direct proportion.

For example, if there is an increase (or decrease) in the quantity of an article, then there will be an increase (or decrease) in the same ratio in their prices.

The arrangement of the terms in the proportion for direct proportion is as follows :

	The quantity of an article		Price of the article
or	Quantity A		Quantity B (in general)
	$a_1 : a_2$	::	$b_1 : b_2$

Q.28. Solve for x :

(i) 4 : 9 : : x : 12

Sol. The product of extremes = The product of means.

$$4 \times 12 = 9 \times x$$

$$48 = 9x$$

$$x = \frac{48}{9} = 5.33$$

Ans.

(ii) x : 8 : : 12 : 16

Sol.

$$x \times 16 = 8 \times 12$$

$$x = \frac{96}{16} = 6$$

Ans.

(iii) 1.5 : 2.5 : : 2 : 5 : x

Sol.

$$1.5 \times x = 2.5 \times 2.5$$

$$x = \frac{5}{1.5} = 3.33$$

Ans.

(iv) 0.03 : x : : x : 0.12

Sol.

$$0.03 \times 0.12 = x \times x$$

$$x^2 = .0036$$

$$x = \sqrt{.0036} = 0.06$$

Ans.

Q.29. If 5 Kg wheat can be bought for Rs. 45, what quantity can be bought for Rs. 95?

Solution: Let the cost of x by wheat be Rs. 95

Quantity of Wheat	*Price*
5 Kg ↓	Rs. 45 ↑
x Kg	Rs. 95

$$5 \times 95 = 45 \times x$$

$$x = \frac{5 \times 95}{45}$$

$$x = \frac{475}{45} = 10.55$$

Hence, 10.55 Kg of wheat can be purchased for Rs. 95 **Ans.**

Q.30. Covers distance of 100 Kms in 3 hours. How much distance will he cover in 5 hours?

Solution: Let the distance covered in 5 hours be x kms.

Time		*Distance*
3 : 5	::	100 : x

$$3x = 100 \times 5$$

$$x = \frac{500}{3} = 166.66$$

Hence, motorcyclist will cover 166.66 kms in 5 hours. **Ans.**

Q.31. A Ship covers a distance in 5 hours @ 60 kmph. In what time that distance can be covered at the speed of 75 kmph.

Solution: Let the required distance be x kms.

Greater the speed, less the time taken (∴ Inversely Proportional)

Speed	*Time*
60 ↓	5 ↑
75	x

∴ The proportion would be :

$$x : 5 : : 60 : 75$$

$$75x = 5 \times 60$$

$$x = \frac{300}{75} = 4$$

Hence, the ship can cover that same distance in 4 hours @ 75 kmph. **Ans.**

Q.32. 25 men can construct a single floor of a multistory building in 15 days. How much days would be taken by 40 men to construct a floor in that building?

Solution: Let the days taken by 40 men be x days.

Greater the men, less the time taken (∴ Inversely Proportional)

Men	*Days*
25 ↓	15 ↑
40	x

∴ The proportion would be :

$$x : 15 : : 25 : 40$$

$$40x = 25 \times 15$$

$$x = \frac{375}{40} = 9.3$$

Hence, 40 men would take 9.3 days to construct a floor. **Ans.**

Q.33. If 4 kg. Ghee can be bought for Rs. 192, what quantity can be bought for Rs. 288?

Solution:

Let the cost of x kg. Ghee be Rs. 288.

Quantity of Ghee	Price
4kg ↓	Rs. 192 ↓
x kg	Rs. 288

Here the arrow indicates in the same direction, because the quantity of ghee is directly proportional to its price.

∴ The arrangement of the terms in the proportion is :

$$\frac{4}{x} = \frac{192}{288}$$

$$\Rightarrow \quad 4 \times 288 = x \times 192$$

$$\therefore \quad x = \frac{4 \times 288}{192} = 6$$

Hence, 6 kg. Ghee can be purchased for Rs. 288.

Q.34. If 5 persons make 15 kilometre road in a certain time, how many kilometre road will be made by 20 persons in the same time?

Solution: Let the required length of the road be x kilometre. Hence, arrangement of the terms in the proportion will be as follows :

Length	Number
15 ↑	5 ↑
x	20

$$x : 15 : : 20 : 5$$

$$\Rightarrow \quad 5x = 15 \times 20$$

$$\therefore \quad x = \frac{15 \times 20}{5} = 60$$

Q.35. A car covers a distance of 60 km in 3 hours. How much distance will it cover in 5 hours ?

Solution: Let the required distance be x km. More time more distance means time and distance are in direct proportion.

Time		Distance
3 : 5	: :	60 : x

$$\Rightarrow \quad 3x = 5 \times 60$$

$$\therefore \quad x = \frac{5 \times 60}{3} = 100$$

Q.36. 12 carpenters make 18 chairs in a certain time. How many chairs will be made by 10 carpenters in the same time.

Solution: Let the required number of chairs be x. Then

Carpenter	Chairs	
12 ↓	18 ↓	
10	x	Direct Proportion

Hence, the arrangement of the terms in the proportion is as follows:

$$12 : 10 : : 18 : x$$

$$\Rightarrow \quad 12 \times x = 10 \times 18$$

$$\therefore \quad x = \frac{10 \times 18}{12} = 15$$

Q.37. 16 men or 28 women can do a work in 40 days. In how many days will 24 men and 14 women complete the same work ?

Solution: 16 men's work = 28 women's work

$\therefore$ 16 men = 28 women (with reference to work)

$\therefore$ 8 men = 14 women (with reference to work)

Now the question,

16 men can do a work in 40 days. In how many days (24 + 8) men complete the same work?

Men	Days
16 ↓	40 ↑
32	x

$$\therefore \quad 16 : 32 : : x : 40$$

$$\Rightarrow \quad 32x = 16 \times 40$$

$$\therefore \quad x = \frac{16 \times 40}{32} = 20 \text{ days.}$$

DISCOUNT

Q.38. What is 'Discount'? Explain its kinds and methods of its calculation.

Or

Define : Marked Price; Trade Discount; Cash Discount.

Ans. Marked Price : Manufacturers, Publishers or Businessmen generally mark their articles at a price higher than actual selling price. This price is known as Marked Price or Market Price or List Price or Catalogue Price etc.

Discount : In business special allowance offered by seller to traders who buy the goods for the purpose of resale is called Discount. As such the buyer has to pay less than the price of the goods. Discount is of two types– (a) Trade discount, and (b) Cash discount.

(a) Trade Discount : It is an allowance which is given by the producer to his whole seller, and by a whole seller to the retailer, on the listed or catalogue price. The purpose of allowing trade discount is that the buyer may sell the goods to the customers at the listed or catalogue price. It is also the purpose of this discount that the buyer may be induced to buy more and more goods. Trade discount is generally allowed at a certain percentage on the listed price and it is deducted from the total amount of the invoice.

(b) Cash Discount : With a view to induce the customer to pay the amount promptly or after a specified period, the seller allows an allowance to the customer is known as cash discount.

A customer can be given both trade discount as well as cash discount at one time.

Methods of Calculation of Discount :

(1) Calculation of the amount of Discount

$$\textit{Discount} = \textit{List Price} \times \frac{\textit{Rate Percent of Discount}}{100}$$

(2) Calculation of Net Selling Price :

$$\textit{Selling Price} = \textit{Marked Price} - \textit{Discount}$$

(3) Calculation of amount of sales when the discount amount and Rate Percent of discount are known :

$$\textit{Amount of Sales} = \frac{\textit{Discount} \times 100}{\textit{Rate Percent of Discount}}$$

(4) Calculation of Rate of Discount, when Sales Amount and Discount Amount are given :

$$\textit{Rate of Discount} = \frac{\textit{Discount} \times 100}{\textit{Amount of Sales}}$$

Q.39. Having marked his article 25% Cost, a trader makes a profit of Rs. 300, after allowing a discount of 10% find the cost price.

Solution: Let the cost price be Rs. 100, then marked price which is 25% above cost will be Rs. 100 + 25% of Rs. 100 i.e., Rs. 100 + 25 = Rs. 125

Discount 10% of Rs. 125 = Rs. 12.50

Selling Price will be Rs. 125 – Rs. 12.50 = Rs.112.50

Profit =Rs. 112.50 – Rs. 100 = Rs. 12.50

When Profit is Rs. 12.50 the cost price is = Rs. 100

When Profit is Rs. 1 the cost price is = Rs. $\frac{100}{12.50}$

When Profit is Rs. 300 the cost price is = Rs. $\frac{100}{12.50} \times 300 = 2{,}400$

∴ Required. Cost Price will be Rs. 2,400 **Ans.**

Q.40. Find a single discount equivalent to the discount series 20%, 10% and 5%.

Solution:

	Let the Selling Price =	Rs. 100.00	
Less :	20% Discount =	Rs. 20.00	
		Rs. 80.00	
Less :	10% Discount =	Rs. 8.00	[10% of Rs. 80]
		Rs. 72.00	
Less :	5% Discount =	Rs. 3.60	[5% of Rs. 72]
∴	Net amount payable =	Rs. 68.40	

Hence single discount equivalent to the given discount series

= Rs. 100.00 – Rs. 68.40 = Rs. 31.60

= Rs. 31.60 on marked price of Rs. 100

= Rs. 31.60% **Ans.**

Q.41. A publication supplies 22 copies for every 20 copies of the book ordered and also allows a cash discount of 10%. What is the net rate of discount to the buyer.

Solution: Let the marked price of a copy = Rs. 1

∴ Marked Price of 22 copies = Rs. 1 × 22 = Rs. 22

and Marked price of 20 copies = 1 × 20 = Rs. 20

Discount = 10% of Rs. 20 = Rs. 2

Selling Price of 20 Copies = 20 – 2 = Rs. 18

Hence, Net Discount = Rs. 22 – Rs. 18 = Rs. 4

∴ Discount on Rs. 22 = Rs. 4

∴ Discount on Rs. 100 = Rs. $\frac{4}{22} \times 100$

$= 18\frac{2}{11}\%$ **Ans.**

Q.42. An importer buys goods at 10% trade discount, but he pays import duty at 5% on cost. Other expenses are 20% of the cost. To earn 15% profit on cost how much percentage above the catalogue price should be quoted?

Solution : Let the Catalogue Price be Rs. 100

Catalogue Price		Rs.	100.00
Less : 10% Trade Discount	=	Rs.	10.00
		Rs.	90.00
Add : 5% Import Duty on Rs. 90.00	=	Rs.	4.50
		Rs.	94.50
Add : Other expenses at 20% on Rs. 94.50	=	Rs.	18.90
Cost Price		Rs.	113.40
Add : Profit @ 15% on Rs. 113.40	=	Rs.	17.01
Sale Price		Rs.	130.41

∴ Difference = Rs. 130.41 – 100 = Rs. 30.41

Hence the price be quoted 30.1% above the catalogue price. **Ans.**

Q.43. A Retailer allows to his customers 10% + 5% trade discount. A customer placed an order for Rs. 7,797.20. Find what amount will be received?

Solution:

Total Price of the Goods =	Rs. 7,797.20
Less : Trade discount 10%	Rs. 779.72
	Rs. 7017.48
	Rs. 350.87
Less : Trade Discount 5% =	Rs. 6666.61
Net amount receivable	Rs. 6656.61

Ans.

Q.44. A Manufacturer after allowing a trade discount of 20% earns profit 25%. If manufacturing cost increases by 25% and no change is made in the rate of discount but the catalogue price is increased by 15%, then find the percentage of revised profit.

Solution: Let the Catalogue Price be Rs. 100

Less : 20% discount on Catalogue Price Rs. 20

Invoice price Rs. 80

On taking 25% profit on cost invoice price will be Rs.100+25= Rs. 125

When invoice price is Rs. 125 the cost is = Rs. 100

∴ When invoice price is Rs. 80 the cost is

$$= \text{Rs. } \frac{100 \times 80}{125} = \text{Rs. } 64$$

On increase in cost 25% $= \frac{64 \times 25}{100} = $ Rs. 16

Revised cost = Rs. 64 + Rs. 16

= Rs. 80

Increased in Catalogue price 15 percent = Rs. 15

New Catalogue price = Rs. 100 + 15 = Rs. 115

Less : Trade discount 20 percent $= \frac{115 \times 20}{100} = 23$

New invoice price = Rs. 92

New invoice price is Rs. 92 and new cost is Rs. 80

∴ Profit = Rs. 92 – Rs. 80 = 12

∴ On cost of Rs. 80 the Profit = 12

∴ On cost of Rs. 1 Profit $= \frac{12}{80}$

∴ On cost of Rs. 100 the Profit $= \frac{12 \times 100}{80} = 15$

New percentage of Profit = 15% **Ans.**

BROKERAGE

Q.45. Who is a Broker? What is Brokerage?

Ans. Broker : He is the agent who establishes dealing between buyer and seller. The remuneration he is paid is called brokerage. His work is finished the moment deal meterialises between buyer and seller. Thus, he has neither the right of maintaining stock nor to realise sale proceeds. Such brokers get brokerage generally from the buyer as well as the seller.

Brokerage : An agent whose main function is to buy or sell Shares, Bonds, Promissory Notes, etc., of the company is called a Broker and the commission of the broker is called 'Brokerage'.

Q.46. An agent sells a house and after deducting $6\frac{2}{3}$% brokerage pay Rs. 28,560 to the owner of the house. Find the cost of the house.

Solution: Let the selling price of the house is Rs. 100

Brokerage is $6\frac{2}{3}$% i.e. Rs. $\frac{20}{3}$

The house owner will get = Price of the house – Brokerage

$$= \text{Rs. } 100 - \frac{20}{3} = \frac{300-20}{3} = \frac{280}{3}$$

When house owner required Rs. $\frac{280}{3}$, the price of the house

= Rs. 100

When house owner required Rs. 1, the price of the house

$$= \text{Rs. } \frac{100 \times 3}{280}$$

When house owner required Rs. 28,560 the price of the house

$$= \text{Rs. } \frac{100 \times 3 \times 28560}{280}$$

= Rs. 30,600 **Ans.**

Q.47. A broker charges brokerage @ 0.75% from the seller and 1.5% from the buyer of a house. If he gets a brokerage of Rs. 4,050, Calculate how much amount of brokerage is received from the buyer as well as seller, and what is the price of the house sold?

Solution: Rate of brokerage from the seller = 0.75%

Rate of brokerage from the buyer =1.5%

Combined rate of brokerage will be (0.75 + 1.5) = 2.25%

(i) $$\text{Price of the house} = \frac{\text{Amount of Brokerage} \times 100}{\text{Combined rate of brokerage}}$$

$$= \text{Rs. } \frac{4050 \times 100}{2.25} = \text{Rs. } 1,80,000 \text{ Ans.}$$

(ii) $$\text{Brokerage received from the buyer} = \frac{\text{Sales} \times \text{Rate}}{100}$$

$$= \text{Rs. } \frac{180000 \times 1.5}{100} = \text{Rs. } 2,700$$

(iii) $$\text{Brokerage received from the seller} = \frac{\text{Sales} \times \text{Rate}}{100}$$

$$= \text{Rs. } \frac{180000 \times 0.75}{100} = \text{Rs. } 1,350$$

Q.48. A person sells his house for Rs. 4,00,000 in all through a broker by giving 2% brokerage, find the brokerage of the broker and the amount received by the house seller.

Solution: Formula :

$$\text{Brokerage} = \frac{\text{Rate of brokerage} \times \text{Amount of sales}}{100}$$

$$= \frac{2 \times 4,00,000}{100} = \text{Rs. } 8,000$$

∴ The amount received by the house seller

$$= 4,00,000 - 8,000$$

$$= \text{Rs. } 3,92,000$$

Q.49. A person sold a house for Rs. 7,00,000 through a broker and gave Rs. 5,000 as brokerage. Find the rate percent of brokerage.

Solution: Formula,

$$\text{Rate of Brokerage} = \frac{\text{Amount of brokerage} \times 100}{\text{Amount of sales}}$$

$$= \frac{5,000 \times 100}{7,00,000}$$

$$= \frac{5}{7} \text{ i.e., brokerage is } \frac{5}{7}\% \text{ Ans.}$$

Q.50. An agent sells a plot and after deducting $6\frac{2}{3}\%$ brokerage pays Rs. 28,560 to the owner of the house. Find the cost of the house.

Solution: Let the selling price of the plot be Rs. 100.

$$\text{Brokerage} = 6\frac{2}{3} = \text{Rs. } \frac{20}{3}$$

The house owner will get = Price of plot – Brokerage

$$= 100 - \frac{20}{3} = \frac{300 - 20}{3}$$

$$= \text{Rs. } \frac{280}{3}$$

When plot owner received Rs. $\frac{280}{3}$,

then the price of the plot = Rs. 100

When plot owner received Rs. 1, then price of the plot $= \frac{100}{280/3}$

When plot owner received Rs. 28,560, then price of the plot

$$= \frac{100 \times 3 \times 28,560}{280}$$

$$= \text{Rs. } 30,600 \text{ Ans.}$$

Q.51. A person sold some part of his total land for Rs. 1,50,000 through a broker. The broker charged $5\frac{1}{2}\%$ brokerage, in relation to the sale. Find the net amount received by the seller.

Solution: Commission or Brokerage = $5\frac{1}{2}\%$ of 1,50,000

$$= \frac{11}{2} \times \frac{1}{100} \times 1,50,000 = \text{Rs. } 8,250$$

The net amount received = Sales – Brokerage

$$= 1,50,000 - 8,250$$

$$= \text{Rs. } 1,14,750 \text{ Ans.}$$

Q.52. A broker charges brokerage @ 1% from the seller and 1.5% from the buyer of a house. If he gets a brokerage of Rs. 5,000, calculate how much amount of brokerage is received from the buyer as well as seller and what is the price of the house sold?

Solution: Rate of brokerage from the seller = 1%

Rate of brokerage from the buyer = 1.5%

Rate of brokerage combined = 1% + 1.5% = 2.50%

$$\text{Price of the house} = \frac{\text{Amount of brokerage} \times 100}{\text{Combined rate of brokerage}}$$

$$= \frac{5{,}000 \times 100}{2.50}$$

$$= \text{Rs. } 2{,}00{,}000 \quad \text{... (1)}$$

$$\text{Brokerage received from the buyer} = \frac{\text{Sales} \times \text{Rate}}{100}$$

$$= \frac{2{,}00{,}000 \times 1.5}{100}$$

$$= \text{Rs. } 3{,}000 \quad \text{... (2)}$$

$$\text{Brokerage received from the seller} = \frac{2{,}00{,}000 \times 1}{100}$$

$$= \text{Rs. } 2{,}000 \quad \text{... (3)}$$

Q.53. A property changed hands three times. Each broker who sold it charged 1% as brokerage. If each time, it was sold for the net amount obtained at the previous sale, calculate its original sale value, if the third sale realised Rs. 1,00,000 net.

Solution: Let the original sale value be Rs. 100.

$$\text{Then Brokerage} = 1\% \text{ of Rs. } 100 = \frac{1}{100} \times 100$$

$$= \text{Rs. } 1$$

$$\text{Second sale value} = 100 - 1 = \text{Rs. } 99$$

$$\text{Brokerage} = 1\% \text{ of Rs. } 99 = \frac{1}{100} \times 99$$

$$= \text{Rs. } 0.99$$

$$\text{Third sale value} = 99 - 0.99 = \text{Rs. } 98.01$$

$$\text{Brokerage} = 1\% \text{ of Rs. } 98.01$$

$$= \frac{1}{100} \times 98.01 = \text{Rs. } 0.9801$$

$$\text{Net Sale} = 98.01 - 0.9801$$

$$= \text{Rs. } 97.0299$$

$$\text{Hence original sale value} = \frac{100 \times 1{,}00{,}000}{97.0299}$$

$$= \text{Rs. } 1{,}03{,}061 \textbf{ Ans.}$$

UNIT 5

Long and Short Answer type Questions :

COMMISSION

Q.1. Write a note on Manager's Commission.

Or

What is Manager's Commission? How is it calculated?

Ans. Manager's Commission : Normally, the employees (i.e., works manager or general manager) of a business are paid salary as their reward, but the payment of salary does not encourage a worker to work with zeal. In order that the employees may put in their best in the performance of their duties, sometimes, they are paid commission over and above their regular salary. This commission is paid either on the basis of sales or on the basis of net profits.

(1) Commission based on sales : A manager may be allowed to get some commission on the basis of sales effected by him or his team workers. In such a case, the calculation of commission is very simple and can be found out with the help of the following formula :

$$\frac{\textit{Sales} \times \textit{Rate of Commission}}{100}$$

(2) Commission based on net profit : This commission is paid out of net profits. Such commission may be :

(a) Fixed percentage on net profits before charging such commission to Profit and Loss Account, or

(b) Fixed percentage on net profits after charging such commission to Profit and Loss Account.

By the term Net Profit means profit which remains after charging all expenses and losses and crediting all gains and profits to Profit & Loss Account.

Manager's Commission will be calculated by finding out profits before commission with the following formula :

(a) If Commission is payable at fixed percentage on net profits before charging such commission :

$$\textit{Manager's Commission} = \frac{\textit{Rate}}{100} \times \textit{Profit before commission}$$

(b) If commission is payable at fixed percentage on net profits after charging such commission :

$$\textit{Manager's commission} = \frac{\textit{Rate}}{(100 + \textit{Rate})} \times \textit{Profit before commission}$$

Q.2. A Manager receives a commission of 10% on the gross sales and bonus of 5% on the sum exceeding Rs. 8,000. If he gets Rs. 3,600 as commission, find the amount of bonus.

Solution: $\text{Total Sales} = \dfrac{\text{Commission} \times 100}{\text{Rate Percent of Commission}}$

$$= \text{Rs. } \frac{3600 \times 100}{10} = \text{Rs. } 36{,}000$$

Sales exceeding Rs. 8,000 = Rs. 36,000 – Rs. 8,000 = Rs. 28,000

$$\therefore \text{Bonus} = \frac{\text{Rate of Bonus} \times \text{Sales exceeding Rs. 8000}}{100}$$

$$= \text{Rs. } \frac{5 \times 28000}{100} = \text{Rs. } 1{,}400$$

Ans.

Q.3. The Gross profit of a Trader for the year ending 31st March, 2004 is Rs. 30,000. Their expenses for the year included Salary Rs. 3,000, office expenses Rs. 2,000, and advertisement Rs. 500. Manager is to be allowed commission at 10% on net profit :

(a) Before charging such commission.

(b) After charging such commission.

Calculate manager's commission under both the circumstances as given above.

Solution: Net Profit before charging Manager's Commission

$$= \text{Gross Profit} - \text{Expenses}$$

$$= \text{Rs. } \{30{,}000 - (3{,}000 + 2000 + 500)\}$$

$$= \text{Rs. } 30{,}000 - 5{,}500 = \text{Rs. } 24{,}500$$

Manager's Commission (a) before charging such commission $= \dfrac{\text{Net Profit} \times \text{Rate Percent of commission}}{100}$

$$= \frac{\text{Rs. } 24500 \times 10}{100}$$

$$= \text{Rs. } 2{,}450$$

(b) Manager's Commission after charging such commission :

Net Profit before commission is Rs. 2,450

The Commission payable is 10% on net profit after charging such commission

$$\text{Hence, Commission} = \frac{\text{Rate}}{(100 + \text{Rate})} \times \text{Profit before commission}$$

$$= \left(\frac{10}{100 + 10}\right) \times \text{Rs. } 24{,}500$$

$$= \frac{10}{110} \times 24{,}500 = \text{Rs. } 2{,}227$$

Manager's Commission Before charging commission = Rs. 2450
Manager's Commission After charging commission = Rs. 2227 **Ans.**

Q.4. A Salesman gets a salary of Rs. 100 per month and a commission at the rate of 4% on sales in excess of Rs. 20,000. If he wants to earn Rs. 500 in a month, how much should be his sales for that month?

Solution: The Salesman wants to earn = Rs. 500 p.m.

Salesman's salary = Rs. 100 p.m.

$\therefore$ Salesman's commission = Rs. 500 – 100 = Rs. 400 p.m.

He gets this commission @ 4% on excess sale of Rs. 20,000

$$\text{Therefore, excess sale than Rs. } 20{,}000 = \frac{\text{Commission} \times 100}{\text{Rate of Commission}}$$

$$= \frac{400 \times 100}{4} = \text{Rs. } 10{,}000$$

$\therefore$ Total sales Rs. 20,000 + 10,000 = Rs. 30,000 **Ans.**

Q.5. An Agent gets 6% commission on total sales, and 2% bonus on sales excess than Rs. 10,000. If the total income of the agent is Rs. 1,080, find the cost of the good sold.

Solution: Agent's total income is = Rs. 1,080

(–) Commission on sale of Rs. 10,000 @ i.e., Rs. 600

(1,080 – 600) = Rs. 480

On sale excess than Rs. 10,000, commission 6% and 2% bonus i.e., Total 8% he gets.

Rs. 8 he gets on sale of Rs. 100

$\therefore$ Re. 1 he gets on sale of Rs. $\dfrac{100}{8}$

$\therefore$ Rs. 480 he gets on sale of Rs. $\dfrac{100 \times 480}{8}$

$$= \text{Rs. } 6{,}000$$

Hence, total Sales = Rs. 10,000 + Rs. 6,000 = Rs. 16,000

Q.6. The terms of 8% commission to an agent on total sales were charged as Rs. 3,500 salary and 3% commission on excess sales over Rs. 20,000. If by this change the income of the agent increases by Rs. 225, find the net price of the goods sold.

Solution: Suppose the total sale is of Rs. x

Income to the agent before change in terms

$$= x \times \frac{8}{100} = \text{Rs. } \frac{8x}{100}$$

Income after change in terms = Salary Rs. 3,500 + Commission

$$3{,}500 + (x - 20{,}000) \times \frac{3}{100} = 35{,}000 + \frac{3x - 60000}{100}$$

Due to change in terms there was an increase of Rs. 225 in income.

$$\text{Therefore, } \frac{3500}{1} + \frac{3x - 60000}{100} - \frac{8x}{100} = \frac{225}{1}$$

Or $3{,}50{,}000 + 3x - 60{,}000 - 8x = 22{,}500$

$$3x - 8x = 22{,}500 + 60{,}000 - 3{,}50{,}000$$

Or $-5x = -2{,}67{,}500$

Or $x = \frac{267500}{5}$; Or $x = 53{,}500$ **Ans.**

Q.7. The manager is paid a commission of 9% of the difference between total profit of the firm and the manager's commission. If the total profit is Rs. 20,000, find manager's commission.

Solution: Let manager's commission = Rs. x.

$$\therefore \quad x = \frac{(\text{Total Profit} - x) \times 9}{100}$$

$$\Rightarrow \quad x + \frac{x \times 8}{109} = \frac{\text{Total Profit} \times 9}{100}$$

$$\Rightarrow \quad x\left(1 + \frac{9}{100}\right) = \frac{\text{Total Profit} \times 9}{100}$$

$$\therefore \quad x = \frac{\text{Total Profit} \times 9}{100 + 9} = \frac{20{,}000 \times 9}{109}$$

$$= \text{Rs. } 1{,}651$$

Verification :

$$\frac{(20{,}000 - 1651) \times 9}{100} = \frac{18{,}349 \times 9}{100} = 1651 \text{ Rs. } \textbf{Ans.}$$

Q.8. The manager of a company is paid a fixed monthly salary plus commission at a fixed rate per ton of output. If the manager receives altogether Rs. 33,000 and Rs. 32,000 on two consecutive years against the output of 43.7 ton and 41.2 ton respectively, find the rate of commission and the monthly salarly paid to him.

Solution: The difference of outputs of two consecutive years

$$= 43.7 - 41.2 = 2.5 \text{ ton}$$

The difference of receipts of two consecutive years

$$= 33{,}000 - 32{,}000$$
$$= \text{Rs. } 1{,}000$$

Thus, the commission on the output of 2.5 ton is Rs. 1,000

$$\therefore \text{ Commission per ton} = \frac{\text{Total commission}}{\text{Total output}}$$

$$= \frac{1{,}000}{2.5} = \text{Rs. } 400$$

$\therefore$ Commission for first year $= 43.7 \times 400 =$ Rs. 17,480

$\therefore$ Salary of one year $= 33{,}000 - 17{,}480$

$=$ Rs. 15,520

and Salary of one month $= \frac{15{,}520}{12} =$ Rs. 1,293

Ans. Rate of commission is Rs. 400 per ton and monthly salary is Rs. 1,293.

Q.9. An agent gets a 10% commission on the full sale and 3% Bonus on the amount of the sale exceeding Rs. 10,000. He gets Rs. 1,200 as commission. Find the amount received by him as Bonus.

Solution: Total sales $= \frac{\text{Amount of commission} \times 100}{\text{Rate of commission}}$

$$= \frac{1{,}200 \times 100}{10} = \text{Rs. } 12{,}000$$

The amount of sales exceeding Rs. 10,000

$$= 12{,}000 - 10{,}000$$
$$= \text{Rs. } 2{,}000$$

$\therefore$

$$\text{Bonus} = \frac{\text{Rate of Bonus} \times \text{The amount of sales exceeding Rs. } 10{,}000}{100}$$

$$= \frac{3 \times 2{,}000}{100} = \text{Rs. } 60 \ \textbf{Ans.}$$

Q.10. An agent is entitled to a commission of 4% on the total turnover and a bonus of 2% on the amount of sale exceeding Rs. 20,000. If his total remuneration Rs. 2,000, find the amount of total sales and bonus.

Solution: Total Sales $= \frac{\text{Commission} \times 100}{\text{Rate of commission}}$

$$= \frac{2{,}000 \times 100}{4} = \text{Rs. } 50{,}000$$

Sale exceeding Rs. 20,000 = 50,000 – 20,000

= Rs. 30,000

$$\therefore \text{Bonus} = \frac{\text{Rate of Bonus} \times \text{Sales exceeding Rs. } 20{,}000}{100}$$

$$= \frac{3 \times 30{,}000}{100} = \text{Rs. } 600 \quad \textbf{Ans.}$$

Q.11. A publisher gives to his agent a fixed amount per year plus a fixed percent on the orders received. He paid Rs. 4,000 in 2017 and Rs. 4,600 in 2018. Find the fixed amount given per year and the rate percent of commission. The amount of orders received on 2017 and 2018 was respectively Rs. 36,000 and Rs. 42,000.

Solution: Difference of the orders received in 2017 and 2018

= 42,000 – 36,000

= Rs. 6,000

Difference of the commission in 2017 and 2018

= 4,600 – 4,000 = Rs. 600

Thus the commission on Rs. 6,000 is Rs. 600.

$$\text{Rate of commission} = \frac{\text{Commission} \times 100}{\text{Sales}}$$

$$= \frac{600 \times 100}{6{,}000} = 10 \qquad \ldots (1)$$

Now, the commission of 2017

$$= \frac{\text{Rate} \times \text{Sales}}{100} = \frac{10 \times 36{,}000}{100}$$

= Rs. 3,600

The fixed amount given in 2017 (i.e., the amount given per annum)

= Total Amount – Commission

= 4,000 – 3,600

= Rs. 400 ... (2)

AVERAGE

Q.12. What do you mean by Statistical Averages? What are its qualities?

Or

Mention the various types of Averages and discuss their advantages and disadvantages.

Or

What do you understand by Dan 'Average'? What are the desirable properties for an Average to possess? Also discuss the objects and functions of Dan Average.

Or

What do you understand by Central tendency? Explain.

Or

What is the concept of measure of central tendency? What are its objectives and functions? Explain the properties of a good average.

Or

Describe various methods of measuring central tendency and point out usefulness of each method.

Ans. Central Tendency Or Statistical Averages : Any number representing the group is known as statistical average. In other words, average is a central tendency, because it tells such point, around which the other units have a central tendency to move.

Simpson and Kafea : 'A measure of central tendency is a typical value around which other figures congregate'.

Spurr, Kellog and Smith : 'An average is a measure of central tendency because individual values of the variable usually cluster around it.'

Objectives of Measures of Central Tendency : The main objectives of a measure of Central tendency are as follows :

(i) To determine a single figure which may be used to represent the whole series involving magnitudes of the same variable.

(ii) To facilitate comparison within one group or between groups of data. In other words, the performance of the members of a group can be compared by relating it to the average performance of that group and the achievement of groups can be compared by a comparing their respective averages.

Functions of Statistical Average (or Measure of Central tendency) :

(1) An average presents the salient feature of numerical data in simple and summarised form.

(2) It (an average) provides a common denominator for comparison of groups of data.

(3) It is valuable in setting standards, estimating and planning.

(4) It gives a picture of a complete group.

(5) It traces mathematical relationship between different groups of data.

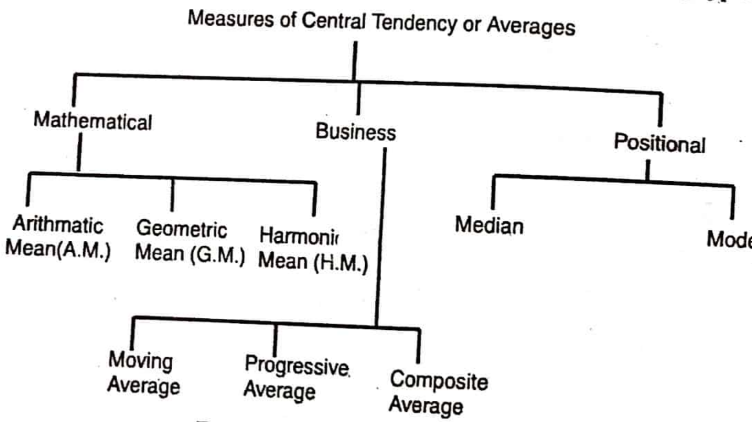
(6) It is a basis for many other techniques of Statistical analysis.

(7) It is helpful in making the decisions or in the process of experimentation and research.

Characteristics of a good Average or Measure of Central tendency : A good average must possess the following characteristics : –

(i) It should be rigidly defined (ii) It should be based on all the items of the series (iii) It should be easily understandable (iv) It should be capable of being calculated easily (v) It should be least affected by the fluctuations of sampling (vi) It should be capable of further algebric treatment.

Kinds or Types of Measures of Central Tendency : Measures of Central tendency which are commonly used are of the following types :



Relative Merits & Demerits

(A) Mean : The arithmetic mean of a series in the figure obtained by dividing the sum of values of all items by their number.

Merits : (1) Simplicity, (2) Certainty, (3) Mathematical interpretation, (4) Based on all values of the series, (5) Systematic arrangements of items, (6) Representative mean, (7) Comparable, (8) Unaffected by the change in samples, (9) Algebrical interpretation.

Demerits : (1) Not known by observation, (2) Calculation not possible when any value is missing, (3) Not possible under graphical method, (4) Influenced by marginal items, (5) Unsuitable by attributes, (6) Mean may be out of the series, (7) Surprising results, (8) Unsuitable for relative measurements, ratios and rates etc.

Use : Mean is used for the studies of economic and social problems. It is popular on account of its simplicity of calculation. It is used for

calculating average production, average expenses, average import-export etc., but it is not useful for studying qualitative attributes.

(B) Median : The median is that value of the variable which divides the group into two equal parts, one part comprising all values greater, and the other all values less than the median.

Merits

1. Can be located by observation,
2. Simple calculations,
3. Calculation possible in unequal internals,
4. Located with the help of graphs,
5. Not affected by marginal items,
6. Certainty,
7. Value found in the series,
8. Useful in qualitative studies also,
9. More useful in the study of skewness.

Demerits

1. Arrangement of series in ascending or descending order is necessary.
2. Not representing the true value of the series.
3. Not suitable for algebrical calculations.
4. Difficulties in even numbers.
5. Total value is not equal when median is multiplied by the number of items.
6. Small change in series may bring remarkable changes in the results.
7. Correct calculation is not possible, if the number of items are less.

Utility : It is useful in qualitative studies, like honesty, clearness, health etc. and other social problems like distribution of income, per capita expenditure etc. But it is unsuitable in business sector.

(C) Mode : 'The value of the variable which occurs most frequently in a distribution is called the mode'. The mode of a distribution is the value at the point around which the items tend to be most heavily concentrated.' It may be regarded as the most typical of a series of values.

Merits

1. Simple and popular,
2. Located by observation only,
3. Possible by graphic method,
4. Least affected by marginal items,

5. Proper representation.
6. Same values in many samples,
7. Popular in business fields,
8. Useful in qualitative studies,
9. Possible if the frequency of any item is missing.

Demerits

1. Need of grouping,
2. Uncertain and not clear,
3. Affected by class intervals,
4. Arrangement of items into ascending or descending order is necessary,
5. Unsuitable for algebrical interpretation,
6. Ignorance of marginal items,
7. Illusory representation,
8. Complex procedure of calculation.

Utility : It is the most effective average to be used in daily routine in business. It is very easy to understand and this is why it is mostly used in business. Business forecasting can be done easily. The popular size of ready-made cloth are determined on the basis of mode only. From scientific point of view, this average is more popular.

(D) Geometric Mean : The geometric mean of a series containing N observations is the N^{th} root of the product of values: The logarithm of the Geometric mean of a set of observation is the arithmetic mean of their logarithms.

Merits

1. Rigidly defined and precised value.
2. Based in all the observations.
3. Biased towards lower values.
4. Not affected by extreme observations.
5. Suitable for further mathematical treatment.
6. Not much affected by the fluctuations of sampling.
7. It gives more weight to smaller items.

Demerits

1. Neither easy nor simple to calculate.
2. Its value may not exist in the series.
3. Imaginary value if any one of the observations is negative.
4. It bring out the property of the ratio of change and not of absolute difference.

5. More weight to smaller items may prove to be a drawback.
6. Calculations are difficult.
7. Lack of popularity.

Utility : It is specially suitable in averaging rates, percentages and rates of increase between two periods. It is used when it is desired to give more weight age to smaller items. It is useful in the construction of index numbers.

(E) Harmonic Mean : Harmonic mean of a series is the reciprocal of the arithmetic average of the reciprocal of the values of its various items.

Merits

1. It is reigidly defined.
2. Capable further algebraic treatment.
3. It is not affected by fluctuations of sampling.
4. Greater importance to small items.
5. Single big figure cannot push its value up.
6. It measures relative changes and useful in averaging certain types of ratios and rates.

Demerits

1. It is not readily understood.
2. Difficult to compute.
3. Not truly representative of the series.
4. Not popular.
5. Calculation is difficult.
6. Not to be followed by an average person.

Utility : It is more useful in comparison to averages, ratios, rates and other types of averages, but it is not widely used.

Which Average is useful? : Each average has got its merits and demerits and it is difficult to say that which average is most suitable in statistical studies. The objects of investigation are kept into mind before selecting any suitable average. Precautions, foresightness, analysis etc. are kept into mind before using any average.

Q.13. Calculate Mean from the following series:

Yields	No. of Plots
Over 0	216
Over 60	210
Over 120	156
Over 180	98
Over 240	57

Over 300	31
Over 360	13
Over 420	7
Over 480	2
Up to 540	216

Solution: First of all convert it into a continuous series as under :

Class interval	Frequency	MV	d(270)	fo(60)	fdx	cf
0 — 60	6	30	–240	–4	–24	6
60 — 120	54	90	–180	–3	–162	60
120 — 180	58	150	–120	–2	–116	118
180 — 240	41	210	–60	–1	–41	159
240 — 300	26	270	0	0	0	185
300 — 360	18	330	60	1	18	203
360 — 420	6	390	120	2	12	209
420 — 480	5	450	180	3	15	214
480 — 540	2	510	240	4	8	216
	216				–343 + 53 –290	

$$a = x + \frac{\Sigma fdx}{n} \times i$$

$$= 270 + \frac{-290}{216} \times 60$$

$$= 270 + \frac{-17400}{216}$$

$$= 270 - 80.55 = 189.45$$ **Ans.**

Q.14. Calculate the Mean from the following data :

Age (In years)	No. of persons
10–25	6
25–40	20
40–55	44
55–70	26
70–85	3
85–100	1

Solution: **Calculation of Mean**

Age (In Years)	Persons f	cf	Midpoint x	$\left(x - \frac{47.5}{15}\right)$ dx	fdx
10–25	6	6	17.5	–2	–12
25–40	20	26	32.5	–1	–20
40–55	44	70	47.5	0	0
55–70	26	96	62.5	+1	+26
70–85	3	99	77.5	+2	+6
85–100	1	100	92.5	+3	+3

Mean $\overline{X} = A + \frac{\Sigma fdx}{N} \times i$

$$= 47.5 + \frac{3}{100} \times 15$$

$$= 47.5 + \frac{45}{100}$$

$$= 47.5 + .45$$

$$= 47.95$$ **Ans.**

Q.15. Calculate arithmetic mean from the following figures:

Marks (Less than)	80	90	60	50	40	30	20	10
No. of Students	100	90	80	60	32	20	13	5

Solution:

Marks (M)	No. of Student (F)	cf	MV	(A) – MV	fdx
0–10	5	5	5	–30	–150
10–20	8	13	15	–20	–160
20–30	7	20	25	–10	–70
30–40	12	32	35(A)	0	0
40–50	28	60	45	+10	280
50–60	20	80	55	+20	400
60–70	10	90	65	+30	300
70–80	10	100	75	+40	400
	100				–380 +1,380 +1,000

$$\bar{X} = A + \frac{\Sigma fdx}{N}$$

$$= 35 + \frac{1000}{100} = 35 + 10$$

$$= 45$$ Ans.

Q.16. Calculate the following :

(a) What pays the highest wages.

(b) What is the combined average wage of both the factories.

Factory	A	B
No. of Workers	250	200
Average Wages	2.00	2.50

Solution:

(a) A — B

$N_1 = 250$ — $N_2 = 200$

$X_1 = 2.00$ — $X_2 = 2.50$

$\Sigma XA = 2 \times 250 = 500$ — $\Sigma XB = 2.50 \times 200 = 500$

Total wages is equal in both the factories.

(b)Combined averages : $X = \frac{X_1 N_1 + X_2 N_2}{N_1 + N_2}$

$$= \frac{(2 \times 250) + (2.50 \times 200)}{250 + 200}$$

$$= \frac{500 + 500}{450} = \frac{1000}{450} = 2.22$$ Ans.

Q.17. Calculate weighted average from the following :

	Monthly Salary	Weight
Principal	**1,800**	**1**
Readers	**1,200**	**10**
Senior Lecturers	**750**	**20**
Lecturers	**600**	**60**
Asstt. Lecturers	**300**	**9**

Solution:

	X	W	WX
Principal	1,800	1	1,800
Reader	1,200	10	12,000
Senior Lecturers	750	20	15,000
Lecturer	600	60	36,000
Asstt. Lecturers	300	9	2700
		100	67,500

$$WA = \frac{\Sigma WX}{\Sigma W} = \frac{67500}{100} = 675$$ Ans.

Q.18. Calculate Geometric mean from the following :

Value	Value
5,470	**687**
92	**8**
0.7	**0.06**
0.004	**0.0003**

Solution:

Value	Logarithm
5,470	3.7355
687	2.8370
92	1.9638
8	0.9031
0.7	1.8451
0.06	2.7782
0.004	3 – 6021
0.0003	4 – 4771
	2.1419

$$\text{G.M.} = \text{Antilog}\left[\frac{\Sigma \text{logs}}{N}\right]$$

$$= \text{A.L.}\left[\frac{2.1419}{8}\right]$$

$$= \text{A.L. } 0.2677$$

$$\text{G.M.} = 1.852$$ Ans.

Q.19. Calculate Harmonic Mean from the following :

15, 250, 15.7, 157, 1.57, 105.7, 10.5, 1.06, 25.7, 0.257

Solution:

Value of Items	Reciprocals
15	0.06667
250	0.00400
15.7	0.06369
157	0.06369

1.57	0.63690
105.7	0.00946
10.5	0.09524
1.06	0.94340
25.7	0.03891
0.257	3.89100
	5.75564

H.M. $= \text{Rec.}\left[\frac{\Sigma \text{Rec.}}{N}\right]$

$= \text{Rec.}\left[\frac{5.75564}{10}\right]$

$= \text{Rec. } 0.5756$

H.M. $= 1.737$ **Ans.**

Q.20. The population of a country increased by 20% in the first decade ; 30% in the second decade, and 45% in the third decade, what is the average rate of increase per decade in the population?

Solution: **Calculation of Geometric Mean**

Population at the beginning of decade	% Increase	Population at the end of decade	Log x
100	20	120	2.0792
100	30	130	2.1139
100	45	145	2.1614
N = 3			Σlogx = 6.3545

$$\text{G.M.} = \text{Antilog}\left(\frac{\Sigma \log x}{N}\right)$$

$$\text{G.M.} = \text{Antilog}\frac{6.3545}{3}$$

$$= \text{Antilog } 2.11817 = 131.2$$

Hence the average rate of increase per decade in the population is (131.2 – 100) = 31.2% **Ans.**

Q.21. A machine is assumed to depreciate 40% in value in the first year, 25% in the second year and 10% per annum for the next three years, each percentage being calculated on the diminishing value. What is the average percentage depreciation for the five years?

Solution:

Year	Depreciated Value	Logs
I	100 – 40 = 60	1.7782
II	100 – 25 = 75	1.8751
III	100 – 10 = 90	1.9542
IV	100 – 10 = 90	1.9542
V	100 – 10 = 90	1.9542
		9.5159

$$\text{G.M.} = \text{A. L.}\left[\frac{\Sigma \text{logs}}{N}\right]$$

$$= \text{A.L.}\left[\frac{9.5159}{5}\right]$$

$$= \text{A.L. } 1.9032 \text{ or } 80$$

∴ Average Depreciation = 100 – 80 = 20% **Ans.**

Q.22. A train starts from rest and travels successive quarters of miles at average speeds of 12, 16, 24 and 48 miles per hour. The average speed over the whole mile is 19.2 m.p.h. and not 25 m.p.h. Explain this statement.

Solution:

$$\text{Average Speed: (H.M.)} = \frac{4}{\frac{1}{12}+\frac{1}{16}+\frac{1}{24}+\frac{1}{48}}$$

$$= \frac{4}{\frac{4+3+2+1}{48}}$$

$$= \frac{4}{\frac{10}{48}} = \frac{4 \times 48}{10}$$

$$= 19.2 \text{ miles per hour}$$ **Ans.**

Q.23. A cyclist covers his first three miles at an average speed of 8 m. p.h. another two miles at 3 m.p.h.and the lost two miles at 2 m.p.h. Find his average speed for the entire journey.

Solution:

$$\text{W.H.M.} = \frac{\Sigma W}{\left(\frac{1}{x_1} \times w_1\right) + \left(\frac{1}{x_2} \times w_2\right) + \left(\frac{1}{x_3} \times w_3\right) + \ldots}$$

$$= \frac{3+2+2}{\left(\frac{1}{8}\times 3\right)+\left(\frac{1}{3}\times 2\right)+\left(\frac{1}{2}\times 2\right)}$$

$$= \frac{7}{\frac{3}{8}+\frac{2}{3}+\frac{2}{2}}$$

$$= \frac{7}{\frac{9+16+24}{24}}$$

$$= \frac{7\times 24}{49} = 3.43 \text{ miles per hour}$$ **Ans.**

Q.24. Calculate Mean from the following :

Wages (Rs.)	No. of Workers
Less than 8	5
Less than 16	12
8 – 24	29
24 and Above	31
32 – 40	8
40 and above	19
48 and Above	5

Solution:

Wages (Rs.)		f	dx	fdx	cf
0 – 8	–	5	–2	– 10	5
8 – 16	(12 – 5)	7	–1	–7	12
16 – 24	(29 – 7)	22	0	0	34
24 – 32	(31 – 27)	4	1	4	38
32 – 40	–	8	2	16	46
40 – 48	(19 –5)	14	3	42	60
48 – 56	–	5	4	20	65

$$\text{Mean} = A + \frac{\Sigma fx}{n}\times i$$

$$= 20 + \frac{65}{65}\times 8 = 28$$ **Ans.**

PROFIT & LOSS

Q.25. What do you understand by Profit & Loss? Write the formulas to find out the Sale Price, Purchase Price, and Profit or Loss.

Ans. Meaning of Profit and Loss : In business, buying and selling of goods is a routine activity. This buying and selling activity often results in some profit or loss. It is very important for every business, whether big or small, to calculate profit or loss. When an article is sold for more than its cost, the article is said to be sold at a profit. Conversely, if the article is sold for less than its cost, the article is said to be sold at a loss.

Cost Price : The price originally paid in buying an article is called its cost price (C.P.). The total price required to prepare (or manufacture) an article is known as its cost. This price includes transport expenses, brokerage etc.

Selling Price : The price that is realised by selling an article is called the selling price (S.P.).

Types of Profit and Loss : There are two types of profit or loss – (i) Gross Profit or Loss, (ii) Net Profit or Loss.

Gross Profit or Gross loss : Gross Profit or Loss is the difference of selling price and the direct expenses.

Net Profit or Loss : It is the difference of Gross Profit and the indirect expenses.

The difference between Selling Price (S.P.) and Cost Price (C.P.) gives Profit or loss :

Profit = S.P. – C. P., i.e., when S.P. > C.P.

Loss = C.P. – S.P., i.e., when C.P. > S.P.

(The symbol > indicates greater than)

Profit and loss are generally calculated or expressed as percentage of the Cost Price, unless it is stated otherwise.

*(**Note :** S.P. of an article of a seller is C.P. of the same article to the purchaser)*

Different Formulas :

(1) To know percentage of profit on cost :

$$\frac{S.P. - C.P.}{C.P.}\times 100 \text{ Or } \frac{Gain}{Cost\ Price}\times 100$$

(2) To know percentage of loss on cost :

$$\frac{C.P. - S.P.}{C.P.}\times 100 \text{ Or } \frac{Loss}{Cost\ Price}\times 100$$

(3) To know percentage of Profit on Selling Price :

$$\frac{S.P.-C.P.}{C.P.} \times 100 \text{ Or } \frac{Gain}{S.P.} \times 100$$

(4) To know percentage of Loss on Selling Price :

$$\frac{C.P.-S.P.}{S.P.} \times 100 \text{ Or } \frac{Loss}{S.P.} \times 100$$

(5) To know Selling Price :

$$S.P. = C.P. + Profit \text{ Or } S.P. = C.P. - Loss$$

(6) To know Cost Price :

$$S.P. - Profit \text{ Or } S.P. - Loss$$

Q.26. By selling 150 mangoes, a fruit seller gain the selling price of 30 apples. Find the gain percent.

Solution: Let Selling Price being Rs. 1

∴ Selling Price of 150 Mangoes = 150 × 1 = 150

∴ Selling Price of 30 Apples = 30 × 1 = 30

Profit being Rs. 30

$$\therefore \quad \text{Cost Price} = \text{Selling Price} - \text{Profit}$$
$$= 150 - 30 = \text{Rs. } 120$$

$$\therefore \quad \text{Gain Percent} = \frac{\text{Profit}}{\text{Cost Price}} \times 100$$
$$= \frac{30}{120} \times 100 = 25\%$$

Ans.

Q.27. A Person purchased a scooter for Rs. 3,000 and sold it for Rs. 3,600. Find his profit and profit percent.

Solution: Cost Price = Rs. 3,000

Selling Price = Rs. 3,600

Profit = 3,600 – 3,000 = 600 Rs.

$$\text{Profit Percent} = \frac{\text{Profit}}{\text{Cost Price}} \times 100$$
$$= \frac{600}{3000} \times 100 = 20\%$$

Ans.

Q.28. Ram purchased a Type writer for Rs. 1,250 and sold it for Rs. 1,100. Calculate his loss or profit in percent.

Solution: Cost Price = Rs. 1,250

Sale Price = Rs. 1,100

Loss : Sale – Cost

= 1, 100 – 1,250 = Rs. 150 Loss

$$\therefore \quad \text{Percentage of loss} = \frac{\text{Loss}}{\text{Cost}} \times 100$$
$$= \frac{150}{1250} \times 100 = 12\%$$

Ans.

Q.29. A Seller buys 11 pens for Rs. 10 and sells 10 pens for Rs. 11. What is his gain percent?

Solution: Selling Price of 10 Pens = Rs. 11

∴ Cost Price of 11 Pens = Rs. 10

$$\therefore \quad \text{Cost of 10 Pens} = \frac{10}{11} \times 10 = \frac{100}{11}$$

$$\therefore \quad \text{Gain} = \text{Selling Price} - \text{Cost Price}$$
$$= \text{Rs. } 11 - \frac{100}{11} = \frac{21}{11}$$

$$\text{Gain Percentage} = \frac{\text{Gain}}{\text{Cost}} \times 100$$
$$= \frac{21/11}{100/11} \times 100 = 21\%$$

Ans.

Q.30. A Trader defrauds by means of a false balance to the extent of 10% in buying and to the same extent in selling goods. What percent does he gain on loss on his outlay by defraud?

Solution: The Trader purchases in Rs. 100 the articles worth Rs. 110.

Again he defrauds 10% in selling the articles, or he sells the article worth Rs. 100 in Rs. 110.

He sells of Rs. 100 in Rs. 110

∴ He will sell the article of Rs. 110 in Rs. $\frac{110}{100} \times 110$

= 121

Gain = 121 – 100 = 21%

Ans.

Q.31. A Dealer marks 30% more than the cost price on his goods and allows a discount of 10%. What is his gain percent?

If he marks 60% more than the cost price and then allows a discount of 40%, what is gain or loss percent?

Solution: Let the Cost Price be Rs. 100

$$\therefore \quad \text{Marked Price} = 100 + 30 = 130$$

$$\therefore \quad \text{Discount at } 10\% = 130 \times \frac{10}{100} = 13$$

$$\therefore \quad \text{Selling Price} = 130 - 13 = 117$$
$$\text{Gain} = 117 - 100 = 17\%$$

When the dealer marks 60% more than the Cost Price, then the marked price

$$= 100 + 60 = 160$$

$$\text{Now Discount at } 40\% = 160 \times \frac{40}{100} = 64$$

$$\text{S.P.} = 160 - 64 = 96$$

$$\text{Loss} = 100 - 96 = 4\%$$

Ans.

Q.32. The selling price of 20 articles is equal to the cost price of 25 articles. Find the gain or loss percent.

Solution: Let the selling price of 20 articles be Rs. 100.

∴ The cost price of 25 articles is Rs. 100 ... (i)

$$\Rightarrow \quad \text{Cost price of an article} = \frac{100}{25} = \text{Rs. } 4$$

and cost price of 20 articles = 4 × 20 = Rs. 80 ... (ii)

Since cost price is less than selling price, hence

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$= 100 - 80 = \text{Rs. } 20$$

$$\text{Profit percent} = \frac{\text{Profit} \times 100}{\text{Cost price}} = \frac{20 \times 100}{80} = 25$$

i.e., Profit = 25% **Ans.**

Q.33. A person bought a cycle for Rs. 360. For what price should he sell it to gain 15%?

Solution: Cost price = Rs. 360

$$\text{Price} = \frac{\text{Cost price} \times \text{Profit percent}}{100}$$

$$= 360 \times \frac{15}{100} = \text{Rs. } 54$$

$$\therefore \quad \text{Selling price} = \text{Cost price} + \text{Profit}$$

$$= 360 + 54 = \text{Rs. } 414$$

Ans.

Q.34. A man purchases 200 kg. rice at the rate of Rs. 10 per kg. He pays Rs. 20 as cartage and Rs. 10 as octroi. He sells the whole rice at 10% gain. Find the selling price per kg. of rice.

Solution: Purchasing price of 200 kg. price = 200 × 10 = Rs. 2,000

Other expenses = 200 + 100 = Rs. 300

∴ Cost price = 2,000 + 300 = Rs. 2,300

Formula : $$\text{Selling Price} = \text{Cost Price}\left(1 + \frac{\text{Profit percent}}{100}\right)$$

$$= 2{,}300\left(1 + \frac{10}{100}\right)$$

$$= 2{,}300\left(1 + \frac{1}{10}\right)$$

$$= 2{,}300 \times \frac{11}{10} = \text{Rs. } 2{,}530$$

Ans.

UNIT-6

Long and Short Answer type Questions :

SIMPLE INTEREST

Q.1. What is the meaning of Simple Interest? Explain the calculation method of simple interest.

Or

What do you mean by Simple Interest? Explain its various elements.

Or

Give the meaning of Simple Interest and discuss how calculation of interest is made?

Or

What do you know by Simple Interest? How would you calculate Simple Interest?

Or

What is Simple Interest? Explain the formulae of calculating Interest. Time, Rate, Principal and Amount. Name the methods of Interest calculations.

Or

What do you mean by Interest? Explain.

Ans. Meaning of Simple Interest: When money is lent, the borrower pays a certain sum of money for the use of money lent. The money lent is called the **Principal** and the sum of money paid for the use of it is called the **Interest.** The sum of the **Principal** and the **Interest** at the end of any time is called the **Amount.**

The **Rate of Interest** is the ratio of the interest in one unit of time to the principal. The unit of time is taken to be one year unless mentioned otherwise. The rate of interest is usually expressed as a percentage. **Simple Interest** (denoted by S.I.) is the interest calculated on the original principal alone for the time during which the money lent is being used.

Elements of Simple Interest : As mentioned above, the various elements of Simple Interest are : (i) Principal, (ii) Time, (iii) Rate of Interest, (iv) Amount, and (v) Interest.

Calculation of Interest : The following are the different problems in respect of the calculation of interest :

(1) Calculation of interest (2) Calculation of period (3) Calculation of rate of interest (4) Calculation of amount (5) Calculation of principal.

Principal (P) : It is the money borrowed.

Rate (r) : It is the rate of calculating interest which is generally per hundred per annum.

Time (t) : It is the period of which the interest is to be calculated.

Amount (A) : It is the total money returned to the money lending agency.

Formula for calculation of simple interest

$$Interest = \frac{Principal \times Rate\ of\ Interest \times Time}{100}$$

$$\text{Or} \quad \frac{P \times r \times t}{100}$$

(i) If the period is given in years the interest

$$= \frac{Principal \times Rate \times Time\ (in\ years)}{100}$$

(ii) If the period is given in months the interest

$$= \frac{Principal \times Rate \times Time\ (in\ months)}{100 \times 12}$$

(iii) If the period is given in days the interest

$$= \frac{Principal \times Rate \times Time\ (in\ days)}{100 \times 365}$$

If we are given any three quantities out of the above four quantities, we can calculate the fourth from the said formula.

Formulae relating to simple interest :

(i) Calculation of Interest $= \dfrac{P \times r \times t}{100}$

(ii) Calculation of Time $= \dfrac{I \times 100}{P \times r}$

(iii) Calculation of Rate $= \dfrac{I \times 100}{P \times t}$

(iv) Calculation of Amount $=$ Principal + Interest

(v) Calculation of Principal :

(a) When amount and the interest is given :

$$Principal = (Amount - Interest)$$

(b) When interest, rate and period of time are given :

$$Principal = \frac{Interest \times 100}{Rate \times Time}$$

(c) When interest is not given but amount, rate and time are given :

Under this situation on the basis of formula for calculating amount as given above the value of principal will be obtained :

$$Amount = P\left(1 + \frac{r \times t}{100}\right)$$

Besides the formula of calculating interest, there are certain other methods by which interest can be calculated, e.g.

(1) Calculation of interest from simple interest tables.

(2) Common multiplier method

(3) Product method

(4) Third, tenth and tenth tool method

(5) With the help of log.

Q.2. Calculate simple interest in the following questions:

(i) Calculate simple interest on Rs. 6000 at the rate of 8 percent per annum for 4 years.

Solution: $$Interest = \frac{P \times r \times t}{100}$$

$$= \frac{6000 \times 8 \times 4}{100} = \text{Rs. } 1920 \quad \textbf{Ans.}$$

(ii) Calculate simple interest on Rs. 6,600 for 3 years and 3 months to the rate of 7.5 percent per annum.

Solution: $$Interest = \frac{P \times r \times t}{100}$$

$$= \frac{6600}{100} \times \frac{15}{2} \times \frac{39\ (months)}{12}$$

$$= 1608.75 \quad \textbf{Ans.}$$

(iii) Calculate simple interest on Rs. 9,000 for 4 years and 4 months at the rate of $6\frac{1}{4}$%.

Solution: $$Interest = \frac{P \times r \times t}{100}$$

$$= \frac{9000 \times 25 \times 52}{1000 \times 4 \times 12}$$

$$= \text{Rs. } 243.75$$

Ans.

(iv) Calculate interest on Rs. 500 at the rate of 4% per annum from 29th August 1998 to 9th March, 1999.

Solution: No. of days from 29th August 1998 to 9th March, 1999 = (2 + 30 + 31 + 30 + 31 + 31 +28 + 9) = 192 days

$$\text{Interest} = \frac{P \times r \times t}{100}$$

$$= \frac{500 \times 4 \times 192}{100 \times 365} = \text{Rs. } 10.52$$

Ans.

Q.3. In the following questions find the rate of Interest :

(i) At what rate percent will Rs. 380 amount to Rs. 893 in 15 years?

Solution: Formula : $r = \frac{100 \times I}{P \times t}$

$$\text{Interest} = \text{Amount} - \text{Principal}$$

$$= \text{Rs. } 893 - \text{Rs. } 380 = \text{Rs. } 513$$

$$\text{Rate} = \text{Rs. } \frac{100 \times 513}{380 \times 15}$$

$$= \text{Rs. } \frac{51300}{5700} = 9\%$$

Ans.

(ii) At what rate percent per annum the interest on Rs. 956 amount to Rs. 119.50 in $2\frac{1}{2}$ years?

Solution: Formula : $r = \frac{100 \times I}{P \times t}$

Hence, I = Rs. 119.50; P = Rs. 956; $t = 2\frac{1}{2}$ years.

$$r = \frac{100 \times 119.50 \times 2}{956 \times 5}$$

$$= \frac{11950 \times 2}{956 \times 5} = \frac{23900}{4780} = 5\%$$

Ans.

(iii) At what rate percent will a sum of money will double of itself in 20 years?

Solution: Let the Principal be Rs. 100. Then,

Simple Interest = Rs. 200 – Rs. 100 = Rs. 100

Formula : $\text{Rate} = \frac{\text{S.I.} \times 100}{P \times t}$

$$= \text{Rs. } \frac{100 \times 100}{100 \times 20} = 5$$

Hence, required rate of Interest = 5%

Ans.

Q.4. In the following questions calculate period of Interest :

(i) In what time the interest at the rate of 6 percent per annum on Rs. 750 will be Rs. 135?

Solution: Formula for calculating time :

$$t = \frac{I \times 100}{P \times r}$$

$$= \frac{135 \times 100}{750 \times 6} = 3 \text{ years}$$

Ans.

(ii) In what time will a sum of money be double of itself at the rate of 10 percent?

Solution: Let the principal be Rs. 100

∴ The Amount = Rs. 200

Interest = Rs. 200 – Rs. 100 = Rs. 100

Rate = 10 Percent per annum

We have formula $t = \frac{100 \times I}{P \times r} = \frac{100 \times 100}{100 \times 10} = 10$

∴ The required time = 10 years

Ans.

(iii) In what time will Rs. 800 yield an interest of Rs. 72 at the rate of 3 percent per annum?

Solution: Formula $t = \frac{100 \times I}{P \times r}$

Here, P = Rs. 800, I = Rs. 72, r = 3 percent

$$t = \frac{100 \times 72}{800 \times 3} = 3 \text{ years}$$

Ans.

Q.5. Calculation of Amount :

What will be the amount of Rs. 1000 at the rate of 8 percent per annum interest after 3 years?

Solution: Formula : Amount = Principal + Interest

$$\text{Amount} = \text{Principal} + \frac{(\text{Principal} \times r \times t)}{100}$$

Here, Principal = Rs. 1,000, r = 8 percent, t = 3 years.

$\therefore$ Amount $= 1,000 + \frac{(1000 + 8 \times 3)}{100}$

Rs. 1,000 + 240 = Rs. 1,240

Ans.

Q.6. In the following Questions calculate Principal :

(i) If the Interest at 3 percent per annum for 3 years is Rs. 27, find the principal.

Solution: Formula : $P = \frac{i \times 100}{r \times t}$

Here, i = Rs. 27, r = 3 percent and t = 3 years

$\therefore$ $P = \frac{27 \times 100}{3 \times 3} = \text{Rs. } 300$ **Ans.**

(ii) What principal will yield Rs. $23\frac{1}{3}$ at the rate of $2\frac{2}{3}$ per annum in $6\frac{2}{3}$ years?

Solution : Formula : $P = \frac{i \times 100}{r \times t}$

Here, $i = \text{Rs. } \frac{70}{3}, r = \frac{8}{3}$ percent, $t = \frac{20}{3}$ years

$\therefore$ $P = \frac{70}{3} \times \frac{100}{1} \times \frac{3}{8} \times \frac{3}{20} = \text{Rs. } 131.25$ **Ans.**

Q.7. A sum of money amount to Rs. 2800 in 2 years and Rs. 3250 in five years. Find the principal and rate percent.

Solution: Since Amount = Principal + Interest

$\therefore$ Principal + 5 years Interest = Rs. 3,250 ....(1)

and Principal + 2 years Interest = Rs. 2,800 ...(2)

Subtracting (2) from (1) we get –

3 years interest = Rs. 3250 – Rs. 2800 = Rs. 450

Here 5 years interest $= \frac{450 \times 5}{3} = \text{Rs. } 750$

$\therefore$ Principal = Rs. 3250 – Rs. 750 = Rs. 2500

$\text{Rate} = \frac{100 \times I}{P \times t} = \frac{100 \times 750}{2500 \times 5} = 6\%$

Principal = Rs. 2,500, Rate 6% **Ans.**

Q.8. A man deposited Rs. 5000 on 20th April in a bank paying interest @ 2% p.a. He withdrew Rs. 3,000 on 15th May and deposited Rs. 5,000 on 15th May and deposited Rs. 4,000 on 6th June. How much interest was due to him on 30th June following.

Solution:

(i) From 21st April to 14th May = (10 + 14 days) = 24 days.

For investment of Rs. 5,000 interest for 24 days @ 2% p. a.

$= \frac{p \times r \times t}{100}$

$= 5,000 \times \frac{24}{365} \times \frac{2}{100} = \text{Rs. } \frac{480}{73}$

(ii) From 15th May to 5th June = (17 + 5) = 22 days.

For investment of Rs. (5,000 – 3,000) = Rs. 2,000 for 22 days @ 2% p.a.

$= \text{Rs. } 2,000 \times \frac{22}{365} \times \frac{2}{100} = \text{Rs. } \frac{176}{73}$

(iii) From 6th June to 30th June = 25 days

for the investment of Rs. (2,000 + 5, 000) = Rs. 7,000)

For 25 days @2% p.a. $= \text{Rs. } 7,000 \times \frac{25}{365} \times \frac{2}{100} = \text{Rs. } \frac{700}{73}$

Total Interest $= \text{Rs. } \frac{480}{73} + \frac{176}{73} + \frac{700}{73} = \text{Rs. } \frac{1356}{73}$

= Rs. 18.57 (Approx.) **Ans.**

Q.9. By mistake a bank clerk calculated the interest on a certain principal for 5 months at $6\frac{1}{2}$ % per annum of interest on certain principal for 6 months at $5^{1/2}$% per annum. He thus made an error of (i) 40 paise and (ii) Rs. 25. What was the Principal?

Solution: Let the principal be Rs. 100

In right situation : Principal Rs. 100; Rate percent $= 6\frac{1}{2}$ or $\frac{13}{2}$;

Time = 5 months i.e., $\frac{5}{12}$ year

$\text{Simple Interest} = \frac{P \times r \times t}{100}$

$= 100 \times \frac{13}{2 \times 100} \times \frac{5}{12} = \text{Rs. } \frac{65}{24}$

In the situation of mistake :

Principal = Rs. 100; Rate Percent $= 5\frac{1}{2} = \frac{11}{2}$

Time = 6 months or $\frac{6}{12}$ year

Simple Interest = $\frac{P \times r \times t}{100}$

$= 100 \times \frac{11}{2 \times 100} \times \frac{6}{12} = \text{Rs. } \frac{33}{12}$

Difference due to mistake = Rs. $\frac{33}{12} - \frac{65}{24}$

$= \frac{66-65}{24} = \text{Rs. } \frac{1}{24}$

When error is of Rs. $\frac{1}{24}$, then Principal = Rs. 100

When error is of Rs. 1, then Principal = Rs. $\frac{100 \times 24}{1}$

When error is of Rs. $\frac{40}{100}$, then Principal = Rs. $\frac{100 \times 24 \times 40}{100}$

= Rs. 960

When the error is Rs. 25 then Principal = $100 \times \frac{24}{1} \times 25$

= Rs. 60,000 **Ans.**

Q.10. Ram lends to Laxman Rs. 500. He charges interest for one part at the rate of 8% per annum and for the remaining part at the rate of 12% per annum for the same period. If he gets equal interest from the above parts of the sum. Find that sum of money which he lent @ 8% per annum.

Solution: Suppose the money lent at 8% p.a. is x and the period is one year.

∴ The money lent @ 12% p.a. = Rs. 500 – x

(i) ∴ $I = \frac{P \times r \times t}{100} = \frac{x \times 8 \times 1}{100} = \frac{8x}{100}$

(ii) $I = \frac{P \times r \times t}{100} = \frac{(500-x) \times 12 \times 1}{100}$

$= \frac{6000 - 12x}{100}$

∴ $\frac{8x}{100} = \frac{6000-12x}{100}$ or $8x + 12x = \text{Rs. } 6{,}000$

or $20x = \text{Rs. } 6{,}000$, Or $x = \frac{6000}{20} = \text{Rs. } 300$

The required sum of money is Rs. 300 **Ans.**

Q.11. Rajkumar borrowed Rs. 2,000 from two different money lenders, one of them charges interest at the rate of 8 percent and the other charges at the rate of 6% per annum. At the end of the year he paid Rs. 136 by way of interest. What sums were borrowed respectively from these money lenders?

Solution: Suppose the sum borrowed at 8 percent was Rs. x. As such the sum borrowed at 6 percent will be Rs. 2,000 – x.

(i) Interest = $\frac{P \times r \times t}{100} = \frac{x \times 8 \times 1}{100} = \frac{8x}{100}$

(ii) $I = \frac{P \times r \times t}{100} = \frac{(200-x) \times 6 \times 1}{100} = \frac{12000-6x}{100}$

∴ $\frac{8x}{100} + \frac{1200-6x}{100} = \text{Rs. } 136$

or $8x + 12{,}000 - 6x = 13{,}600$ or $2x$ or $x = 800$

The sum borrowed from the other lender = Rs. 2,000–800= Rs.1,200

From one lender Rs. 800, From the second lender Rs. 1,200 **Ans.**

Q.12. A trader borrows Rs. 1,00,000 from a money lender and a bank. If the rate of interest are 18% and 16% per annum respectively and the trader had paid Rs. 16,600 as interest for one year, find the sum borrowed from the money lender and bank respectively.

Solution: Let the sum borrowed from the moneylender be x. Hence, the sum borrowed from the bank = (Rs. 1,00,000–x)

S.I. on Rs. x for 1 year = $\frac{x \times 18 \times 1}{100} = \text{Rs. } \frac{18x}{100}$

S.I. on Rs. (1,00,000 – x) for 1 year

$= \text{Rs. } \frac{(1{,}00{,}000 - x) \times 16 \times 1}{100}$

According to the condition given,

Total S.I. = $\frac{18x}{100} + \frac{16\,(1{,}00{,}000 - x)}{100} = \text{Rs. } 16{,}600$

$18x - 16x + 16{,}00{,}000 = \text{Rs. } 16{,}60{,}000$

$2x = 16{,}60{,}000 - 16{,}00{,}000$

$2x = \text{Rs. } 60{,}000$

$x = \text{Rs. } \frac{60000}{2} = \text{Rs. } 30{,}000$

Thus, the sum borrowed from moneylender is Rs. 30,000 **Ans.**

and the sum borrowed from bank (Rs. 1,00,000 – Rs. 30,000)

$= \text{Rs. } 70{,}000$ **Ans.**

Q.13. A gave a Loan of Rs. 8,000 @4% per annum to B and Rs. 10,000 @5% per annum to C. The sum paid to C is for 2 years more than to B. If he gets from both B and C Rs. 3460 by way of simple interest. Find the time for which the loans were provided.

Solution: Suppose the sum was lent to B for x years. As such the loan was provided to c for x + 2 years.

(i) Interest $= \dfrac{P \times r \times t}{100} = \dfrac{8000 \times 4 \times x}{100} = \text{Rs. } 320x$

(ii) Interest $= \dfrac{P \times r \times t}{100} = \dfrac{10000 \times 5 \times (x+2)}{100} = \text{Rs. } 500x + 1{,}000$

Total Interest $= 320x + 500x + 1{,}000 = 3{,}460$

or $820x = 2{,}460$; or $x = \dfrac{2460}{820} = 3$ years

Hence the sum was lent to B for 3 years and to C (3 + 2) = 5 years **Ans.**

Q.14. A father deposits a total sum of Rs. 8,000 at 5% in the names of his two sons of 15 years and 12 years respectively in such a way that at the age of 18 years both of them get equal amount. What are the sum's in the names of each son?

Solution: Let the father deposit Rs. x in the name of his elder son. Then the sum deposited in the name of his younger son is Rs. (8,000 – x)

Interest of Rs. x at 5% for (18 – 15) years i.e., for 3 years

$$= \frac{x \times 5 \times 3}{100} = \text{Rs. } \frac{3x}{20}$$

∴ Amount in the name of the eldest son

$$= \text{Rs. } x + \frac{3x}{20} = \text{Rs. } \frac{23x}{20}$$

Again, the interest of Rs. (8,000 – x) at 5% for (18–12 years) i.e., for 6

$$\text{years} = \frac{\text{Rs. } (8{,}000 - x) \times 5 \times 6}{100}$$

$$= \frac{\text{Rs. } 3(8{,}000 - x)}{10}$$

∴ Amount in the name of younger son

$$= \text{Rs. } (8{,}000 - x) + \frac{3(8{,}000 - x)}{10}$$

$$= \text{Rs. } (8{,}000 - x)\left(1 + \frac{3}{10}\right)$$

$$= \text{Rs. } \frac{13(8{,}000 - x)}{10}$$

According to the question $\dfrac{23x}{20} = \dfrac{13(8{,}000 - x)}{10}$

$$\Rightarrow \quad 23x = 26(8{,}000 - x)$$

$$\Rightarrow \quad 23x + 26x = 26 \times 8{,}000$$

$$\Rightarrow \quad 49x = 2{,}08{,}000$$

$$\Rightarrow \quad x = \frac{208000}{49} = \text{Rs. } 4{,}244.90$$

∴ The sum in the name of eldest son is Rs. 4,244.90

∴ The sum in the name of younger son = Rs. 8,000 – Rs. 4,244.90

= Rs. 3,755.10 **Ans.**

COMPOUND INTEREST

Q.15. What do you understand by compound interest? Differentiate between simple interest and compound interest. Write the formula of calculating compound interest and also state the names of the methods of calculating compound interest.

Ans. Meaning of Compound Interest : After the specified period the interest is not paid but immediately invested to earn further interest, and thus it is added to the original principal to form a new principal which may be re-loaned or re-invested. This can be repeated for several time periods. The difference between the original principal and the amount at the end of the last period is known as the compound interest.

Interest is generally paid at the end of a year, half year or quarter year. We say that interest is compounded annually, half yearly or quarterly.

Difference Between Simple Interest and Compound Interest :

(1) Simple interest is always calculated on the original sum lent whereas compound interest is calculated on different principals according to the time period intervals.

(2) To calculate simple interest the principal is always be the same but in case of compound interest it changes with every time period.

(3) The amount of compound interest is always more than its corresponding simple interest.

(4) The simple interest money for any part of the whole time period is always in proportion but the compound interest money is different in different time periods.

(5) For the first time period both simple interest and compound interest are equal.

Methods of Calculating Compound interest : The following methods are used for calculating compound interest :

(1) By Simple interest method (2) By use of tables (3) By formula (4) By the use of logarithms.

Formula of Calculating Compound Interest :

$$A = P + \left(1 + \frac{R}{100}\right)^t$$

Where, p = Principal ; R = Rate ; t = time ; and A = Amount

Q.16. A Man took a sum of Rs. 8000 on loan at 10% simple interest and gave the whole sum to another person at 12% compound interest, find after two years what gain will he get?

Solution: Simple interest on Rs. 8,000 at 10% per annum for 2 years

$$= \frac{8000 \times 10 \times 2}{100} = \text{Rs. } 1,600$$

Compound interest on Rs. 8,000 at 12% per annum for 2 years

$$= 8,000\left[\left(1 + \frac{12}{100}\right)^2 - 1\right]$$

$$= 8,000 \times \left[\frac{28}{25} \times \frac{28}{25} - 1\right]$$

$$= 8,000 \times \frac{159}{625} = \text{Rs. } 2,035.20$$

Gain = Rs. 2,035.20 – 1600 = Rs. 435.20 **Ans.**

Q.17. The difference between compound interest and simple interest on a certain sum of money for 2 years at 10% per annum is Rs. 42. Find the sum.

Solution: Let the sum be Rs. 100

Simple interest on Rs. 100 for 2 years at 10% $= \frac{100 \times 2 \times 10}{100}$

= Rs. 20

and compound $= 100\left(1 + \frac{10}{100}\right)^2 - 100$

= Rs. 121 – 100 = Rs. 21

Difference between C.I. and S.I. = Rs. 21 – 20 = Re. 1.00

If difference is Re. 1 then principal = Rs. 100

$\therefore$ If difference is Rs. 42 then principal = Rs. $\frac{100 \times 42}{1}$

= Rs. 4,200 **Ans.**

Q.18. Find compound interest on Rs. 8,000. Time (n) = 3 years and interest rate (r) = 5%.

Solution:
$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$= 8,000\left(1 + \frac{5}{100}\right)^3$$

$$= 8,000\left(\frac{105}{100}\right)^3$$

$$= 8,000\left(\frac{21}{20} \times \frac{21}{20} \times \frac{21}{20}\right) = \text{Rs. } 9,261$$

Compound Interest = Amount – Principal

= Rs. 9,261 – Rs. 8,000 = Rs. 1,261 **Ans.**

Q.19. What will be the compound interest on Rs. 1,000 for 20 years at 5% per annum? How much would be the simple interest on the same principal for the same period?

Solution: 1st Part :

Here P = Rs. 1,000, t = 20 years, $i = \frac{5}{100} = .05$

C.I. = A – P = ?

$A = P(1 + i)^t = 1,000(1 + .05)^{20} = 1,000(1.05)^{20}$

$\therefore$ log A = log 1,000 + 20 log 1.05 = 3 + 20 × 0.0212

= 3 + .424 = 3.424

A = Antilog (3.424) = Rs. 2,655

$\therefore$ Compound interest = A – P = Rs. 2,655 – Rs. 1,000 = Rs. 1,655

2nd Part :

$$\text{Simple Interest} = \frac{P \times r \times t}{100}$$

$$= \frac{1000 \times 5 \times 20}{100}$$

= Rs. 1,000 **Ans.**

Q.20. Amar borrowed Rs. 6,000 from a money–lender but he could not pay any amount in a period of 4 years. Accordingly the money lender demanded now Rs. 7,500 from him. What rate

percent per annum compound interest did the latter required for lending his money?

Solution:

Here Amount = Rs. 7,500 ; Principal = Rs. 6,000

Number of years =4 ; Rate = ?

$A = P(1+i)^t$ or Rs. 7,500 = Rs. 6,000 $(1+i)^4$

Or $\text{Log } 7{,}500 = \text{Log } 6{,}000 + 4 \log(1+i)$

Or $3.8751 = 3.7782 + 4\log(1+i)$

Or $4\log(1+i) = 3.8751 - 3.7782 = 0.0969$

Or $\text{Log}(1+i) = 0.02422$

Or $1 + i = \text{antilog}\ .02422 - 1.057$

Or $i = 1.057 - 1 = 0.57$

$\therefore$ $r = .057 \times 100 = 5.7\%$

$\therefore$ Hence the required rate is 5.7% **Ans.**

Q.21. In what time would a sum of money triple itself at 8% compound interest?

[log 1.08 = .03342, log 3 = .47712]

Solution: We have $A = P(1+i)^n$

Here if P = Rs. 100, then A = Rs. 300, $i = \frac{8}{100} = .08$

$\therefore$ $300 = 100(1+.08)^n = 100(1.08)^n$;

$\therefore$ $\log 300 = \log 100 + n \log 1.08$

or $2.47712 = 2 + n \times 0.03342$

or $n \times 0.03342 = 2.47712 - 2 = 0.47712$

or $n = \frac{0.47712}{0.03342} = \frac{47712}{3342} = 14.28$

Hence, the required time is 14.28 years. **Ans.**

Q.22. In what time will a sum of money double itself at 4% p.a. compound interest payable half yearly?

Solution: Let P be the principal, then A = 2P, $i = \frac{4}{100} = .04$, n = ?

We have in this case, $A = P\left(1+\frac{i}{2}\right)^{2n} = (1.02)^{2n}$

$\therefore$ $2P = P\left(1+\frac{.04}{2}\right)^{2n}$, or $2 = (1+.02)^{2n} = (1.02)^{2n}$,

$\therefore$ $\log 2 = 2n \log 1.02$

or $2n = \frac{\log_2}{\log 1.02} = \frac{.3010}{.0086} = \frac{3010}{86}$

or $n = \frac{3010}{86 \times 2} = 17.5$ years **Ans.**

Q.23. Rs. 10,000 were given as loan on compound interest. The rate of interest is 14% for the first year, 15% for the second year, and 16% for the third year. Find the amount and compound interest after 3 years.

Solution: Formula : $A = P\left(1+\frac{R_1}{100}\right)^1\left(1+\frac{R_2}{100}\right)^1\left(1+\frac{R_3}{100}\right)^1$

$\Rightarrow A = \text{Rs.}10{,}000\left(1+\frac{14}{100}\right)^1\left(1+\frac{15}{100}\right)^1\left(1+\frac{16}{100}\right)^1$

$\Rightarrow$ $A =$ Rs. 10,000 (1.14) (1.15) (1.16)

$\therefore$ $\log A = \log 10{,}000 + \log 1.14 + \log 1.15 + \log 1.16$

$\log A = 4.000 + 0.0569 + 0.0607 + 0.0645$

$= 4.1821$

$A = \text{Antilog } 4.1821$

A = Rs. 15,210

Interest = Rs. 15,210 – Rs. 10,000 = Rs. 5210

Hence, the amount = Rs. 15,210 and the interest = Rs. 5,210 **Ans.**

Q.24. Find Compound interest on Rs. 1,500 for 2 years at 12% per annum, when interest is compounded :

(1) Yearly (2) Half yearly (3) Quarterly, and (4) Monthly.

Solution: (1) When interest is compounded yearly :

P =Rs. 1,500 ; r = 12%; n = 2 years ; A = ? ; I = ?

$A = P(1+i)^n$

$\log A = \log P + n\log(1+i)$

$= \log 1{,}500 + 2\log(1+.12)$

$= 3.1761 \times 2 \times .0492$

$= 3.2745$

A = Rs. 1,881 ; A – P = I

Rs. 1,881 – Rs. 1500 = Rs. 381 **Ans.**

(ii) When interest is compounded half yearly :

P = 1,500 ; n = 2 years (4 times in 2 years) ;

$r = \frac{12}{2} = 6\%$ A = ?, I =?

$\text{Log } A = \log P + n\log(1+I)$

$= \log 1{,}500 + 4 \log (1 + .06)$
$= 3.1761 + 4 \times .0255$
$= 31.761 + .1012$

A – P i.e., Rs. 1,893 – Rs. 1,500 = Rs. 393

A = Rs. 1893 ; interest is compounded quarterly **Ans.**

(iii) P = Rs. 1,500 ; n = 2 years (8 times in 2 years)

$$R = \frac{12}{4} = 3\%, A = ? ; I = ?$$

$$A = P(1 + i)^n$$

$$\log A = \log P + n \log (1 + i)$$
$$= \log 1{,}500 + 8 \log (1 + .03)$$
$$= 3.1{,}761 + 8 \times .0128$$
$$= 3.1{,}761 + .1028$$
$$A = AL\ 3.2785$$
$$A = 1{,}899$$

A – P = I i.e., Rs. 1,899 – 1,500 = Rs. 399

A = Rs. 1,899 and Interest = Rs. 399 **Ans.**

(iv) When interest is compounded monthly :

$$P = Rs.\ 1{,}500 ; n = 2 \times 12 = 24; r = \frac{12}{12} = 1$$

$$A = ? ; I = ?$$
$$A = P(1 + i)^n$$
$$\log A = \log P + n \log (I + i)^n$$
$$\log 1{,}500 + 24 \log \left(1 + \frac{12}{12} \times \frac{1}{100}\right)$$
$$= 3.1761 + 24 \log \frac{1212}{1200}$$
$$= 3.1761 + 24 \times \log 1.01$$
$$= 3.1761 + 24 \times .0043$$
$$= 3.1761 + .1032$$
$$A = AL\ 3.2793$$
$$A = Rs.\ 1{,}902$$

A – P = I. i.e., Rs. 1,902 – Rs. 1,500 = Rs. 402

A = Rs. 1,902 ; Interest = Rs. 402 **Ans.**

□□□

Objective Type Questions

(1) Vedic Mathematics was rediscovered and popularized by:

(a) Aryabhata (b) Swami Bharati Krishna Tirtha
(c) Ramanujan (d) Bhaskara II

Ans. (b) Swami Bharati Krishna Tirtha

(2) Which Vedic Sutra is used for multiplication of numbers close to a base like 10, 100, or 1000?

(a) EkadhikenaPurvena
(b) NikhilamNavatashcaramamDashatah
(c) UrdhvaTiryagbhyam (d) ParavartyaYojayet

Ans. (b) NikhilamNavatashcaramamDashatah

(3) The Sutra "UrdhvaTiryagbhyam" refers to:

(a) Crosswise and vertical multiplication
(b) Division by parts
(c) Addition shortcut (d) Verification of sums

Ans. (a) Crosswise and vertical multiplication

(4) The Sutra "EkadhikenaPurvena" is mainly used to find:

(a) Squares of numbers ending with 5
(b) Cubes of numbers ending with 5
(c) Division shortcuts (d) HCF and LCM

Ans. (a) Squares of numbers ending with 5

(5) The digit sum method is commonly used for:

(a) Approximations (b) Verification of calculations
(c) Finding LCM (d) Prime number testing

Ans. (b) Verification of calculations

(6) The square of 75 can be quickly calculated using Vedic method as:

(a) 5625 (b) 5525
(c) 5725 (d) 5620

Ans. (a) 5625

(7) Using Ekadhikena Purvena, the square of 125 is:

(a) 15625 (b) 15225
(c) 15825 (d) 16225

Ans. (a) 15625

(8) Which Sutra helps in solving division problems quickly?

(a) ParavartyaYojayet (b) EkadhikenaPurvena
(c) UrdhvaTiryagbhyam (d) Anurupyena

Ans. (a) Paravartya Yojayet

(9) In digit sum verification, the digit sum of 678 is:

(a) 21 (b) 12
(c) 6 (d) 3

Ans. (c) 6

(10) By Nikhilam method, 97 × 96 =
(a) 9212 (b) 9312
(c) 9216 (d) 9316

Ans. (c) 9216

(11) "All from 9 and the last from 10" is the English translation of which Sutra?
(a) Ekanyunena Purvena (b) UrdhvaTiryagbhyam
(c) Nikhilam Navatashcaramam Dashatah
(d) ParavartyaYojayet

Ans. (c) Nikhilam Navatashcaramam Dashatah

(12) Which of these is NOT a part of Vedic Mathematics?
(a) UrdhvaTiryagbhyam (b) EkadhikenaPurvena
(c) Nikhilam Sutra (d) Simpson's Rule

Ans. (d) Simpson's Rule

(13) Using digit sum, verify if 248 × 3 = 744 is correct.
(a) Correct (b) Incorrect
(c) Cannot be verified (d) Correct only for even numbers

Ans. (a) Correct

(14) The crosswise method (UrdhvaTiryagbhyam) is especially useful for:
(a) Multiplying large numbers (b) Dividing decimals
(c) Finding averages (d) Finding HCF

Ans (a) Multiplying large numbers

(15) The square root of 2025 using Vedic methods is:
(a) 55 (b) 45
(c) 65 (d) 75

Ans. (b) 45

(16) Which Vedic method can be used to simplify long division by 9?
(a) Ekadhikena Purvena (b) Nikhilam Sutra
(c) ParavartyaYojayet (d) UrdhvaTiryagbhyam

Ans. (c) ParavartyaYojayet

(17) If a number is divisible by 9, its digit sum must be:
(a) 3 (b) 6
(c) 9 (d) 0

Ans. (c) 9

(18) The square of 95 using Vedic method is:
(a) 9025 (b) 9020
(c) 8525 (d) 9625

Ans. (a) 9025

(19) The digit sum of 12,345 is:
(a) 6 (b) 7
(c) 9 (d) 8

Ans. (b) 6

(20) Which is a key advantage of Vedic Mathematics in modern context?
(a) Increases use of calculators (b) Reduces reliance on algebra
(c) Speeds up calculations and enhances mental math
(d) Eliminates probability concepts

Ans. (c) Speeds up calculations and enhances mental math

(21) The cube of 25 using Vedic approximation methods is closest to:
(a) 15625 (b) 15650
(c) 15675 (d) 15700

Ans. (a) 15625

(22) In Vedic multiplication near base, 101 × 109 =
(a) 11009 (b) 11109
(c) 11101 (d) 11001

Ans. (b) 11109

(23) Which sutra is helpful in factorization and solving quadratic equations?
(a) Vilokanam (b) ParavartyaYojayet
(c) ShunyamSamyasamuccaye (d) UrdhvaTiryagbhyam

Ans. (c) ShunyamSamyasamuccaye

(24) The square of 35 using Vedic method is:
(a) 1225 (b) 1155
(c) 1355 (d) 1055

Ans. (a) 1225

(25) The digit sum of the product 123 × 9 = 1107 verifies as:
(a) Correct (b) Incorrect
(c) Correct only for odd numbers (d) Cannot be checked

Ans. (a) Correct

(26) In elections one candidate gets 42% votes but he defeats by 64 votes. Find the total number of votes polled :
(a) 200 (b) 300
(c) 400 (d) 500

Ans. (c) 400

(27) Find the number whose 20% is $\frac{1}{8}$:

(a) 0.625 (b) 1.50
(c) 6.25 (d) 62.5

Ans. (a) 0.625

(28) $12\frac{1}{2}$% of Rs. 500 will be :

(a) 60 (b) 52
(c) 62 (d) 62.50

Ans. (d) 62.50

(29) The height of six buildings is 13m, 14m, 15m, 16m, 17m and 27m respectively, find the average height of the buildings :

(a) 15 metres (b) 17 metres
(c) 18 metres (d) 20 metres

Ans. (b) 17 metre

(30) In a school, there are 7 boys of 12 years, 8 boys of 15 years and 18 boys of 14 years. Find the average age of the boys :

(a) 13.61 years (b) 13.81 years
(c) 14.81 years (d) 14.61 years

Ans. (b) 13.81 years

(31) The average of three numbers is 14. If the first two numbers has an average 8. Find the third number :

(a) 34 (b) 56
(c) 22 (d) 26

Ans. (d) 26

(32) The average of 11 numbers is 63. If average of first six numbers is 60 and last 6 numbers is 65, find the value of sixth number :

(a) 64 (b) 61
(c) 59 (d) 57

Ans. (d) 57

(33) A,B,C, have an average age 15 years. If the average age of A and B is 18 years. The age of C will be :

(a) 9 years (b) 12 years
(c) 18 years (d) 33 years

Ans. (a) 9 years

(34) The average of 9 numbers is 301, if the average of first five number is 163 and last 5 numbers is 430. Find the value of 5th number.

(a) 78 (b) 215
(c) 256 (d) 312

Ans. (c) 256

(35) 5% of discount is provided on an amount of Rs. 5,048/– the actual amount will be :

(a) Rs. 4795.60 (b) Rs. 252.40
(c) Rs. 225.40 (d) Rs. 4975.60

Ans. (a) Rs. 4795.60

(36) Find the single discount equivalent to the discount series 20%, 10% and 5%.

(a) 20% (b) 35%
(c) 15% (d) 31.6%

Ans. (d) 31.6%

(37) An agent gets a commission of 6.25%. If the sale is Rs. 14,000/– what will be the commission?

(a) 875 Rs. (b) 785 Rs.
(c) 575 Rs. (d) 872 Rs.

Ans. (a) 875 Rs.

(38) If an agent gets 15% of 4500/–. What will be the total sale :

(a) 3,000 Rs. (b) 10,000 Rs.
(c) 30,000 Rs. (d) 50,000 Rs.

Ans. (c) 30,000 Rs.

(39) An agent sells goods of Rs. 24,000 and gets a commission of Rs. 18000. Find the rate of his commission?

(a) 7.5 (b) 6.5
(c) 5.7 (d) 5.6

Ans. (a) 7.5

(40) A person buys a Scooter in Rs. 8,000 and sell at Rs. 9,600 what is the profit percentage?

(a) 16 (b) 20
(c) 25 (d) 30

Ans. (b) 20

(41) A trader defrauds by means of a false balance to extent of 20% in buying and in selling good. What percent does he gain on loss on his outlay by defraud?

(a) 40% (b) 44%
(c) 32% (d) None of above

Ans. (b) 44%

(42) The characteristic of log 0.00704 will be :

(a) $\bar{1}$ (b) $\bar{3}$
(c) $\bar{4}$ (d) $\bar{5}$

Ans. (b) $\bar{3}$

(43) **Antilog of 1.9820 will be :**

(a) 95.94 (b) 9.954

(c) 959.4 (d) .9594

Ans. (a) 95.94

(44) **The simple interest of Rs. 450 at rate of 4% for $2\frac{1}{2}$ years will be :**

(a) Rs. 54 (b) Rs. 22.50

(c) Rs. 90 (d) Rs. 45

Ans. (d) Rs. 45

(45) **Find the principal amount if the interest at 5 percent for 4 years is Rs. 90 :**

(a) Rs. 375 (b) Rs. 450

(c) Rs. 1,500 (d) Rs. 1,800

Ans. (b) Rs. 450

(46) **At what rate percent will a sum of money will triple of itself in 25 years?**

(a) 4% (b) 6%

(c) 8% (d) 10%

Ans. (c) 8%

(47) **The compound interest of sum Rs. 400 for 5% per annum for 2 years will be :**

(a) Rs. 51 (b) Rs. 31

(c) Rs. 51 (d) Rs. 41

Ans. (d) Rs. 41

(48) **Find the principal amount, if the compound interest is Rs. 126.10 at 5% per annum for three years :**

(a) Rs. 1,260 (b) Rs. 900

(c) Rs. 1,000 (d) Rs. 800

Ans. (d) Rs. 800

(49) **Find the sum if in 3 years at rate of 5% the compound interest will be Rs. 261 :**

(a) Rs. 7,000 (b) Rs. 8,000

(c) Rs. 9,000 (d) Rs. 5,000

Ans. (d) Rs. 5,000

(50) **The sum of two numbers is 40 and difference is 4. What is the ratio between the numbers?**

(a) 5 : 6 (b) 11 : 9

(c) 3 : 4 (d) 9 : 10

Ans. (b) 11: 9

(51) **Total salary of two employees is Rs. 345. The ratio of salaries is 7 : 8. Find the salary of each.**

(a) 161, 184 (b) 190, 130

(c) 140, 160 (d) 100, 120

Ans. (a) 161, 184

(52) **Gross Profit Ratio = $\frac{?}{\text{Net Sales}} \times 100$. Here missing word is :**

(a) Net profit (b) Profit

(c) Gross Profit (d) Loss

Ans. (c) Gross Profit

(53) **Net Profit Ratio = $\frac{\text{Net Profit}}{?} \times 100$. Find the missing link?**

(a) Profit (b) Loss

(c) Both (d) Net Sales

Ans. (d) Net Sales

(54) **An agent gets 6% commission on total sales and 2% bonus on sales excess than Rs. 10,000. If his total income is Rs. 1,080, find the cost of goods sold?**

(a) 16,000 (b) 12,000

(c) 13,000 (d) 14,000

Ans. (a) Rs. 16,000

(55) **By the help of log table, find the value of 7.2×8.3 ×.94 :**

(a) 54.4 (b) 56.00

(c) 56.16 (d) 60.00

Ans. (c) 56.16

(56) **Find the value of – $\sqrt[7]{0.00487}$**

(a) 4.675 (b) 0.4675

(c) 46.75 (d) 467.5

Ans. (b) 0.4675

(57) **At what rate % will be Rs. 380 amount to Rs. 893 in 15 years?**

(a) 5% (b) 10%

(c) 7% (d) 9%

Ans. (d) 9%

(58) **A sum of money amount to Rs. 2,800 in 2 years and Rs. 3450 in five years with rate of interest 6%. Find the Principal :**

(a) Rs. 1,000 (b) Rs. 2,500

(c) Rs. 5,000 (d) Rs. 500

Ans. (b) Rs. 2,500

(59) **Formula for calculating compound interest is :**

(a) $\frac{P \times R \times T}{100}$
(b) $\left(P + \frac{R}{100}\right)^t$
(c) $P + \left(1 + \frac{R}{100}\right)^t$
(d) $(P \times R)^t \times 100$

Ans. (c) $P + \left(1 + \frac{R}{100}\right)^t$

(60) Find compound interest on Rs. 8,000, if Time = 3 years, Rate = 5% :
(a) Rs. 1,300
(b) Rs. 1,200
(c) 11,000
(d) 1,261

Ans. (d) 1,261

(61) In what time will a sum of money double itself at 4% p. a. compound interest payable half yearly?
(a) 17 years
(b) 17.5 years
(c) 1.7 years
(d) 1.75 years

Ans. (b) 17.5 years

(62) In a class of 42 students, 6 students are girls. The ratio of boys and girls in class is :
(a) 7 : 1
(b) 7 : 6
(c) 6 : 7
(d) 6 : 1

Ans. (d) 6 : 1

(63) The ratio of pass and fail students is 9 : 2. If 20 students fail, number of pass students is :
(a) 180
(b) 220
(c) 110
(d) 90

Ans. (d) 90

(64) Two amounts have a ratio 4 : 3. If the first one is worth Rs. 28, find the other one :
(a) 12
(b) 21
(c) 4
(d) 3

Ans. (b) 21

(65) The sum of two number is 12 and their difference is 8. Find bigger number :
(a) 12
(b) 10
(c) 8
(d) 4

Ans. (b) 10

(66) At what per cent of simple interest a sum would double itself after 10 years?
(a) 5%
(b) 10%
(c) $12\frac{1}{2}$ %
(d) 20%

Ans. (b) 10%

(67) '4/5 of price' mean:
(a) 20%
(b) 40%
(c) 60%
(d) 80%

Ans. (d) 80%

(68) The value of $12\frac{1}{2}$ % of Rs. 500 is :
(a) Rs. 60
(b) Rs. 52
(c) Rs. 62
(d) Rs. 62.50

Ans. (d) Rs. 62.50

(69) What percent of 1 kilogram is 5 gram?
(a) 0.5
(b) 5
(c) 0.005
(d) 5.5

Ans. (a) 0.5

(70) What percent of a day is 3 hours?
(a) 12.5
(b) 22.5
(c) 12
(d) 12.25

Ans. (a) 12.5

(71) A's income is 20% more than B's income. How much less in B's income that A's income?
(a) 20%
(b) 83%
(c) $16\frac{2}{3}$ %
(d) 80%

Ans. (c) $16\frac{2}{3}$ %

(72) 5 litre of 100% acid is added to 5 litre of 20% acid mixture. The quantity of acid in the mixture is :
(a) 50%
(b) 52.2%
(c) 57.5%
(d) 60%

Ans. (d) 60%

(73) If 2x + 20 = 0, then the value of x will be :
(a) –20
(b) –10
(c) 20
(d) 10

Ans. (b) –10

(74) If 3x + 4 = 34, then the value of x is :
(a) 7
(b) 30
(c) 10
(d) 4

Ans. (c) 10

(75) The sum of two numbers is 20 and difference is 4. The numbers are :
(a) 10, 6
(b) 12, 8
(c) 4, 16
(d) 13, 7

Ans. (b) 12, 8

(76) The sum of the digits of a two digit number is 12 and product of the digits is 27. One-third of that number is :

(a) 31 (b) 13
(c) 43 (d) 34

Ans. (a) 31

(77) Fathers age is nine year less than four times the age of his son. If the age of son is 12 years, fathers age will be (in years) :

(a) 40 (b) 39
(c) 37 (d) 35

Ans. (b) 39

(78) A and B have in all 17 shirts. A has 5 shirts more than B. The number of shirts for each is respectively :

(a) 6, 11 (b) 11, 6
(c) 10, 7 (d) 7, 10

Ans. (b) 11, 6

(79) If $\log \frac{x}{y} = x + y$, then the value of $\frac{dy}{dx}$ is :

(a) $\frac{1-x}{x} \cdot \frac{y}{y+1}$ (b) $\frac{xy+1}{x(y+1)}$
(c) $\frac{y(1-y)}{x(1-x)}$ (d) 2

Ans. (a) $\frac{1-x}{x} \cdot \frac{y}{y+1}$

(80) An agent is entitled to a commission of Rs. 2,750@ $5\frac{1}{2}$ % on turnover. Find the turnover:

(a) Rs. 1,000 (b) Rs. 10,000
(c) 5,000 (d) Rs. 50,000

Ans. (d) Rs. 50,000

(81) An agent sold goods worth Rs. 3,640 on which he is given $2\frac{1}{2}$% commission. The amount of his commission is :

(a) Rs. 19 (b) Rs. 91
(c) Rs. 16 (d) Rs. 61

Ans. (b) Rs. 91

(82) An agent sold goods worths Rs. 7,280 on which he is given @ 5% commission. The amount of his commission is :

(a) 728 (b) 346

(c) 364 (d) 264

Ans. (c) 364

(83) For the commission of Rs. 4,500 at 4%, the amount of sales is :

(a) Rs. 1,15,000 (b) Rs. 1,12,500
(c) Rs. 75,000 (d) None of these

Ans. (b) Rs. 1,12,500

(84) An agent charges 5% commission plus 1% del-credere commission. If he sells for Rs. 1,240, his total earning is :

(a) Rs. 62 (b) Rs. 74.40
(c) Rs. 85.20 (d) Rs. 49.69

Ans. (b) Rs. 74.40

(85) An agent received Rs. 600 as commission on the sale of a certain property. If he charged $2\frac{1}{2}$% commission, the amount for which the property is sold, is :

(a) Rs. 12,000 (b) Rs. 48,000
(c) Rs. 15,000 (d) Rs. 24,000

Ans. (d) Rs. 24,000

(86) A person received Rs. 1,000 as commission on the sale of certain property. If the rate of commission is $2\frac{1}{2}$%, the selling price will be :

(a) Rs. 10,000 (b) Rs. 40,000
(c) Rs. 25,000 (d) Rs. 1,00,000

Ans. (b) Rs. 40,000

(87) Commission on sale of Rs. 8,000 @ 8% general commission and 1% del-credere commission is:

(a) Rs. 1,680 (b) Rs. 640
(c) Rs. 1,040 (d) Rs. 1,600

Ans. (a) Rs. 1,680

(88) If a broker gets Rs. 120 as brokerage on a transaction of Rs. 4,000, then rate of brokerage is :

(a) 3% (b) 2%
(c) 1.2% (d) 4%

Ans. (a) 3%

(89) Brokerage on Rs. 8,000 at the rate of 3.5% is (in rupees) :

(a) 280 (b) 208
(c) 118 (d) 108

Ans. (a) 280

(90) Principal got Rs. 202 on the sale of suitcase in Rs. 240. The rate of commission is :

(a) $15\frac{5}{6}\%$ (b) $15\frac{5}{8}\%$

(c) $18\frac{41}{50}5$ (d) $18\frac{5}{6}\%$

Ans. (a) $15\frac{5}{6}\%$

(91) The simple form of ratio of Rs. 4 . 25 : 8 . 50 will be :

(a) 425 : 850 (b) 1 : 2

(c) 4.25 : 825 (d) 17 : 34

Ans. (b) 1 : 2

(92) The greatest ratio among 3 : 4, 13 : 14, 7 : 7 and 15 : 16 is :

(a) 13 : 14 (b) 7 : 8

(c) 15 : 16 (d) 3 : 4

Ans. (c) 15 : 16

(93) Two persons jointly donated 196 grams gold in Prime Minister Relief fund. If one person donated 66 gram gold. What was the ratio of their donation :

(a) 3 : 1 (b) 1 :2

(c) 1 : 3 (d) 2 : 3

Ans. (b) 1 : 2

(94) The $\frac{3}{5}$ of the cost means :

(a) 20% (b) 40%

(c) 60% (d) 80%

Ans. (c) 60%

(95) In an election, one candidate gets 65% votes and he wins by 273 votes. Find the number of votes polled.

(a) 420 (b) 240

(c) 338 (d) 910

Ans. (d) 910

(96) For the commission of Rs. 1,000 at 10% the amount is:

(a) Rs. 9,000 (b) Rs. 10,000

(c) Rs. 11,000 (d) Rs. 10,500

Ans. (b) Rs. 10,000

(97) The arithmetic mean of 9, 18, 27, 36, 54 is :

(a) 18 (b) 27

(c) 36 (d) 45

Ans. (b) 27

□□□